

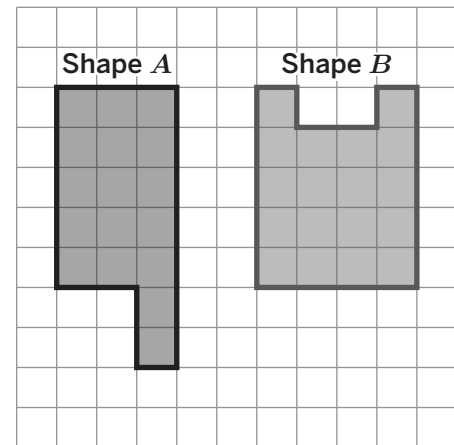
Additional Practice

ACC6.1.01

1. Which shape has a greater area?
Show or explain your thinking.

- A. Shape *A*
- B. Shape *B***
- C. They have the same area

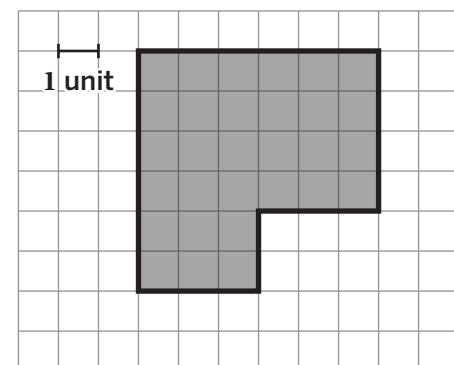
Explanations vary. Shape A has an area of 17 square units. Shape B has an area of 18 square units.



Problems 2–4: Here is a new shape.

2. Determine the area of the shape.
Show or explain your thinking.

30 square units. Explanations vary. I split the shape into two rectangles by drawing a line across it. The top rectangle has an area of $6 \cdot 4 = 24$ square units. The bottom rectangle has an area of $3 \cdot 2 = 6$ square units. $24 + 6 = 30$ square units



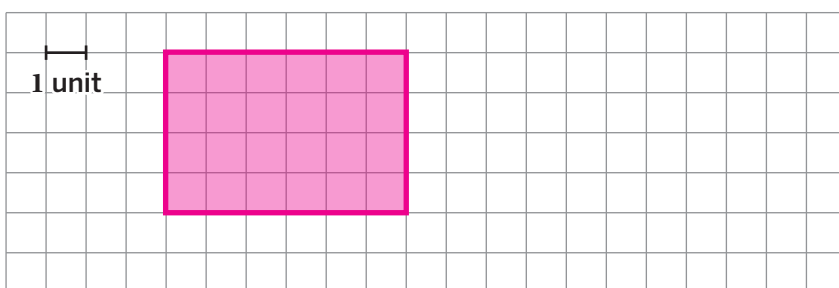
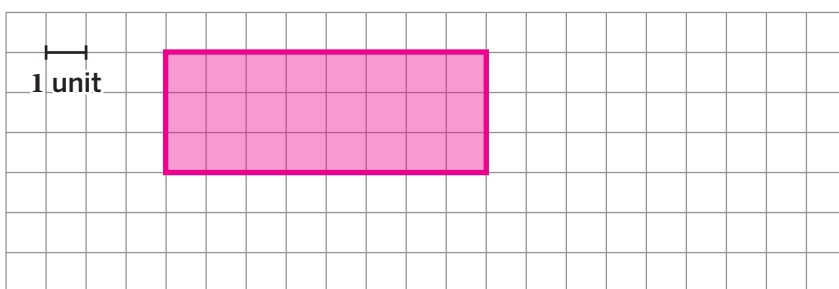
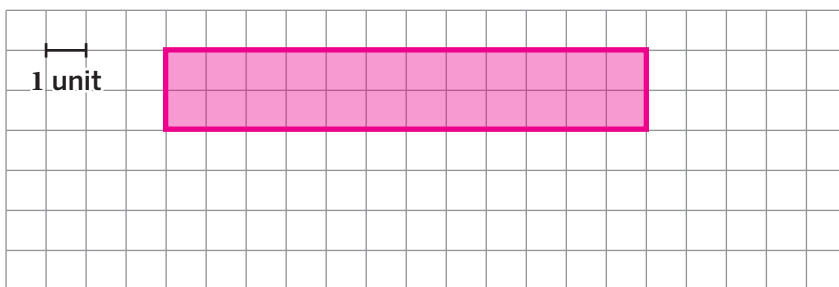
3. Show or describe another way to determine the area of this shape.

Responses vary. I can draw a large square around the shape that has an area of $6 \cdot 6 = 36$ square units, then subtract the unshaded area. $36 - 6 = 30$ square units

4. Show or describe how you would change this shape so it has an area of 20 square units.

Responses vary. I would cut out a rectangle with a height of 5 units and a base of 2 units. Then the area would be $30 - 10 = 20$ square units.

5. Draw *three* different quadrilaterals, each with an area of 24 square units.
 Each square in this grid has an area of 1 square unit. **Responses vary. Samples shown on grids.**



Name: _____ Date: _____ Period: _____

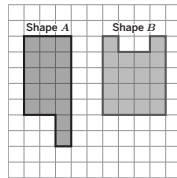
Additional Practice

ACC6.1.01

1. Which shape has a greater area?
Show or explain your thinking.

- A. Shape A
B. Shape B
C. They have the same area

Explanations vary. Shape A has an area of 17 square units. Shape B has an area of 18 square units.

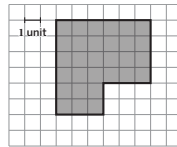


Problems 2–4: Here is a new shape.

2. Determine the area of the shape.

Show or explain your thinking.

30 square units. Explanations vary. I split the shape into two rectangles by drawing a line across it. The top rectangle has an area of $6 \cdot 4 = 24$ square units. The bottom rectangle has an area of $3 \cdot 2 = 6$ square units. $24 + 6 = 30$ square units.



3. Show or describe another way to determine the area of this shape.

Responses vary. I can draw a large square around the shape that has an area of $6 \cdot 6 = 36$ square units, then subtract the unshaded area. $36 - 6 = 30$ square units.

4. Show or describe how you would change this shape so it has an area of 20 square units.

Responses vary. I would cut out a rectangle with a height of 5 units and a base of 2 units. Then the area would be $30 - 10 = 20$ square units.

Unit 1 Lesson 1

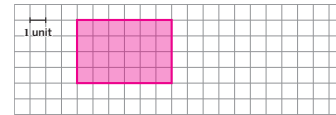
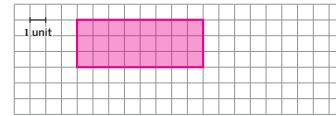
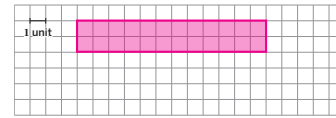
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5. Draw three different quadrilaterals, each with an area of 24 square units.

Each square in this grid has an area of 1 square unit. Responses vary. Samples shown on grids.



Unit 1 Lesson 1

4

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	6.G.A.1
2	2	6.G.A.1
3	2	6.G.A.1
4	2	6.G.A.1
5	2	6.G.A.1

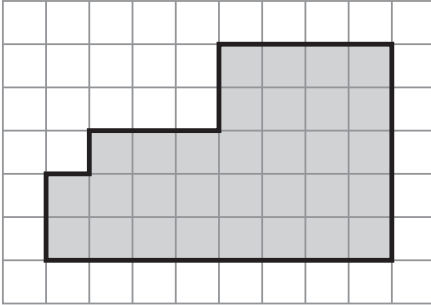
Notes:

Additional Practice

ACC6.1.02

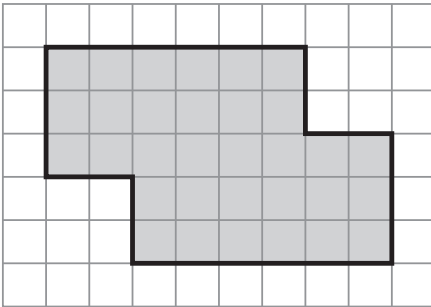
1. Determine the total area of each shaded region.

a



31 square units

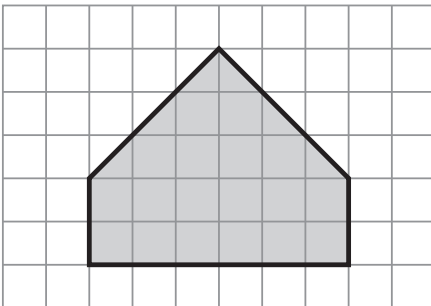
b



32 square units

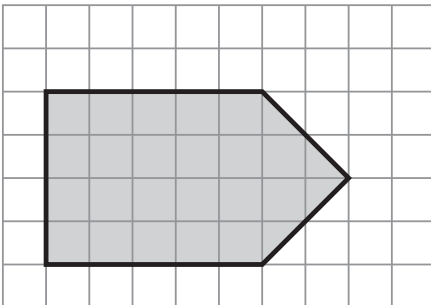
2. Determine the total area of each shaded region.

a



21 square units

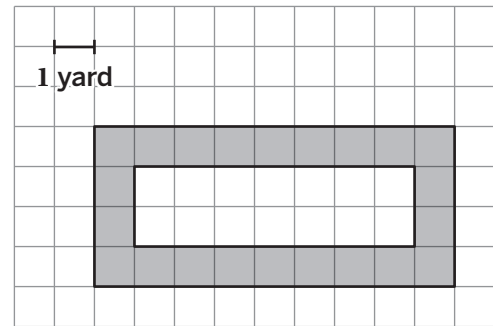
b



24 square units

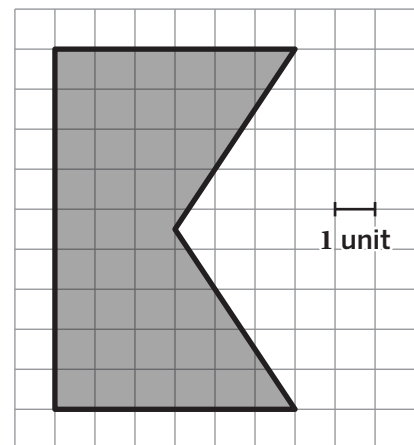
3. Determine the total area of the shaded region.

22 square yards



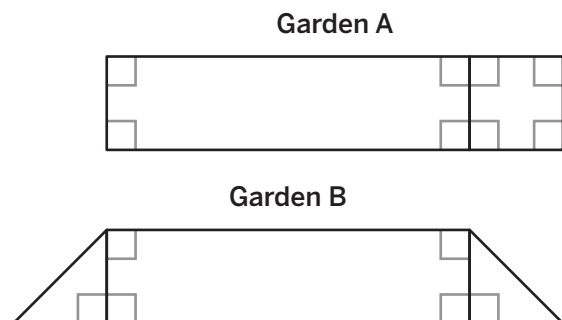
4. Determine the area of the shape.

40.5 square units



5. Two gardens have very different shapes. Clare said that both gardens have the same area. Is Clare correct? Explain your thinking.

Yes. Explanations vary. The two triangles from Garden B form the same-sized square as the one attached to Garden A. So, the area of the two gardens is the same.

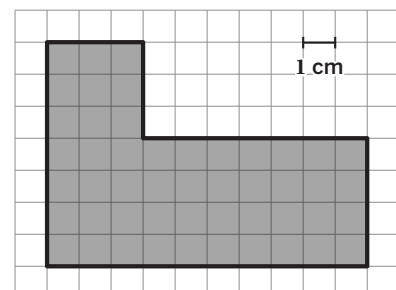


6. Clare calculated the area of this figure. Her work is shown. Is Clare correct? Explain your thinking.

Clare

$$\begin{aligned} (10 \times 4) - (3 \times 3) \\ = 40 - 9 \\ = 31 \text{ sq. cm} \end{aligned}$$

No. Explanations vary. Clare should have added the areas of each rectangle; $40 + 9 = 49$ square centimeters.

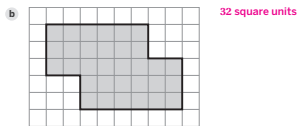


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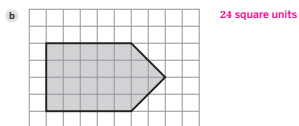
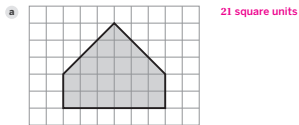
Additional Practice

ACC6.1.02

1. Determine the total area of each shaded region.



2. Determine the total area of each shaded region.



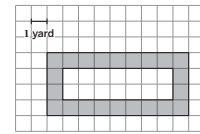
Unit 1 Lesson 2

5

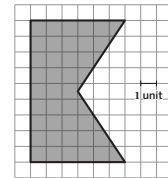
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Name: _____ Date: _____ Period: _____

3. Determine the total area of the shaded region.
22 square yards



4. Determine the area of the shape.
40.5 square units



5. Two gardens have very different shapes. Clare said that both gardens have the same area. Is Clare correct? Explain your thinking.



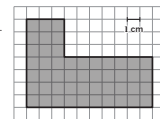
Yes. **Explanations vary.** The two triangles from Garden B form the same-sized square as the one attached to Garden A. So, the area of the two gardens is the same.

6. Clare calculated the area of this figure. Her work is shown. Is Clare correct? Explain your thinking.

Clare

$$\begin{aligned} (10 \times 4) - (3 \times 3) \\ = 40 - 9 \\ = 31 \text{ sq. cm} \end{aligned}$$

No. **Explanations vary.** Clare should have added the areas of each rectangle; $40 + 9 = 49$ square centimeters.



Unit 1 Lesson 2

6

Additional Practice

Practice Problem Analysis

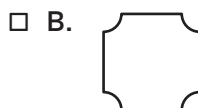
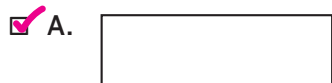
Problem	DOK	Standard(s)
1	2	6.G.A.1
2	2	6.G.A.1
3	2	6.G.A.1
4	2	6.G.A.1
5	2	6.G.A.1
6	2	6.G.A.1

Notes:

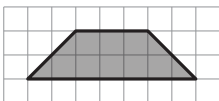
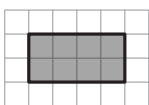
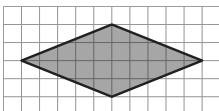
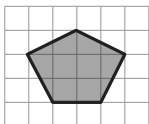
Additional Practice

ACC6.1.03

1. Select *all* of the shapes that appear to be parallelograms.



2. Determine whether each figure is a parallelogram. Explain your thinking.

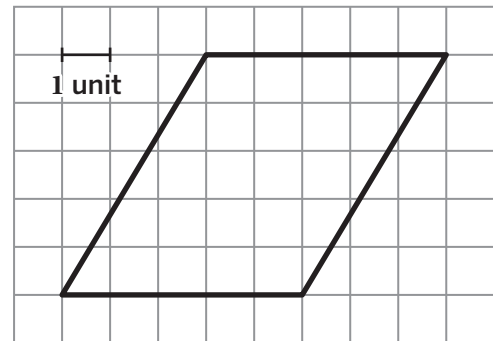
Figure	Parallelogram? (yes/no)	Explanation
a 	No	<i>Explanations vary. This quadrilateral does not have two pairs of parallel sides, so it is not a parallelogram.</i>
b 	Yes	<i>Explanations vary. This quadrilateral is a rectangle. A rectangle is a parallelogram because it has two pairs of parallel sides, and each pair of opposite sides has the same length.</i>
c 	Yes	<i>Explanations vary. This quadrilateral has two pairs of parallel sides that have the same length.</i>
d 	No	<i>Explanations vary. This polygon has 5 sides, and parallelograms must have four sides.</i>

3. Select *all* the true statements.

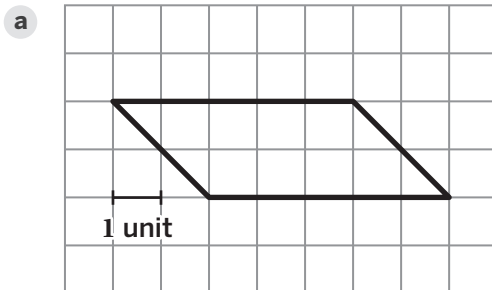
- ☒ A. A parallelogram can be rearranged to form a rectangle.
- ☐ B. A parallelogram has exactly one pair of parallel sides.
- ☐ C. A parallelogram can have more than four sides.
- ☒ D. In a parallelogram, opposite sides have the same length.
- ☒ E. A square is a parallelogram.

4. Use any strategy to determine the area of the parallelogram.

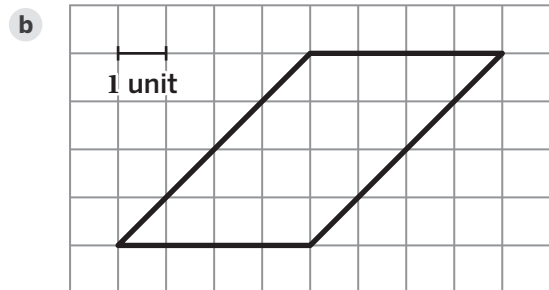
25 square units



5. Use any strategy to determine the area of each parallelogram.



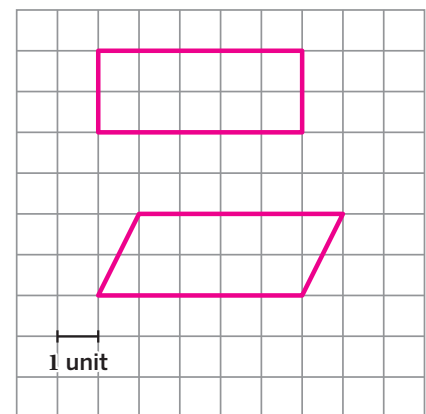
10 square units



16 square units

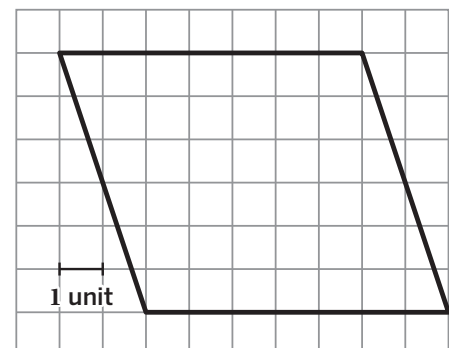
6. Draw two different parallelograms on the grid, each with an area of 10 square units.

Responses vary. Sample shown on grid.



7. Han says he can determine the area of this parallelogram by cutting part of the left side and moving it to the right side to form a rectangle, and that the area of this parallelogram is 36 square units. Is Han correct? Explain your thinking.

No. Explanations vary. The area of this parallelogram is 42 square units, not 36 square units, because $7 \times 6 = 42$. Han is correct that he can cut part of the left side and move it to the right side to form a rectangle.

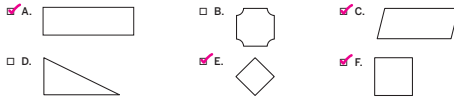


Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.1.03

1. Select *all* of the shapes that appear to be parallelograms.



2. Determine whether each figure is a parallelogram. Explain your thinking.

Figure	Parallelogram? (yes/no)	Explanation
a. 	No	Explanations vary. This quadrilateral does not have two pairs of parallel sides, so it is not a parallelogram.
b. 	Yes	Explanations vary. This quadrilateral is a rectangle. A rectangle is a parallelogram because it has two pairs of parallel sides, and each pair of opposite sides has the same length.
c. 	Yes	Explanations vary. This quadrilateral has two pairs of parallel sides that have the same length.
d. 	No	Explanations vary. This polygon has 5 sides, and parallelograms must have four sides.

3. Select *all* the true statements.

- ☒ A. A parallelogram can be rearranged to form a rectangle.
☐ B. A parallelogram has exactly one pair of parallel sides.
☐ C. A parallelogram can have more than four sides.
☒ D. In a parallelogram, opposite sides have the same length.
☒ E. A square is a parallelogram.

Unit 1 Lesson 3

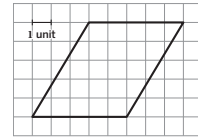
7

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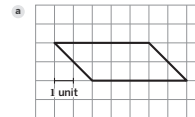
Name: _____ Date: _____ Period: _____

4. Use any strategy to determine the area of the parallelogram.

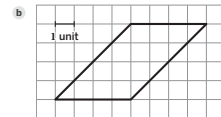
25 square units



5. Use any strategy to determine the area of each parallelogram.



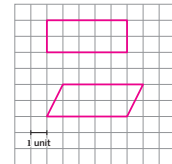
10 square units



16 square units

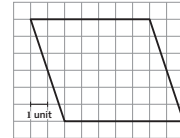
6. Draw two different parallelograms on the grid, each with an area of 10 square units.

Responses vary. Sample shown on grid.



7. Han says he can determine the area of this parallelogram by cutting part of the left side and moving it to the right side to form a rectangle, and that the area of this parallelogram is 36 square units. Is Han correct? Explain your thinking.

No. Explanations vary. The area of this parallelogram is 42 square units, not 36 square units, because $7 \times 6 = 42$. Han is correct that he can cut part of the left side and move it to the right side to form a rectangle.



Unit 1 Lesson 3

8

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	6.G.A.1
2	1	6.G.A.1
3	1	6.G.A.1
4	1	6.G.A.1
5	1	6.G.A.1
6	2	6.G.A.1
7	2	6.G.A.1

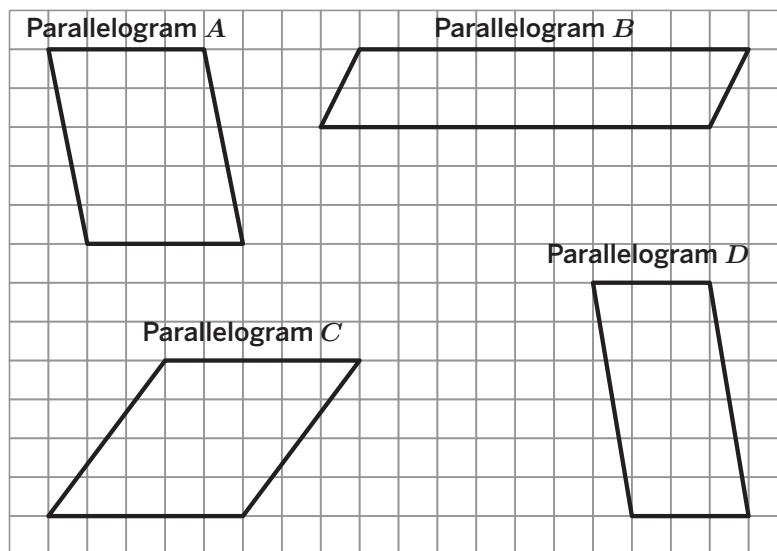
Notes:

Additional Practice

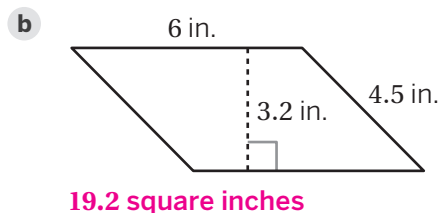
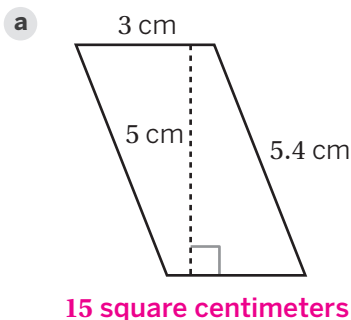
ACC6.1.04

1. Three of these parallelograms have the same area. Which parallelogram has a *different* area than the others?

- A. Parallelogram A
- B. Parallelogram B
- C. Parallelogram C
- D. Parallelogram D**



2. Determine the area of each parallelogram.



3. The base lengths, b , and corresponding heights, h , are listed for two different parallelograms. Determine the area of each parallelogram.

a $b = 12.5$ in., $h = 9$ in.
112.5 square inches

b $b = 6$ cm, $h = 5.5$ cm
33 square centimeters

4. The base lengths, b , and corresponding heights, h , of four different parallelograms are listed. Which base-height pair represents the parallelogram with the greatest area?

- A. $b = 8$ units, $h = 3.2$ units
- B. $b = 5.6$ units, $h = 4$ units
- D. $b = 7.4$ units, $h = 4$ units**
- C. $b = 10$ units, $h = 2.6$ units

5. The base lengths, b , and corresponding areas, A , of four different parallelograms are listed. Determine the height of each parallelogram.

a $b = 12$ units, $A = 84$ square units
 $h = 7$ units

b $b = 6$ units, $A = 54$ square units
 $h = 9$ units

c $b = 5.5$ units, $A = 33$ square units
 $h = 6$ units

d $b = 8.2$ units, $A = 28.7$ square units
 $h = 3.5$ units

6. The heights, h , and corresponding areas, A , of four different parallelograms are listed. Determine the length of the base of each parallelogram.

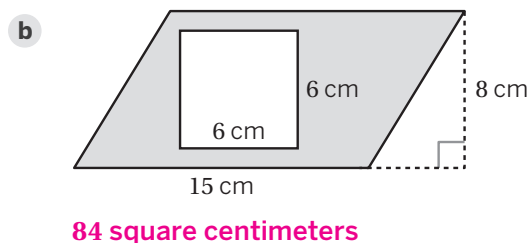
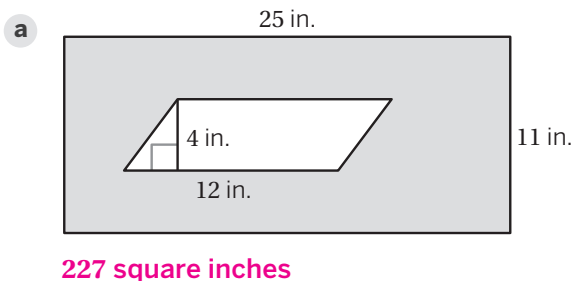
a $h = 4$ units, $A = 16$ square units
 $b = 4$ units

b $h = 7$ units, $A = 35$ square units
 $b = 5$ units

c $h = 2.5$ units, $A = 10$ square units
 $b = 4$ units

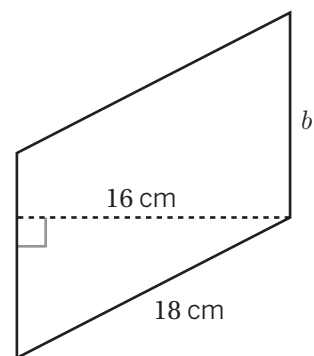
d $h = 8$ units, $A = 28$ square units
 $b = 3.5$ units

7. Determine the area of the shaded region in each figure.



8. The parallelogram shown has an area of 144 square centimeters. Mai claims the length of the base, b , is 8 centimeters. Is Mai correct? Explain your thinking.

No. Explanations vary. Mai did not divide the area of the parallelogram by the height, 16 centimeters. Because $144 \div 16 = 9$, the length of the base b is 9 centimeters.



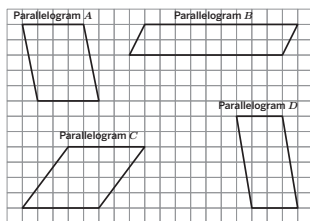
Name: _____ Date: _____ Period: _____

Additional Practice

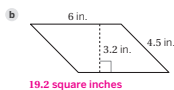
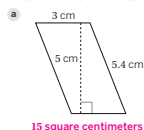
ACC6.1.04

1. Three of these parallelograms have the same area. Which parallelogram has a different area than the others?

- A. Parallelogram A
B. Parallelogram B
C. Parallelogram C
D. Parallelogram D



2. Determine the area of each parallelogram.



3. The base lengths, b , and corresponding heights, h , are listed for two different parallelograms. Determine the area of each parallelogram.

a $b = 12.5$ in., $h = 9$ in.
112.5 square inches

b $b = 6$ cm, $h = 5.5$ cm
33 square centimeters

4. The base lengths, b , and corresponding heights, h , of four different parallelograms are listed. Which base-height pair represents the parallelogram with the greatest area?

A. $b = 8$ units, $h = 3.2$ units
C. $b = 10$ units, $h = 2.6$ units

B. $b = 5.6$ units, $h = 4$ units
D. $b = 7.4$ units, $h = 4$ units

Unit 1 Lesson 4

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Name: _____ Date: _____ Period: _____

5. The base lengths, b , and corresponding areas, A , of four different parallelograms are listed. Determine the height of each parallelogram.

a $b = 12$ units, $A = 84$ square units
 $h = 7$ units

b $b = 6$ units, $A = 54$ square units
 $h = 9$ units

c $b = 5.5$ units, $A = 33$ square units
 $h = 6$ units

d $b = 8.2$ units, $A = 28.7$ square units
 $h = 3.5$ units

6. The heights, h , and corresponding areas, A , of four different parallelograms are listed. Determine the length of the base of each parallelogram.

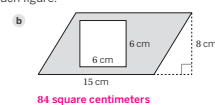
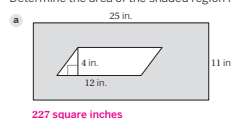
a $h = 4$ units, $A = 16$ square units
 $b = 4$ units

b $h = 7$ units, $A = 35$ square units
 $b = 5$ units

c $h = 2.5$ units, $A = 10$ square units
 $b = 4$ units

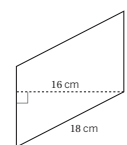
d $h = 8$ units, $A = 28$ square units
 $b = 3.5$ units

7. Determine the area of the shaded region in each figure.



8. The parallelogram shown has an area of 144 square centimeters.

Mai claims the length of the base, b , is 8 centimeters. Is Mai correct? Explain your thinking.
No. Explanations vary. Mai did not divide the area of the parallelogram by the height, 16 centimeters. Because $144 \div 16 = 9$, the length of the base b is 9 centimeters.



Unit 1 Lesson 4

10

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	6.G.A.1
2	1	6.G.A.1
3	2	6.G.A.1
4	2	6.G.A.1
5	2	6.G.A.1
6	2	6.G.A.1
7	2	6.G.A.1
8	2	6.G.A.1

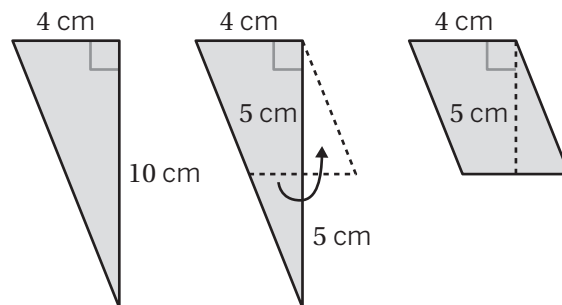
Notes:

Additional Practice

ACC6.1.05

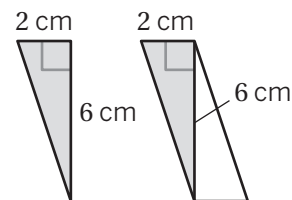
- To determine the area of this triangle, Jada decomposed and rearranged it into a parallelogram. Explain how Jada could use her parallelogram to determine the area of the triangle.

Explanations vary. Jada could find the area of the parallelogram by multiplying the base by the height. The area of the parallelogram is $4 \cdot 5 = 20$ square centimeters. The triangle is composed of the same pieces, so its area is also 20 square centimeters.

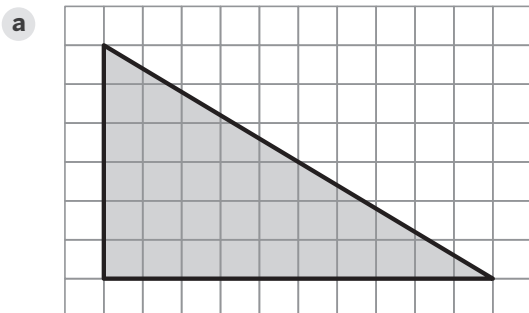


- To determine the area of this triangle, Lin used two identical copies of the triangle to create a parallelogram. Explain how Lin could use her parallelogram to determine the area of the triangle.

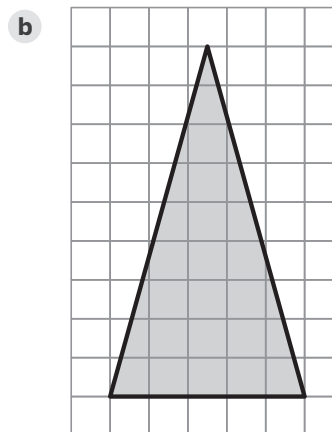
Explanations vary. The area of the parallelogram is $2 \cdot 6 = 12$ square centimeters. The triangle's area is exactly half the area of the parallelogram. So, the area of the triangle is $\frac{1}{2} \cdot 12 = 6$ square centimeters.



- Determine the area of each triangle. Each small square in the grid has an area of 1 square unit.



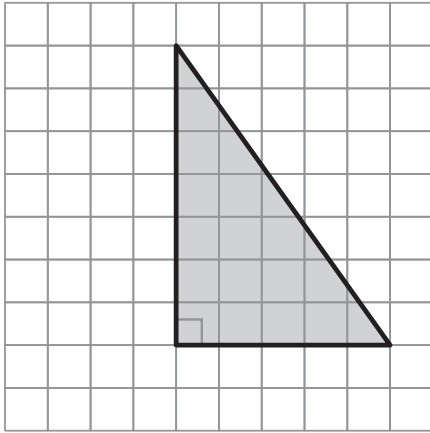
30 square units



22.5 square units

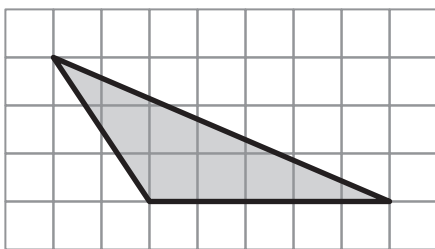
Problems 4–6: Determine the area of each triangle. Show or explain your thinking. Each small square in the grid has an area of 1 square unit.

4.



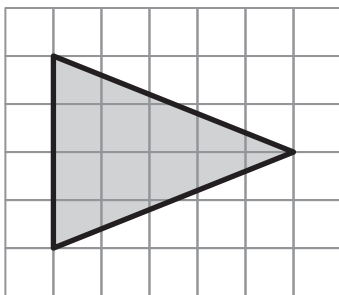
17.5 square units. *Explanations vary.* I composed a parallelogram using two identical copies of the triangle. The area of the parallelogram is $5 \cdot 7 = 35$ square units. Each triangle is half the area of the parallelogram, and $35 \div 2 = 17.5$ square units.

5.



7.5 square units. *Explanations vary.* I composed a parallelogram using two identical copies of the triangle. $b \cdot h = 5 \cdot 3 = 15$ square units. Each triangle is half the area of the parallelogram. Because half of 15 is 7.5, the area of the triangle is 7.5 square units.

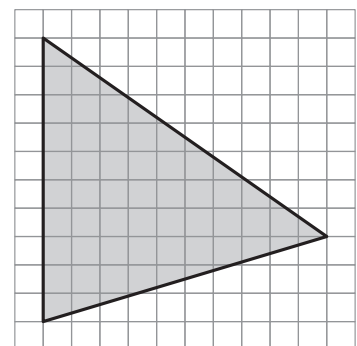
6.



10 square units. *Explanations vary.* I decomposed the triangle by taking the part of the triangle from its three rightmost columns and placing it underneath the remaining triangle on the left side to compose a parallelogram. The area of the parallelogram is 10 square units, which has the same area of the triangle. So, the area of the triangle is 10 square units.

7. Shawn says the area of this triangle is 100 square units. Is Shawn correct? Explain your thinking.

No. *Explanations vary.* Shawn found the area of a parallelogram but should have divided by 2 to find the area of the triangle. The area is $100 \div 2 = 50$ square units.

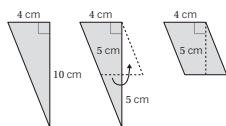


Name: _____ Date: _____ Period: _____

Additional Practice

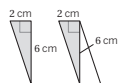
ACC6.1.05

1. To determine the area of this triangle, Jada decomposed and rearranged it into a parallelogram. Explain how Jada could use her parallelogram to determine the area of the triangle.



Explanations vary. Jada could find the area of the parallelogram by multiplying the base by the height. The area of the parallelogram is $4 \cdot 5 = 20$ square centimeters. The triangle is composed of the same pieces, so its area is also 20 square centimeters.

2. To determine the area of this triangle, Lin used two identical copies of the triangle to create a parallelogram. Explain how Lin could use her parallelogram to determine the area of the triangle.

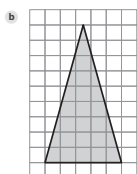


Explanations vary. The area of the parallelogram is $2 \cdot 6 = 12$ square centimeters. The triangle's area is exactly half the area of the parallelogram. So, the area of the triangle is $\frac{1}{2} \cdot 12 = 6$ square centimeters.

3. Determine the area of each triangle. Each small square in the grid has an area of 1 square unit.



30 square units



22.5 square units

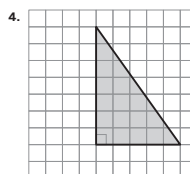
Unit 1 Lesson 5

11

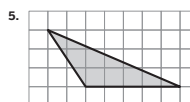
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Name: _____ Date: _____ Period: _____

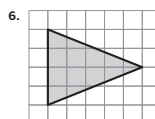
Problems 4–6: Determine the area of each triangle. Show or explain your thinking. Each small square in the grid has an area of 1 square unit.



17.5 square units. *Explanations vary.* I composed a parallelogram using two identical copies of the triangle. The area of the parallelogram is $5 \cdot 7 = 35$ square units. Each triangle is half the area of the parallelogram, and $35 \div 2 = 17.5$ square units.



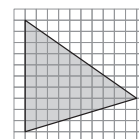
7.5 square units. *Explanations vary.* I composed a parallelogram using two identical copies of the triangle. $5 \cdot 3 = 15$ square units. Each triangle is half the area of the parallelogram. Because half of 15 is 7.5, the area of the triangle is 7.5 square units.



10 square units. *Explanations vary.* I decomposed the triangle by taking the part of the triangle from its three rightmost columns and placing it underneath the remaining triangle on the left side to compose a parallelogram. The area of the parallelogram is 10 square units, which has the same area of the triangle. So, the area of the triangle is 10 square units.

7. Shawn says the area of this triangle is 100 square units. Is Shawn correct? Explain your thinking.

No. Explanations vary. Shawn found the area of a parallelogram but should have divided by 2 to find the area of the triangle. The area is $100 \div 2 = 50$ square units.



Unit 1 Lesson 5

12

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	6.G.A.1
2	2	6.G.A.1
3	2	6.G.A.1
4	2	6.G.A.1
5	2	6.G.A.1
6	2	6.G.A.1
7	2	6.G.A.1

Notes:

Additional Practice

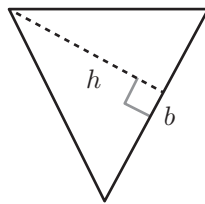
ACC6.1.06

1. Select *all* the statements that are true about triangles.

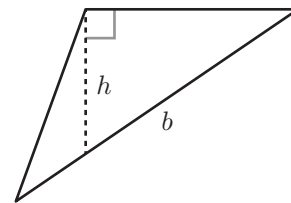
- ☒ A. Any side of a triangle can be a base.
- ☐ B. The height must be drawn inside the triangle.
- ☒ C. A height that corresponds to the base of a triangle is always perpendicular to the base.
- ☒ D. A height that corresponds to the base of a triangle is always drawn at a right angle to the base.
- ☒ E. The height is sometimes drawn outside the triangle.

2. Select *all* of the triangles that have a correct height labeled for the given base.

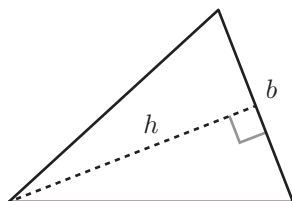
☒ A. Triangle A



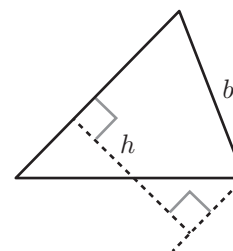
☐ B. Triangle B



☒ C. Triangle C

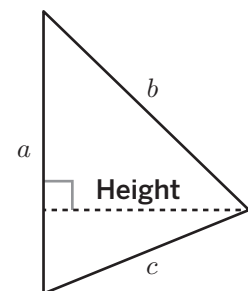


☐ D. Triangle D



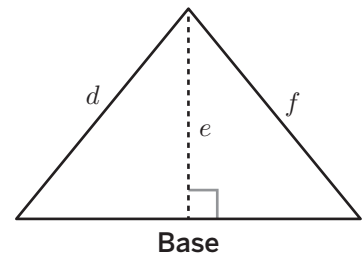
3. Which is a matching base for the indicated height of the triangle?

- ☒ A. Side a
- ☐ B. Side b
- ☐ C. Side c



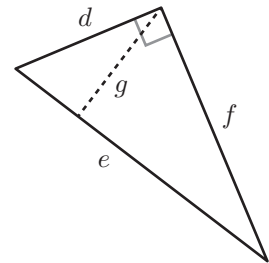
4. Which is a height that matches the given base of the triangle?

- A. Side d
B. Length e
 C. Side f



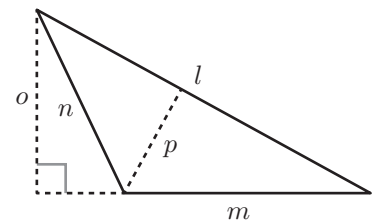
5. Name a matching height for each indicated base.

- a Side d **Side f is a corresponding height.**
 b Side e **Length g is a corresponding height.**
 c Side f **Side d is a corresponding height.**



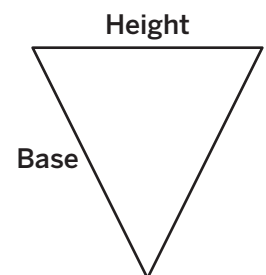
6. Name a matching height for each indicated base.

- a Side l **Length p is a corresponding height.**
 b Side m **Length o is a corresponding height.**



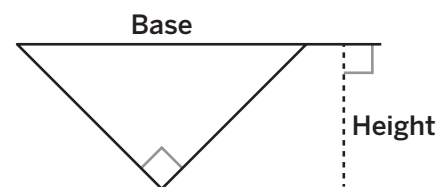
7. Tyler labeled the base and height of the triangle. Are Tyler's labels correct? Explain your thinking.

No. Explanations vary. Tyler did not correctly draw a corresponding height to the base that he identified. The height needs to be drawn perpendicular to the identified base.



8. Elena labeled the base and height of the triangle. Are Elena's labels correct? Explain your thinking.

Yes. Explanations vary. Elena correctly labeled a base and its corresponding height because the height is drawn perpendicular to the base.



Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.1.06

1. Select *all* the statements that are true about triangles.

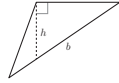
- ☒ A. Any side of a triangle can be a base.
- ☐ B. The height must be drawn inside the triangle.
- ☒ C. A height that corresponds to the base of a triangle is always perpendicular to the base.
- ☒ D. A height that corresponds to the base of a triangle is always drawn at a right angle to the base.
- ☒ E. The height is sometimes drawn outside the triangle.

2. Select *all* of the triangles that have a correct height labeled for the given base.

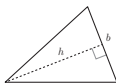
☒ A. Triangle A



☐ B. Triangle B



☒ C. Triangle C

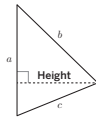


☐ D. Triangle D



3. Which is a matching base for the indicated height of the triangle?

- ☒ A. Side a
- ☐ B. Side b
- ☐ C. Side c



Unit 1 Lesson 6

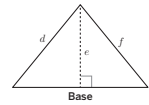
13

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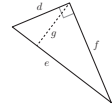
4. Which is a height that matches the given base of the triangle?

- ☐ A. Side d
- ☒ B. Length e
- ☐ C. Side f



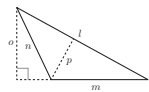
5. Name a matching height for each indicated base.

- ☐ a. Side d **Side f is a corresponding height.**
- ☐ b. Side e **Length g is a corresponding height.**
- ☐ c. Side f **Side d is a corresponding height.**



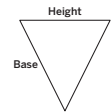
6. Name a matching height for each indicated base.

- ☐ a. Side l **Length p is a corresponding height.**
- ☐ b. Side m **Length o is a corresponding height.**



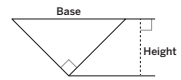
7. Tyler labeled the base and height of the triangle. Are Tyler's labels correct? Explain your thinking.

No. Explanations vary. Tyler did not correctly draw a corresponding height to the base that he identified. The height needs to be drawn perpendicular to the identified base.



8. Elena labeled the base and height of the triangle. Are Elena's labels correct? Explain your thinking.

Yes. Explanations vary. Elena correctly labeled a base and its corresponding height because the height is drawn perpendicular to the base.



Unit 1 Lesson 6

14

Additional Practice

Practice Problem Analysis

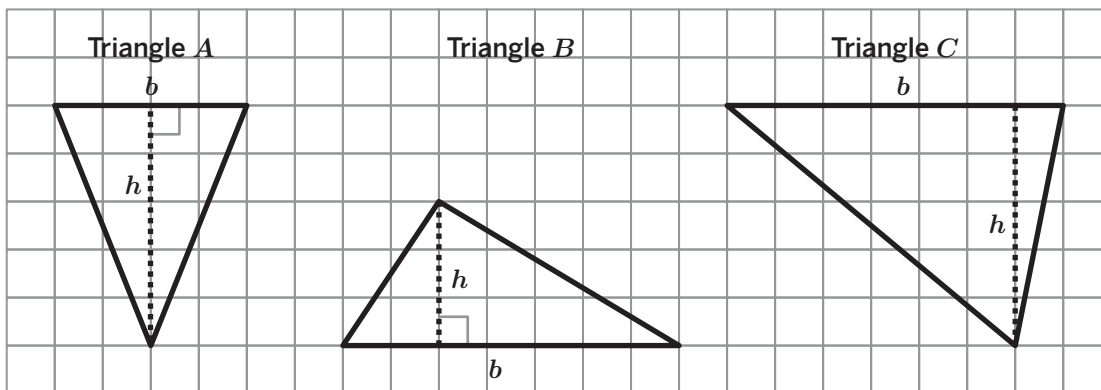
Problem	DOK	Standard(s)
1	1	6.G.A.1
2	2	6.G.A.1
3	2	6.G.A.1
4	2	6.G.A.1
5	2	6.G.A.1
6	2	6.G.A.1
7	2	6.G.A.1
8	2	6.G.A.1

Notes:

Additional Practice

ACC6.1.07

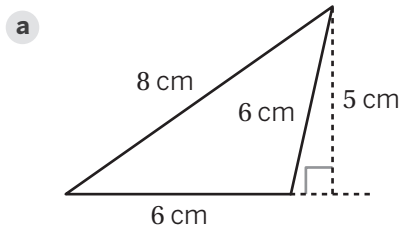
1. For each triangle, a base, b , and its corresponding height, h , are labeled. Each small square in the grid has an area of 1 square unit.



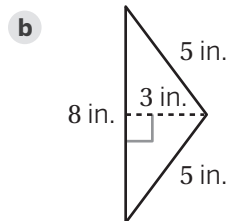
Complete the table.

	Base (units)	Height (units)	Area (square units)
Triangle A	4	5	10
Triangle B	7	3	10.5
Triangle C	7	5	17.5
Any Triangle	b	h	$\frac{1}{2} \cdot b \cdot h$

2. Determine the area of each triangle.



15 square centimeters



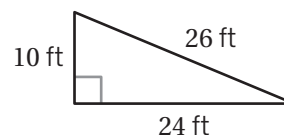
12 square inches

3. Use the given bases and heights to determine the area of each triangle in the table.

	Base (units)	Height (units)	Area (square units)
Triangle A	8	6	24
Triangle B	16	5	40
Triangle C	3	3	4.5
Triangle D	10.5	5	26.25

4. Determine the area of the triangle. Show your thinking.

120 square feet. Explanations vary. The area of a triangle is $\frac{1}{2} \cdot b \cdot h$. I substituted 24 for the base and 10 for the height, and the product is 240. Half of the product is 120, so the area is 120 square feet.



5. A triangle has a base with a length of 24.8 centimeters and a height of 16 centimeters. Determine its area.

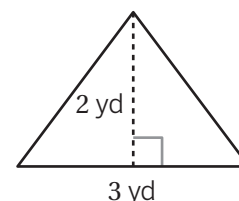
198.4 square centimeters

6. A triangle has a base with a length of 7 inches and a height of 12 inches. Determine its area.

42 square inches

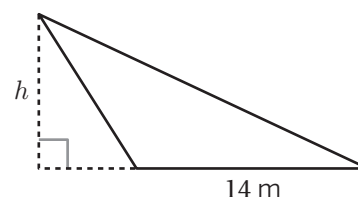
7. A herb garden in Bard's backyard is shaped like a triangle with the dimensions shown. Determine the area of the herb garden.

3 square yards



8. A triangle has a base of 14 meters and an area of 70 square meters. What is the height of this triangle? Explain your thinking.

10 meters. Explanations vary. If the area of the triangle is 70 square meters, then I can use the area formula and substitute 70 for A and substitute 14 for the base, and solve for h : $70 = \frac{1}{2} \cdot 14 \cdot h$; the height of the triangle is 10 meters.

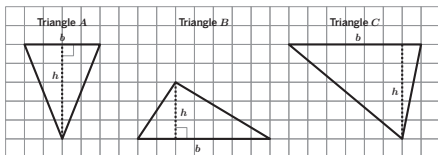


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Additional Practice

ACC6.1.07

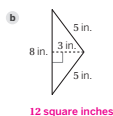
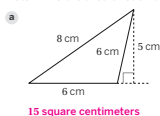
1. For each triangle, a base, b , and its corresponding height, h , are labeled. Each small square in the grid has an area of 1 square unit.



Complete the table.

	Base (units)	Height (units)	Area (square units)
Triangle A	4	5	10
Triangle B	7	3	10.5
Triangle C	7	5	17.5
Any Triangle	b	h	$\frac{1}{2} \cdot b \cdot h$

2. Determine the area of each triangle.



Unit 1 Lesson 7

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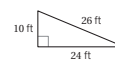
Name: _____ Date: _____ Period: _____

3. Use the given bases and heights to determine the area of each triangle in the table.

	Base (units)	Height (units)	Area (square units)
Triangle A	8	6	24
Triangle B	16	5	40
Triangle C	3	3	4.5
Triangle D	10.5	5	26.25

4. Determine the area of the triangle. Show your thinking.

120 square feet. *Explanations vary. The area of a triangle is $\frac{1}{2} \cdot b \cdot h$. I substituted 24 for the base and 10 for the height, and the product is 240. Half of the product is 120, so the area is 120 square feet.*



5. A triangle has a base with a length of 24.8 centimeters and a height of 16 centimeters. Determine its area.

198.4 square centimeters

6. A triangle has a base with a length of 7 inches and a height of 12 inches. Determine its area.

42 square inches

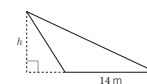
7. A herb garden in Bard's backyard is shaped like a triangle with the dimensions shown. Determine the area of the herb garden.

3 square yards



8. A triangle has a base of 14 meters and an area of 70 square meters. What is the height of this triangle? Explain your thinking.

10 meters. *Explanations vary. If the area of the triangle is 70 square meters, then I can use the area formula and substitute 70 for A and substitute 14 for the base, and solve for h : $70 = \frac{1}{2} \cdot 14 \cdot h$; the height of the triangle is 10 meters.*



Unit 1 Lesson 7

16

Additional Practice

Practice Problem Analysis

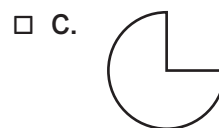
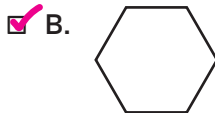
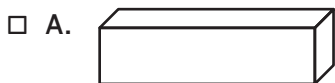
Problem	DOK	Standard(s)
1	2	6.G.A.1
2	2	6.G.A.1
3	2	6.G.A.1
4	2	6.G.A.1
5	2	6.G.A.1
6	2	6.G.A.1
7	2	6.G.A.1
8	2	6.G.A.1

Notes:

Additional Practice

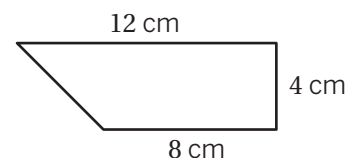
ACC6.1.08

1. Select *all* the polygons.



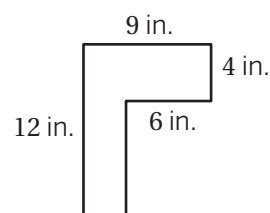
2. Use any strategy to determine the area of the polygon.
Explain your thinking.

40 square centimeters. Explanations vary. I split this shape into a right triangle and a rectangle. The area of the triangle is $\frac{1}{2} \cdot 4 \cdot 4 = 8$, and the area of the rectangle is $8 \cdot 4 = 32$. Then I combined the areas, $8 + 32 = 40$ square centimeters.



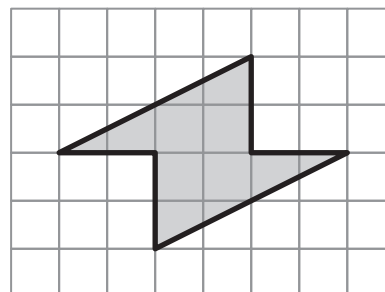
3. Use any strategy to determine the area of the polygon.
Explain your thinking.

60 square inches. Explanations vary. I partitioned the polygon into two rectangles. The first rectangle has a length of 9 and a width of 4, with an area of 36. The second rectangle has a length of 8 and a width of 3, with an area of 24. Then I combined the areas, $36 + 24 = 60$ square inches.



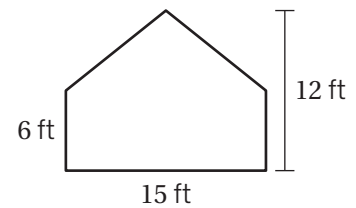
4. Use any strategy to determine the area of the polygon.
Explain your thinking.

8 square units. Explanations vary. I decomposed this shape into two triangles. Each triangle has a base length of 4 units and a height of 2 units. I used the area formula to find the area of each triangle, then I multiplied by two because there are two triangles. $\frac{1}{2} \cdot 4 \cdot 2 = 4$; $2 \cdot 4 = 8$ square units



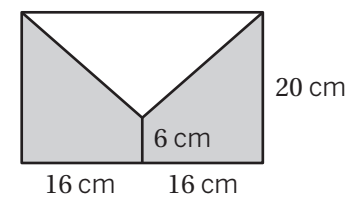
5. Use any strategy to determine the area of the polygon.
Explain your thinking.

135 square feet. Explanations vary. I decomposed the polygon into a triangle and a rectangle. The base length of the triangle is 15 and its height is 6. I found the area of the triangle using $\frac{1}{2} \cdot 15 \cdot 6 = 45$ square feet. The area of the rectangle is $15 \cdot 6 = 90$ square feet. The total area is 135 square feet.



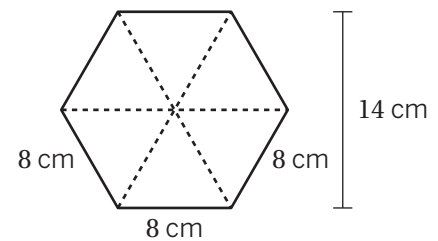
6. Determine the area of the shaded region.
Explain your thinking.

416 square centimeters. Explanations vary. First, I found the area of the rectangle: $32 \cdot 20 = 640$. Next, I found the area of the triangle: $\frac{1}{2} \cdot 32 \cdot 14 = 224$. Finally, I subtracted the area of the triangle from the area of the rectangle to find the area of the shaded region: $640 - 224 = 416$ square centimeters.



7. The hexagon has a side length of 8 centimeters. Diego says: *The area is 28 square centimeters because each triangle has a base of 8 centimeters and a height of 7 centimeters. If I multiply 8 by 7 and divide the product by 2, I get 28 square centimeters. Is Diego correct?*
Explain your thinking.

No. Explanations vary. Diego found the area of one of the triangles. He needs to multiply the area of one of the triangles by 6 to find the total area: $28 \cdot 6 = 168$ square centimeters.

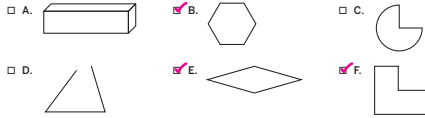


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Additional Practice

ACC6.1.08

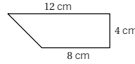
1. Select *all* the polygons.



2. Use any strategy to determine the area of the polygon.

Explain your thinking.

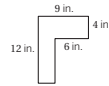
40 square centimeters. *Explanations vary.* I split this shape into a right triangle and a rectangle. The area of the triangle is $\frac{1}{2} \cdot 4 \cdot 4 = 8$, and the area of the rectangle is $8 \cdot 4 = 32$. Then I combined the areas, $8 + 32 = 40$ square centimeters.



3. Use any strategy to determine the area of the polygon.

Explain your thinking.

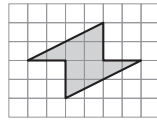
60 square inches. *Explanations vary.* I partitioned the polygon into two rectangles. The first rectangle has a length of 9 and a width of 4, with an area of 36. The second rectangle has a length of 8 and a width of 3, with an area of 24. Then I combined the areas, $36 + 24 = 60$ square inches.



4. Use any strategy to determine the area of the polygon.

Explain your thinking.

8 square units. *Explanations vary.* I decomposed this shape into two triangles. Each triangle has a base length of 4 units and a height of 2 units. I used the area formula to find the area of each triangle, then I multiplied by two because there are two triangles. $\frac{1}{2} \cdot 4 \cdot 2 = 4$; $2 \cdot 4 = 8$ square units.



Unit 1 Lesson 8

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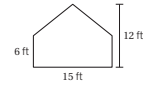
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Name: _____ Date: _____ Period: _____

5. Use any strategy to determine the area of the polygon.

Explain your thinking.

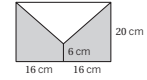
135 square feet. *Explanations vary.* I decomposed the polygon into a triangle and a rectangle. The base length of the triangle is 15 and its height is 6. I found the area of the triangle using $\frac{1}{2} \cdot 15 \cdot 6 = 45$ square feet. The area of the rectangle is $15 \cdot 6 = 90$ square feet. The total area is 135 square feet.



6. Determine the area of the shaded region.

Explain your thinking.

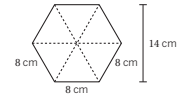
416 square centimeters. *Explanations vary.* First, I found the area of the rectangle: $32 \cdot 20 = 640$. Next, I found the area of the triangle: $\frac{1}{2} \cdot 32 \cdot 14 = 224$. Finally, I subtracted the area of the triangle from the area of the rectangle to find the area of the shaded region: $640 - 224 = 416$ square centimeters.



7. The hexagon has a side length of 8 centimeters. Diego says: The area is 28 square centimeters because each triangle has a base of 8 centimeters and a height of 7 centimeters. If I multiply 8 by 7 and divide the product by 2, I get 28 square centimeters. Is Diego correct?

Explain your thinking.

No. *Explanations vary.* Diego found the area of one of the triangles. He needs to multiply the area of one of the triangles by 6 to find the total area: $28 \cdot 6 = 168$ square centimeters.



Unit 1 Lesson 8

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Additional Practice

Practice Problem Analysis

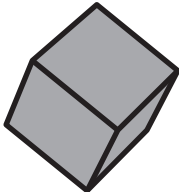
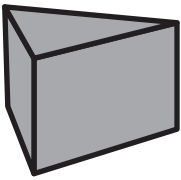
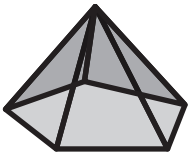
Problem	DOK	Standard(s)
1	1	6.G.A.1
2	2	6.G.A.1
3	2	6.G.A.1
4	2	6.G.A.1
5	2	6.G.A.1
6	2	6.G.A.1
7	2	6.G.A.1

Notes:

Additional Practice

ACC6.1.09

Problems 1–5: Here are some polyhedra.

Figure A	Figure B	Figure C	Figure D	Figure E
				

1. Which polyhedra are prisms?

Figures A, B, and D

2. Which polyhedra are pyramids?

Figures C and E

3. What type of polyhedron is figure D?

Rectangular prism

4. What type of polyhedron is figure C?

Triangular pyramid

5. Which of these polyhedra could be created from this net?

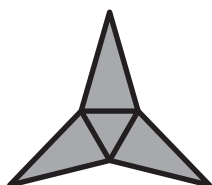
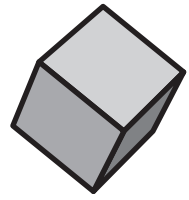


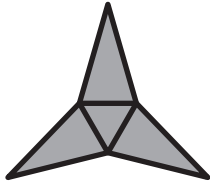
Figure C

6. Here is a polyhedron.

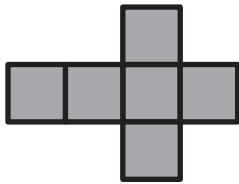
Which net could create this polyhedron?



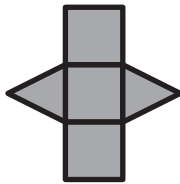
A.



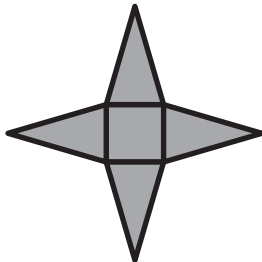
B.



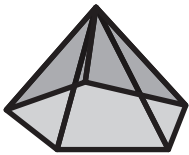
C.



D.



7. What polygons make up the faces of this three-dimensional figure?



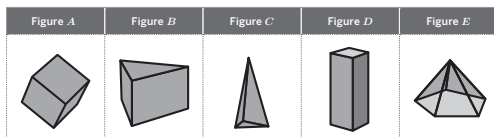
Five triangles and one pentagon

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.1.09

Problems 1–5: Here are some polyhedra.



1. Which polyhedra are prisms?

Figures A, B, and D

2. Which polyhedra are pyramids?

Figures C and E

3. What type of polyhedron is figure D?

Rectangular prism

4. What type of polyhedron is figure C?

Triangular pyramid

5. Which of these polyhedra could be created from this net?



Figure C

Unit 1 Lesson 9

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Name: _____ Date: _____ Period: _____

6. Here is a polyhedron.

Which net could create this polyhedron?



A.



B.



C.



D.



7. What polygons make up the faces of this three-dimensional figure?



Five triangles and one pentagon

Unit 1 Lesson 9

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	6.G.A.4
2	1	6.G.A.4
3	1	6.G.A.4
4	1	6.G.A.4
5	1	6.G.A.4
6	2	6.G.A.4
7	2	6.G.A.4

Notes:

Additional Practice

ACC6.1.10

1. Select *all* the units that can be used to describe surface area.

- | | |
|--|--|
| ☒ A. Square inches | ☐ B. Meters |
| ☐ C. Centimeters | ☐ D. Cubic feet |
| ☒ E. Square meters | ☐ F. Cubic inches |

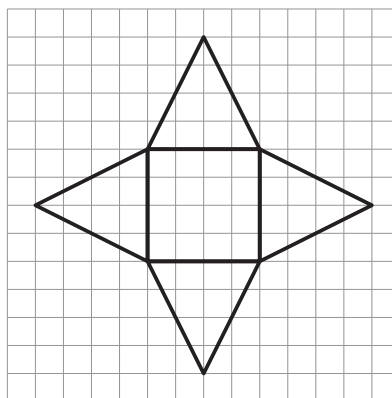
2. Select *all* the units that can be used to describe volume.

- | | |
|---|---|
| ☐ A. Square inches | ☐ B. Meters |
| ☐ C. Centimeters | ☒ D. Cubic feet |
| ☐ E. Square meters | ☒ F. Cubic inches |

3. What is the difference between the units used to measure surface area and the units used to measure volume? Look back at your work on Problems 1–2 if it is helpful.

Explanations vary. Surface area is measured in square units, like square inches or square meters. Whereas, volume is measured in cubic units, like cubic feet or cubic inches.

4. Here is a net.

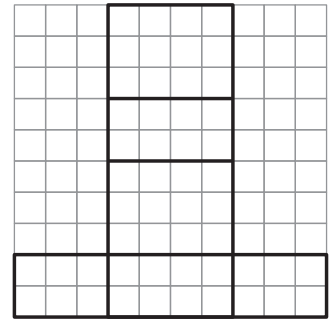


Determine the surface area, in square units, of the polyhedron that could be created using this net.

48 square units

5. Determine the surface area of the rectangular prism that could be created from this net.

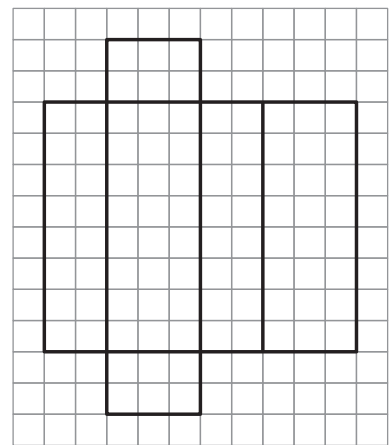
52 square units



6. Determine the surface area, in square units, of the rectangular prism that would be formed by this net. Show or explain your thinking.

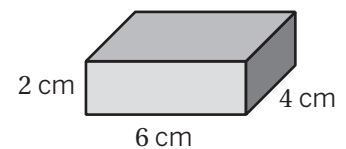
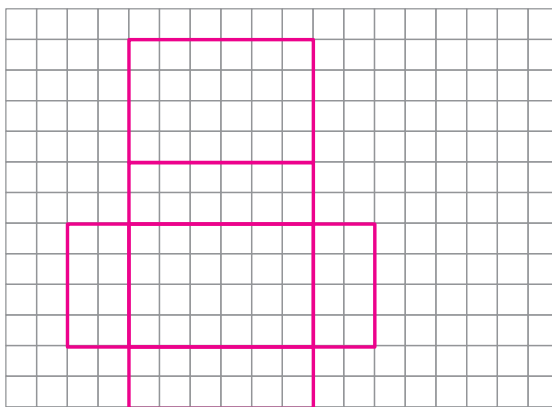
92 square units. Work varies.

$$2 \cdot (3 \cdot 8) + 2 \cdot (2 \cdot 8) + 2 \cdot (2 \cdot 3) = 92 \text{ square units}$$



7. Here is a rectangular prism.

- a Use the grid to draw a net for the prism. The length of one grid square is 1 centimeter.



- b Determine the surface area of the prism.

88 square units

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.1.10

1. Select *all* the units that can be used to describe surface area.

- ☒ A. Square inches ☐ B. Meters
☐ C. Centimeters ☐ D. Cubic feet
☒ E. Square meters ☐ F. Cubic inches

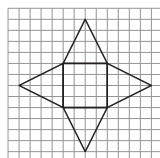
2. Select *all* the units that can be used to describe volume.

- ☐ A. Square inches ☐ B. Meters
☐ C. Centimeters ☒ D. Cubic feet
☐ E. Square meters ☒ F. Cubic inches

3. What is the difference between the units used to measure surface area and the units used to measure volume? Look back at your work on Problems 1–2 if it is helpful.

Explanations vary. Surface area is measured in square units, like square inches or square meters. Whereas, volume is measured in cubic units, like cubic feet or cubic inches.

4. Here is a net.



Determine the surface area, in square units, of the polyhedron that could be created using this net.

48 square units

Unit 1 Lesson 10

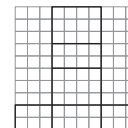
21

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Name: _____ Date: _____ Period: _____

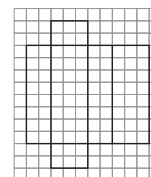
5. Determine the surface area of the rectangular prism that could be created from this net.

52 square units



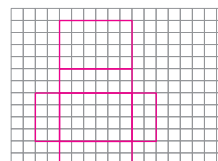
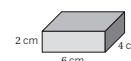
6. Determine the surface area, in square units, of the rectangular prism that would be formed by this net. Show or explain your thinking.

92 square units. Work varies.
 $2 \cdot (3 \cdot 8) + 2 \cdot (2 \cdot 8) + 2 \cdot (2 \cdot 3) = 92$ square units



7. Here is a rectangular prism.

- a. Use the grid to draw a net for the prism. The length of one grid square is 1 centimeter.



- b. Determine the surface area of the prism.

88 square units

Unit 1 Lesson 10

22

Additional Practice

Practice Problem Analysis

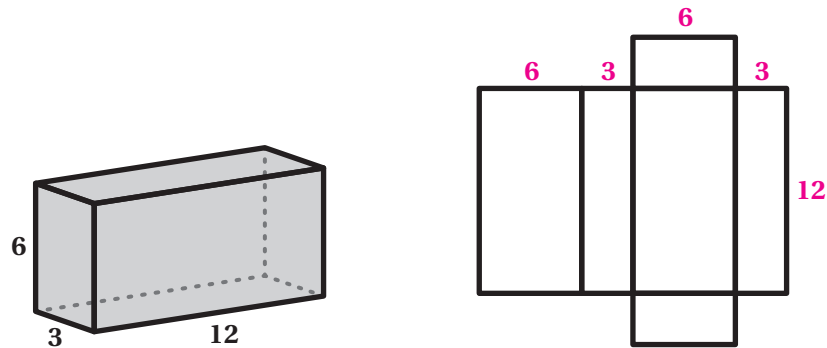
Problem	DOK	Standard(s)
1	1	6.G.A.4
2	1	6.G.A.4
3	2	6.G.A.4
4	2	6.G.A.4
5	2	6.G.A.4
6	2	6.G.A.4
7	2	6.G.A.4

Notes:

Additional Practice

ACC6.1.11

Problems 1–2: Here is a rectangular prism and its matching net.



1. Use the rectangular prism to label all the lengths in this net.

Responses shown on diagram.

2. Use the net to calculate the surface area in square units. Show or explain your thinking.

252 square units. Explanations vary. I can split the net into a large 12-by-18 rectangle and two 6-by-3 rectangles, which gives a total surface area of 252 square units.

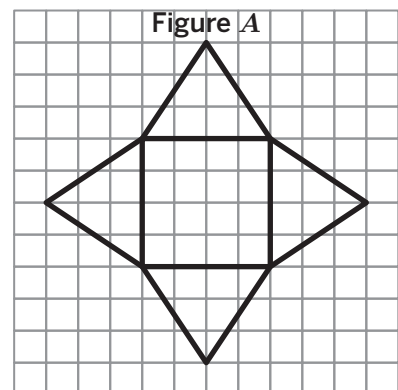
3. The net for figure A is shown.

- a Name the type of polyhedron that the net would form when assembled.

Rectangular pyramid

- b Determine the surface area of the polyhedron.

40 square units



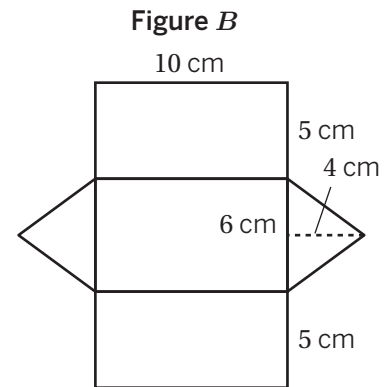
4. The net for figure *B* is shown.

- a Name the type of polyhedron that the net would form when assembled.

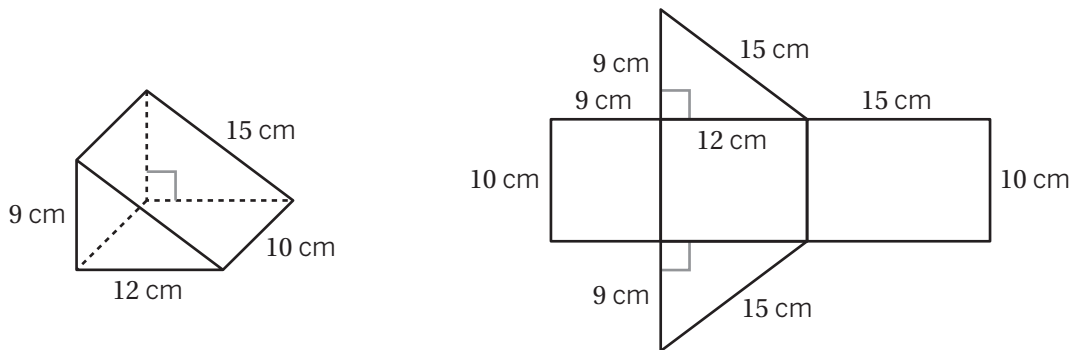
Triangular prism

- b Determine the surface area of the polyhedron.

184 square centimeters



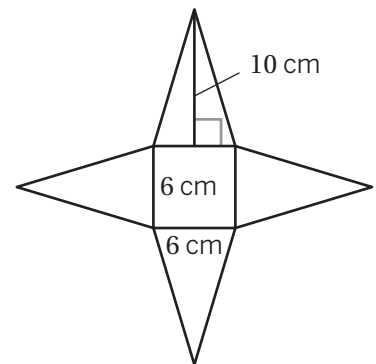
5. Here is a triangular prism and its matching net. Determine the surface area.



468 square centimeters

6. Bard claims the surface area of this square pyramid is 192 square centimeters. Is Bard correct? Explain your thinking.

No. Explanations vary. Bard mistook this pyramid for a prism and calculated the surface area with two bases. The surface area of this square pyramid is $(6 \cdot 6) + 4 \cdot \left(\frac{1}{2} \cdot 6 \cdot 10\right) = 156$ square centimeters.

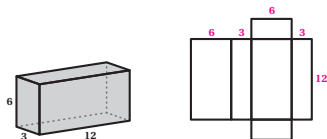


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Additional Practice

ACC6.1.11

Problems 1-2: Here is a rectangular prism and its matching net.



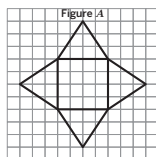
1. Use the rectangular prism to label all the lengths in this net.

Responses shown on diagram.

2. Use the net to calculate the surface area in square units. Show or explain your thinking.
252 square units. *Explanations vary. I can split the net into a large 12-by-18 rectangle and two 6-by-3 rectangles, which gives a total surface area of 252 square units.*

3. The net for figure A is shown.

- a. Name the type of polyhedron that the net would form when assembled.
Rectangular pyramid
- b. Determine the surface area of the polyhedron.
40 square units



Unit 1 Lesson 11

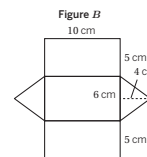
23

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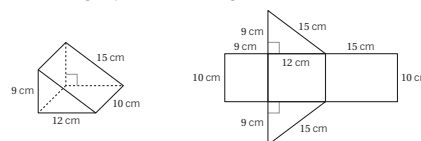
Name: _____ Date: _____ Period: _____

4. The net for figure B is shown.

- a. Name the type of polyhedron that the net would form when assembled.
Triangular prism
- b. Determine the surface area of the polyhedron.
184 square centimeters



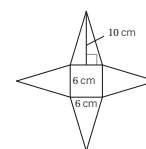
5. Here is a triangular prism and its matching net. Determine the surface area.



468 square centimeters

6. Bard claims the surface area of this square pyramid is 192 square centimeters. Is Bard correct? Explain your thinking.

No. Explanations vary. Bard mistook this pyramid for a prism and calculated the surface area with two bases. The surface area of this square pyramid is $(6 \cdot 6) + 4 \cdot \left(\frac{1}{2} \cdot 6 \cdot 10\right) = 156$ square centimeters.



Unit 1 Lesson 11

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	6.G.A.4
2	2	6.G.A.4
3	2	6.G.A.4
4	2	6.G.A.4
5	2	6.G.A.4
6	2	6.G.A.4

Notes:

Additional Practice

ACC6.2.01

1. Here are sets of shapes. Complete each statement.

a Triangles  

Circles  

There are 3 triangles for every 2 circles. (or equivalent)

b Squares  

Diamonds  

The ratio of diamonds to squares is 4 to 1. (or equivalent)

2. Select *all* the statements that correctly describe the relationship between the almonds and pretzels in this snack mix.

Almonds  
Pretzels  

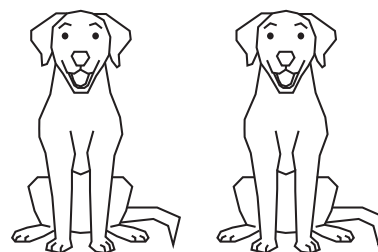
- ☐ A. The ratio of almonds to pretzels is 3 to 2.
- ☐ B. For every 2 pretzels, there are 3 almonds.
- ☒ C. The ratio of almonds to pretzels is 2 : 3.
- ☐ D. For every 3 almonds, there are 2 pretzels.
- ☐ E. For every 6 pretzels, there are 2 almonds.
- ☒ F. The ratio of almonds to pretzels is 4 : 6.

3. A recipe for banana bread uses 4 bananas, 2 eggs, and 1 cup of sugar.

Bananas  
Eggs  
Sugar (cups)  

- a The ratio of eggs to sugar is 2 : 1. (or equivalent)
- b The ratio of sugar to bananas is 1 to 4. (or equivalent)
- c There are 2 bananas for every 1 egg.

Problems 4–5: Here is a picture of two dogs.



4. Which diagram represents the ratio relationship between the number of eyes, tails, and feet?

A. Eyes ☐ ☐
Tails ☐☐☐☐☐☐☐☐
Feet ☐☐ ☐☐

B. Eyes ☐☐ ☐☐
Tails ☐ ☐
Feet ☐☐☐☐☐☐

C. Eyes ☐☐ ☐☐
Tails ☐ ☐
Feet ☐☐☐☐☐☐☐☐

D. Eyes ☐ ☐
Tails ☐☐ ☐☐
Feet ☐☐☐☐☐☐☐☐

5. Complete each statement.

- a The ratio of eyes to tails is 4 : 2 (or equivalent)
b The ratio of feet to tails is 4 to 1 (or equivalent)
c There are 2 eyes for every tail.
d There are 4 feet for every tail.

6. The diagram represents the number of percussion players and woodwind players in a school band.

Percussion



Woodwind



Clare says the ratio of woodwind players to percussion players is 2 : 6. Andre says the ratio of woodwind players to percussion players is 1 : 3. Who is correct? Explain your thinking.

Neither. Explanations vary. Clare is incorrect because she reversed the ratio of woodwind players to percussion players. The ratio is 12 : 4 or 6 : 2. Andre also reversed the ratio. The ratio of woodwind players to percussion players is also 3 : 1.

7. The diagram represents the number of cups of water and tablespoons of loose tea to make iced tea. Mai says the ratio of water to tea is 4 : 2. What is another ratio that Mai could write for the ratio of water to tea? Explain your thinking.

Water (cups)



Tea (tbsp)



2 : 1 (or equivalent). Explanations vary. If the ratio of water to tea is 4 : 2, that also means for every 2 cups of water, 1 tablespoon of tea is used.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.2.01

1. Here are sets of shapes. Complete each statement.

a. Triangles  

Circles  

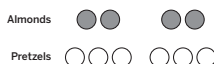
There are 3 triangles for every 2 circles. (or equivalent)

b. Squares  

Diamonds  

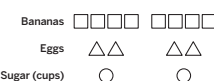
The ratio of diamonds to squares is 4 to 1. (or equivalent)

2. Select all the statements that correctly describe the relationship between the almonds and pretzels in this snack mix.



- ☐ A. The ratio of almonds to pretzels is 3 to 2.
☐ B. For every 2 pretzels, there are 3 almonds.
☒ C. The ratio of almonds to pretzels is 2 : 3.
☐ D. For every 3 almonds, there are 2 pretzels.
☐ E. For every 6 pretzels, there are 2 almonds.
☒ F. The ratio of almonds to pretzels is 4 : 6.

3. A recipe for banana bread uses 4 bananas, 2 eggs, and 1 cup of sugar.



- a. The ratio of eggs to sugar is 2 : 1. (or equivalent)
 b. The ratio of sugar to bananas is 1 to 4. (or equivalent)
 c. There are 2 bananas for every 1 egg.

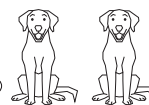
Unit 2 Lesson 1

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Name: _____ Date: _____ Period: _____

Problems 4–5: Here is a picture of two dogs.



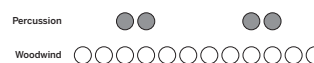
4. Which diagram represents the ratio relationship between the number of eyes, tails, and feet?

A. Eyes 	B. Eyes 
Tails 	Tails 
Feet 	Feet 
C. Eyes 	D. Eyes 
Tails 	Tails 
Feet 	Feet 

5. Complete each statement.

- a. The ratio of eyes to tails is 4 : 2. (or equivalent)
 b. The ratio of feet to tails is 4 to 1. (or equivalent)
 c. There are 2 eyes for every tail.
 d. There are 4 feet for every tail.

6. The diagram represents the number of percussion players and woodwind players in a school band.



Clare says the ratio of woodwind players to percussion players is 2 : 6. Andre says the ratio of woodwind players to percussion players is 1 : 3. Who is correct? Explain your thinking.

Neither. Explanations vary. Clare is incorrect because she reversed the ratio of woodwind players to percussion players. The ratio is 12 : 4 or 6 : 2. Andre also reversed the ratio. The ratio of woodwind players to percussion players is also 3 : 1.

7. The diagram represents the number of cups of water and tablespoons of loose tea to make iced tea. Mai says the ratio of water to tea is 4 : 2. What is another ratio that Mai could write for the ratio of water to tea? Explain your thinking.



2 : 1 (or equivalent). Explanations vary. If the ratio of water to tea is 4 : 2, that also means for every 2 cups of water, 1 tablespoon of tea is used.

Unit 2 Lesson 1

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Additional Practice

Practice Problem Analysis

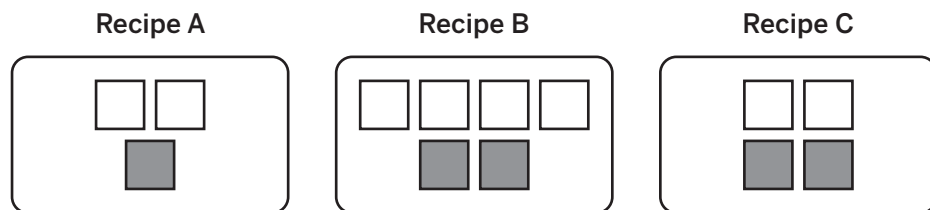
Problem	DOK	Standard(s)
1	1	6.RP.A.1
2	1	6.RP.A.1
3	1	6.RP.A.1
4	1	6.RP.A.1
5	1	6.RP.A.1
6	2	6.RP.A.1
7	2	6.RP.A.1

Notes:

Additional Practice

ACC6.2.02

Problems 1–4: Here are three recipes for green dye using yellow and blue food coloring.



Key:

 = 1 drop yellow food coloring  = 1 drop blue food coloring

1. Use the diagrams to complete each pair of statements.

- a** Recipe A uses 2 drop(s) of yellow food coloring and 1 drop(s) of blue food coloring.

The ratio of drops of yellow to blue in Recipe A is 2 : 1.

- b** Recipe B uses 4 drop(s) of yellow food coloring and 2 drop(s) of blue food coloring.

The ratio of drops of yellow to blue in Recipe B is 4 : 2 (or equivalent)

- c** Recipe C uses 2 drop(s) of yellow food coloring and 2 drop(s) of blue food coloring.

The ratio of drops of yellow to blue in Recipe C is 2 : 2 (or equivalent)

2. Which statement is *true* about the ratios of the recipes?

- A. The quantities in Recipe A are double those of Recipe B.
- B. The quantities in Recipe B are double those of Recipe C.
- C. The quantities in Recipe C are double those of Recipe A.
- D.** The quantities in Recipe B are double those of Recipe A.

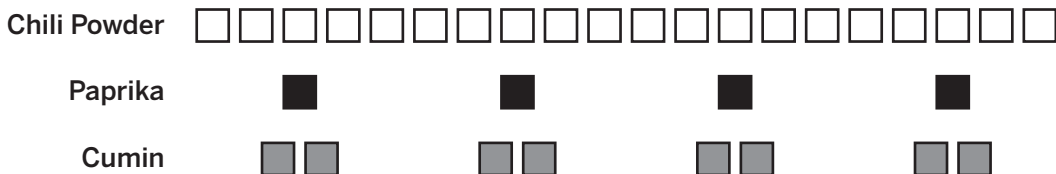
3. To darken the green dye, you can add more drops of blue. Which recipe has the darkest shade of green dye? Explain your thinking.

Recipe C. Explanations vary. Recipe C will give the darkest shade of green because the ratio of yellow to blue is 1 : 1, while the other two recipes both have a ratio of 2 drops of yellow for every 1 drop of blue. Recipe C has the greatest amount of blue.

4. What could be added to Recipe B so that the color will be the same as Recipe C?

2 drops of blue food coloring

5. A recipe for 1 batch of taco seasoning says, “Mix 5 parts chili powder, 1 part paprika, and 2 parts cumin.” How many batches are represented by the diagram? Explain your thinking.



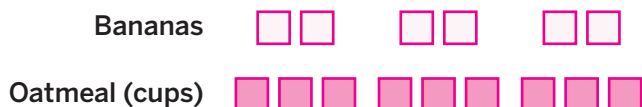
4 batches. Explanations vary. When I group them, I can see that there are 4 groups of 5 for the chili powder, 4 groups of 1 for the paprika, and 4 groups of 2 for the cumin.

6. Kiran makes banana oatmeal pancakes by mixing 2 mashed bananas with 3 cups of oatmeal (along with some other ingredients).

- a Draw a diagram that represents doubling the recipe for banana oatmeal pancakes.



- b Draw a diagram that represents tripling the recipe for banana oatmeal pancakes.



7. In a recipe for sparkling orange juice, the ratio of cups of orange juice to cups of sparkling water is 2 : 1.

- a Write the ratio of cups of orange juice to cups of sparkling water that represents a double batch.

4 : 2

- b Write the ratio of cups of orange juice to cups of sparkling water that represents a quadruple batch.

8 : 4

8. When Elena makes one batch of hummingbird food, she mixes 4 cups of water with 1 cup of sugar, which is enough for one hummingbird feeder.

- a If Elena has 5 hummingbird feeders in her yard, what is the ratio of cups of water to cups of sugar that she would use to fill all 5 feeders?

20 : 5

- b Elena says that if she has 3 hummingbird feeders in her yard, she would need 2 cups of sugar and 8 cups of water. Is Elena correct? Explain your thinking.

No. Explanations vary. If Elena has three hummingbird feeders in her yard, she needs to multiply both the cups of sugar and water from the original recipe by 3, which means that she will need 3 cups of sugar and 12 cups of water.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.2.02

Problems 1–4: Here are three recipes for green dye using yellow and blue food coloring.



Key:

□ = 1 drop yellow food coloring ■ = 1 drop blue food coloring

1. Use the diagrams to complete each pair of statements.

- a Recipe A uses 2 drop(s) of yellow food coloring and 1 drop(s) of blue food coloring.

The ratio of drops of yellow to blue in Recipe A is 2:1.

- b Recipe B uses 4 drop(s) of yellow food coloring and 2 drop(s) of blue food coloring.

The ratio of drops of yellow to blue in Recipe B is 4:2 (or equivalent)

- c Recipe C uses 2 drop(s) of yellow food coloring and 2 drop(s) of blue food coloring.

The ratio of drops of yellow to blue in Recipe C is 2:2 (or equivalent)

2. Which statement is true about the ratios of the recipes?

- A. The quantities in Recipe A are double those of Recipe B.
B. The quantities in Recipe B are double those of Recipe C.
C. The quantities in Recipe C are double those of Recipe A.
D. The quantities in Recipe B are double those of Recipe A.

3. To darken the green dye, you can add more drops of blue. Which recipe has the darkest shade of green dye? Explain your thinking.

Recipe C. *Explanations vary.* Recipe C will give the darkest shade of green because the ratio of yellow to blue is 1:1, while the other two recipes both have a ratio of 2 drops of yellow for every 1 drop of blue. Recipe C has the greatest amount of blue.

4. What could be added to Recipe B so that the color will be the same as Recipe C?

2 drops of blue food coloring

Unit 2 Lesson 2

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Name: _____ Date: _____ Period: _____

5. A recipe for 1 batch of taco seasoning says, "Mix 5 parts chili powder, 1 part paprika, and 2 parts cumin." How many batches are represented by the diagram? Explain your thinking.

Chili Powder □□□□□□□□□□□□□□□□

Paprika ■ ■ ■ ■

Cumin ■■ ■■ ■■ ■■

4 batches. *Explanations vary.* When I group them, I can see that there are 4 groups of 5 for the chili powder, 4 groups of 1 for the paprika, and 4 groups of 2 for the cumin.

6. Kiran makes banana oatmeal pancakes by mixing 2 mashed bananas with 3 cups of oatmeal (along with some other ingredients).

a Draw a diagram that represents doubling the recipe for banana oatmeal pancakes.

Bananas □□ □□

Oatmeal (cups) □□□□□□□□

b Draw a diagram that represents tripling the recipe for banana oatmeal pancakes.

Bananas □□ □□ □□

Oatmeal (cups) □□□□□□□□□□

7. In a recipe for sparkling orange juice, the ratio of cups of orange juice to cups of sparkling water is 2:1.

a Write the ratio of cups of orange juice to cups of sparkling water that represents a double batch.

4:2

b Write the ratio of cups of orange juice to cups of sparkling water that represents a quadruple batch.

8:4

8. When Elena makes one batch of hummingbird food, she mixes 4 cups of water with 1 cup of sugar, which is enough for one hummingbird feeder.

a If Elena has 5 hummingbird feeders in her yard, what is the ratio of cups of water to cups of sugar that she would use to fill all 5 feeders?

20:5

b Elena says that if she has 3 hummingbird feeders in her yard, she would need 2 cups of sugar and 8 cups of water. Is Elena correct? Explain your thinking.

No. *Explanations vary.* If Elena has three hummingbird feeders in her yard, she needs to multiply both the cups of sugar and water from the original recipe by 3, which means that she will need 3 cups of sugar and 12 cups of water.

Unit 2 Lesson 2

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	6.RP.A.1
2	1	6.RP.A.1
3	2	6.RP.A.1
4	1	6.RP.A.1
5	2	6.RP.A.1
6	2	6.RP.A.1
7	2	6.RP.A.1
8	2	6.RP.A.1

Notes:

Additional Practice

ACC6.2.03

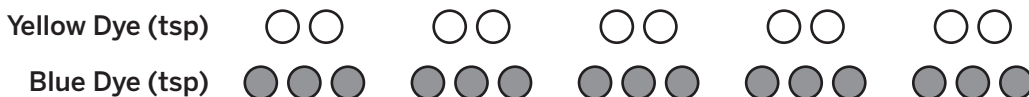
1. The diagram shows a mixture of black paint and white paint needed for 3 batches of a gray paint.



What is the ratio of black paint to white paint needed for 1 batch?

5 : 3

2. The diagram shows a mixture of yellow dye and blue dye needed for 5 batches of a green dye.



What is the ratio of blue dye to yellow dye needed for 1 batch?

3 : 2

3. Bard makes pink paint by mixing 16 tablespoons of white paint and 4 tablespoons of red paint. Select *all* of the mixtures that produce the same pink paint as Bard's mixture, but in a smaller amount.

- ☐ A. Mix 15 tablespoons of white paint and 3 tablespoons of red paint.
- ☒ B. For every tablespoon of red paint, mix 4 tablespoons of white paint.
- ☐ C. Mix tablespoons of red paint and white paint in the ratio 4 : 1.
- ☒ D. For every 2 tablespoons of red paint, mix 8 tablespoons of white paint.
- ☐ E. Mix 8 tablespoons of white paint with 4 tablespoons of red paint.

4. Lin makes sparkling lemonade by mixing 12 cups of lemonade and 16 cups of sparkling water.

- a What is the ratio of sparkling water to lemonade?

16 : 12 (or equivalent)

- b If Lin uses 6 cups of lemonade, how many cups of sparkling water should she use so the taste remains the same?

8 cups

- c If Lin uses 4 cups of sparkling water, how many cups of lemonade should she use so the taste remains the same?

3 cups

5. To make 1 batch of pale yellow paint, Diego mixes 3 cups of yellow paint with 1 gallon of white paint. Select *all* the mixtures that are a darker tint of yellow than the pale yellow.

- ☐ A. Add more white paint to the mixture.
- ☒ B. Mix 5 cups of yellow paint with 1 gallon of white paint.
- ☐ C. Mix 2 gallons of white paint with 3 cups of yellow paint.
- ☒ D. Add more yellow paint to the mixture.
- ☒ E. Mix 3 cups of yellow paint with $\frac{1}{2}$ gallon of white paint.

6. When dyeing yarn, use 750 milliliters of water for every 25 grams of yarn fibers. Complete the table.

Water (mL)	Yarn Fibers (g)
750	25
150	5
30	1

7. To make a large batch of pancake mix, the directions say to use 40 cups of water and 28 cups of pancake mix.

- a Diego only needs half the amount of pancake mix. What ratio would represent half of the recipe?

20 cups of water to 14 cups of mix, or 20 : 14

- b Lin wants to use 8 cups of water and 7 cups of pancake mix. Is her ratio equivalent to the ratio in the directions? Explain your thinking.

No. Explanations vary. To get from 40 cups of water to 8 cups of water, I would divide by 5. To keep the ratio equivalent, I would also divide 28 by 5 to get 5.6 cups of pancake mix, which does not equal 7 cups.

8. Orange paint can be made by mixing 35 teaspoons of red paint and 14 teaspoons of yellow paint. Kiran and Priya each attempted to make a smaller amount of the same orange paint color.

Kiran		Priya	
Red (tsp)	☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐	Red (tsp)	☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐
Yellow (tsp)	☐ ☐	Yellow (tsp)	☐ ☐ ☐ ☐ ☐ ☐

Does either person's color mixture make the same orange color as the original orange paint? Explain your thinking.

Yes. Explanations vary. Priya's orange mixture is the same as the original paint color because the ratio of 15 : 6 is the same as 35 : 14.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.2.03

1. The diagram shows a mixture of black paint and white paint needed for 3 batches of a gray paint.

Black Paint (tsp) ●●●●●● ●●●●●● ●●●●●●
White Paint (tsp) ○○○ ○○○ ○○○

What is the ratio of black paint to white paint needed for 1 batch?

5:3

2. The diagram shows a mixture of yellow dye and blue dye needed for 5 batches of a green dye.

Yellow Dye (tsp) ○○ ○○ ○○ ○○ ○○
Blue Dye (tsp) ●●● ●●● ●●● ●●● ●●●

What is the ratio of blue dye to yellow dye needed for 1 batch?

3:2

3. Bard makes pink paint by mixing 16 tablespoons of white paint and 4 tablespoons of red paint. Select *all* of the mixtures that produce the same pink paint as Bard's mixture, but in a smaller amount.

- ☐ A. Mix 15 tablespoons of white paint and 3 tablespoons of red paint.
☒ B. For every tablespoon of red paint, mix 4 tablespoons of white paint.
☐ C. Mix tablespoons of red paint and white paint in the ratio 4:1.
☒ D. For every 2 tablespoons of red paint, mix 8 tablespoons of white paint.
☐ E. Mix 8 tablespoons of white paint with 4 tablespoons of red paint.

4. Lin makes sparkling lemonade by mixing 12 cups of lemonade and 16 cups of sparkling water.

- a. What is the ratio of sparkling water to lemonade?
16:12 (or equivalent)
b. If Lin uses 6 cups of lemonade, how many cups of sparkling water should she use so the taste remains the same?
8 cups
c. If Lin uses 4 cups of sparkling water, how many cups of lemonade should she use so the taste remains the same?
3 cups

Unit 2 Lesson 3

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Name: _____ Date: _____ Period: _____

5. To make 1 batch of pale yellow paint, Diego mixes 3 cups of yellow paint with 1 gallon of white paint. Select *all* the mixtures that are a darker tint of yellow than the pale yellow.

- ☐ A. Add more white paint to the mixture.
☒ B. Mix 5 cups of yellow paint with 1 gallon of white paint.
☐ C. Mix 2 gallons of white paint with 3 cups of yellow paint.
☒ D. Add more yellow paint to the mixture.
☒ E. Mix 3 cups of yellow paint with $\frac{1}{2}$ gallon of white paint.

6. When dyeing yarn, use 750 milliliters of water for every 25 grams of yarn fibers. Complete the table.

Water (mL)	Yarn Fibers (g)
750	25
150	5
30	1

7. To make a large batch of pancake mix, the directions say to use 40 cups of water and 28 cups of pancake mix.

- a. Diego only needs half the amount of pancake mix. What ratio would represent half of the recipe?
20 cups of water to 14 cups of mix, or 20:14
b. Lin wants to use 8 cups of water and 7 cups of pancake mix. Is her ratio equivalent to the ratio in the directions? Explain your thinking.
No. Explanations vary. To get from 40 cups of water to 8 cups of water, I would divide by 5. To keep the ratio equivalent, I would also divide 28 by 5 to get 5.6 cups of pancake mix, which does not equal 7 cups.

8. Orange paint can be made by mixing 35 teaspoons of red paint and 14 teaspoons of yellow paint. Kiran and Priya each attempted to make a smaller amount of the same orange paint color.

Kiran
Red (tsp) □□□□□□□□
Yellow (tsp) □□
Priya
Red (tsp) □□□□□□□□□□
Yellow (tsp) □□□□□□

Does either person's color mixture make the same orange color as the original orange paint? Explain your thinking.
Yes. Explanations vary. Priya's orange mixture is the same as the original paint color because the ratio of 15:6 is the same as 35:14.

Unit 2 Lesson 3

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	6.RP.A.3.A
2	1	6.RP.A.3.A
3	2	6.RP.A.3.A
4	1	6.RP.A.3.A
5	2	6.RP.A.3.A
6	1	6.RP.A.3.A
7	2	6.RP.A.3.A
8	2	6.RP.A.3.A

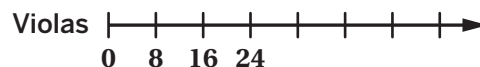
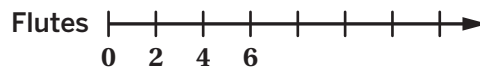
Notes:

Additional Practice

ACC6.2.04

1. In an orchestra, the ratio of flutes to violas is 2 : 8.

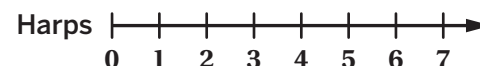
List two other combinations of flutes and violas for a different orchestra with the same ratio of instruments. *Responses vary.*



Flutes	Violas
8	32
10	40

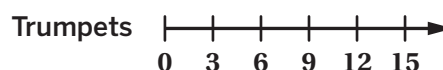
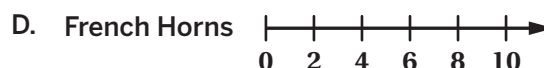
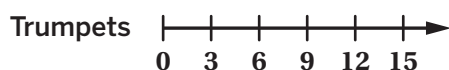
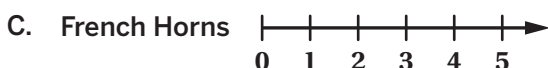
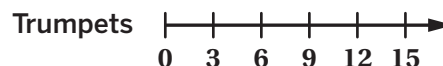
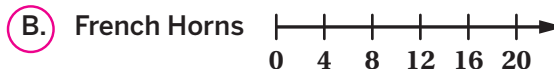
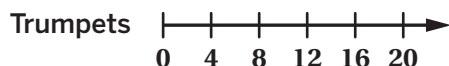
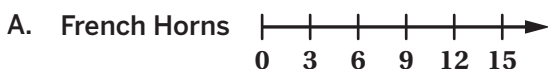
2. In an orchestra, the ratio of first violins to harps is 3 : 1.

List two other combinations of first violins and harps for a *larger* orchestra with the same ratio of instruments. *Responses vary.*



First Violins	Harps
24	8
36	12

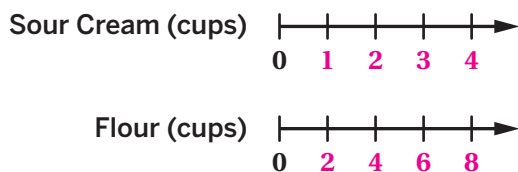
3. In an orchestra, the ratio of French horns to trumpets is 4 : 3.
Which double number line correctly shows this ratio?



Problems 4–7: Here are the ingredients of a recipe for one batch of blueberry muffins.

Ingredient	Amount
Blueberries	$2\frac{1}{4}$ cups
Sour cream	1 cup
Salt	$\frac{1}{2}$ tsp
Flour	2 cups
Baking powder	3 tsp
Sugar	1 cup

4. Complete the double number line representing the amounts of sour cream and flour.



5. Bard plans on making several batches of blueberry muffins.

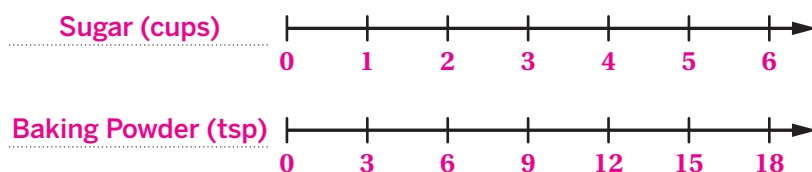
- a If Bard uses $2\frac{1}{2}$ teaspoons of salt, how many batches could be made?

5 batches

- b If Bard uses 12 teaspoons of baking powder, how many batches could be made?

4 batches

6. Draw a double number line to represent the amounts of sugar and baking powder in the recipe.



7. Diego wants to make 3 batches of blueberry muffins. He says he will need $6\frac{1}{2}$ cups of blueberries, 6 cups of flour, and 3 teaspoons of salt. Is Diego correct? Explain your thinking.

No. Explanations vary. If Diego is making 3 batches of blueberry muffins, he will need $2\frac{1}{4} \cdot 3 = 6\frac{3}{4}$ cups of blueberries, $2 \cdot 3 = 6$ cups of flour, and $\frac{1}{2} \cdot 3 = 1\frac{1}{2}$ teaspoons of salt.

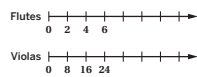
Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.2.04

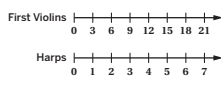
1. In an orchestra, the ratio of flutes to violas is 2 : 8.
List two other combinations of flutes and violas for a different orchestra with the same ratio of instruments. **Responses vary.**

Flutes	Violas
8	32
10	40

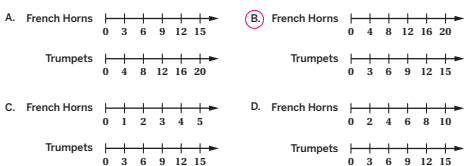


2. In an orchestra, the ratio of first violins to harps is 3 : 1.
List two other combinations of first violins and harps for a larger orchestra with the same ratio of instruments. **Responses vary.**

First Violins	Harps
24	8
36	12



3. In an orchestra, the ratio of French horns to trumpets is 4 : 3.
Which double number line correctly shows this ratio?



Additional Practice

ACC6.2.07

1. Determine *all* the factors of each number.

a 16 **1, 2, 4, 8, 16**

b 25 **1, 5, 25**

c 100 **1, 2, 4, 5, 10, 20, 25, 50, 100**

d 37 **1, 37**

2. For the numbers 20 and 24, determine:

a The factors of 20.

1, 2, 4, 5, 10, 20

b The factors of 24.

1, 2, 3, 4, 6, 8, 12, 24

c The common factors of 20 and 24.

1, 2, 4

d The greatest common factor (GCF) of 20 and 24.

4

3. For the numbers 18 and 27, determine:

a The factors of 18.

1, 2, 3, 6, 9, 18

b The factors of 27.

1, 3, 9, 27

c The common factors of 18 and 27.

1, 3, 9

d The greatest common factor (GCF) of 18 and 27.

9

4. Determine the greatest common factor for each pair of numbers.

a 33 and 15 **3**

b 8 and 32 **8**

c 45 and 54 **9**

d 35 and 21 **7**

5. Consider the numbers 42 and 96.

- a How many common factors do 42 and 96 have? Explain your thinking.
4. Explanations vary. The common factors of 42 and 96 are 1, 2, 3, and 6.

- b What is the greatest common factor of 42 and 96? Explain your thinking.
6. Explanations vary. 6 is the largest number that goes into both 42 and 96.

6. A teacher is making gift bags. Each bag will have pencils and erasers. The teacher has 40 pencils and 60 erasers to use. Each bag will have the same number of each item, with no items left over. Select *all* of the possible numbers of pencils and erasers in the gift bags.

- ☒ A. 10 bags with 4 pencils and 6 erasers
☐ B. 4 bags with 6 pencils and 4 erasers
☐ C. 20 bags with 2 pencils and 6 erasers
☒ D. 2 bags with 20 pencils and 30 erasers
☒ E. 5 bags with 8 pencils and 12 erasers

7. A middle school band has 72 sixth grade students and 54 seventh grade students. The band director wants to make groups of performers, with the same combination of sixth grade and seventh grade students in each group.

- a What is the greatest number of groups that could be formed? Explain your thinking.
18. Explanations vary. The greatest common factor of 72 and 54 is 18.
- b Using your answer from Problem 7a, how many sixth graders would be in each group?
4
- c Using your answer from Problem 7a, how many seventh graders would be in each group?
3

8. Jada, Shawn, and Noah were all asked to determine the greatest common factor of 16, 72, and 80. Their responses are shown.

- Jada says that the greatest common factor is 16.
- Shawn says that the greatest common factor is 8.
- Noah says that the greatest common factor is 9.

Who is correct? Explain your thinking.

Shawn. Explanations vary. Shawn is correct that the greatest common factor of 16, 72, and 80 is 8. The number 16 is a common factor of 16 and 80, but not of 72. The number 9 is a factor of 72, but not of 16 or 80.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.2.07

1. Determine all the factors of each number.

- a 16 **1, 2, 4, 8, 16** b 25 **1, 5, 25**
c 100 **1, 2, 4, 5, 10, 20, 25, 50, 100** d 37 **1, 37**

2. For the numbers 20 and 24, determine:

- a The factors of 20.
1, 2, 4, 5, 10, 20
b The factors of 24.
1, 2, 3, 4, 6, 8, 12, 24
c The common factors of 20 and 24.
1, 2, 4
d The greatest common factor (GCF) of 20 and 24.
4

3. For the numbers 18 and 27, determine:

- a The factors of 18.
1, 2, 3, 6, 9, 18
b The factors of 27.
1, 3, 9, 27
c The common factors of 18 and 27.
1, 3, 9
d The greatest common factor (GCF) of 18 and 27.
9

4. Determine the greatest common factor for each pair of numbers.

- a 33 and 15 **3** b 8 and 32 **8**
c 45 and 54 **9** d 35 and 21 **7**

Unit 2 Lesson 7

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Name: _____ Date: _____ Period: _____

5. Consider the numbers 42 and 96.

- a How many common factors do 42 and 96 have? Explain your thinking.
4. Explanations vary. The common factors of 42 and 96 are 1, 2, 3, and 6.
b What is the greatest common factor of 42 and 96? Explain your thinking.
6. Explanations vary. 6 is the largest number that goes into both 42 and 96.

6. A teacher is making gift bags. Each bag will have pencils and erasers. The teacher has 40 pencils and 60 erasers to use. Each bag will have the same number of each item, with no items left over. Select all of the possible numbers of pencils and erasers in the gift bags.

- ☒ A. 10 bags with 4 pencils and 6 erasers
☐ B. 4 bags with 6 pencils and 4 erasers
☐ C. 20 bags with 2 pencils and 6 erasers
☒ D. 2 bags with 20 pencils and 30 erasers
☒ E. 5 bags with 8 pencils and 12 erasers

7. A middle school band has 72 sixth grade students and 54 seventh grade students. The band director wants to make groups of performers, with the same combination of sixth grade and seventh grade students in each group.

- a What is the greatest number of groups that could be formed? Explain your thinking.
18. Explanations vary. The greatest common factor of 72 and 54 is 18.
b Using your answer from Problem 7a, how many sixth graders would be in each group?
4
c Using your answer from Problem 7a, how many seventh graders would be in each group?
3

8. Jada, Shawn, and Noah were all asked to determine the greatest common factor of 16, 72, and 80. Their responses are shown.

- Jada says that the greatest common factor is 16.
- Shawn says that the greatest common factor is 8.
- Noah says that the greatest common factor is 9.

Who is correct? Explain your thinking.

Shawn. Explanations vary. Shawn is correct that the greatest common factor of 16, 72, and 80 is 8. The number 16 is a common factor of 16 and 80, but not of 72. The number 9 is a factor of 72, but not of 16 or 80.

Unit 2 Lesson 7

42

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	6.NS.B.4
2	1	6.NS.B.4
3	1	6.NS.B.4
4	1	6.NS.B.4
5	2	6.NS.B.4
6	2	6.NS.B.4
7	2	6.NS.B.4
8	2	6.NS.B.4

Notes:

Additional Practice

ACC6.2.08

1. Priya has 2 white marbles and 4 blue marbles. Lin has 4 white marbles and 16 blue marbles. Who has the greater ratio of white marbles to blue marbles? Explain your thinking.

A. Priya **B.** Lin **C.** They have the same ratio of white to blue marbles.

Explanations vary. If Priya had $4 \cdot 4 = 16$ blue marbles, she would have $2 \cdot 4 = 8$ white marbles, which is more white marbles than what Lin has.

2. Kiran shops at a bulk store where he can purchase dry goods by the ounce.

a Complete the table.

Item	Price (\$)	Number of Ounces	Price per Ounce (\$)
Banana chips	9	36	0.25
Macadamia nuts	16	8	2
Dried mango	8	8	1
Cinnamon pecans	18	24	0.75
Pistachios	3.75	5	0.75

b Which item has the lowest price per ounce? Explain your thinking.

Banana chips. Explanations vary. The ratio of price to ounces of banana chips gives the lowest price per ounce out of all of the items.

3. Here are 3 possible mixtures to make pink paint.

- Mixture A is made with 18 teaspoons of white paint and 12 teaspoons of red paint.
- Mixture B is made with 6 teaspoons of white paint and 5 teaspoons of red paint.
- Mixture C is made with 12 teaspoons of white paint and 10 teaspoons of red paint.

Which mixture is the *lightest* shade of pink paint? Explain your thinking.

Mixture A. Explanations vary. Mixtures B and C are equivalent ratios, so their color is the same. I made equivalent ratios to compare Mixture A with Mixture B. When they both have 18 teaspoons of white paint, Mixture A has 12 teaspoons of red paint and Mixture B has 15 teaspoons of red paint. The mixture with less red paint makes a lighter pink.

4. Noah's team won 10 games and lost 5 games. Tyler's team won 12 games and lost 4 games. Whose team had the greater ratio of wins to losses? Explain your thinking.

A. Noah's team **B.** Tyler's team C. Both teams have the same ratio of wins to losses.

Explanations vary. $\frac{12}{4} = 3$ is greater than $\frac{10}{5} = 2$.

5. Movie tickets cost different amounts depending on the day and time of the show.

- Bard paid \$39 for 6 tickets.
- Andre paid \$21.75 for 3 tickets.
- Mai paid \$31.25 for 5 tickets.

Who paid the lowest price per ticket? Explain your thinking.

A. Bard B. Andre **C.** Mai

Explanations vary. Bard paid $\frac{39}{6} = \$6.50$ per ticket. Andre paid $\frac{21.75}{3} = \$7.25$ per ticket, and Mai paid $\frac{31.25}{5} = \$6.25$ per ticket.

6. Diego swam 200 meters in 58 seconds. Han swam 300 meters in 87 seconds. Both swam at a constant speed. Who swam at a faster constant speed? Explain your thinking.

A. Diego B. Han **C.** They swam at the same speed.

Explanations vary. $\frac{200}{58} = \frac{300}{87}$

7. Clare ran 3 miles in 27 minutes. Jada ran 5 miles in 40 minutes. Both ran at a constant speed. Did they run at the same constant speed? Explain your thinking.

No. Explanations vary. Clare and Jada did not run at the same constant speed. If I use the LCM of 3 and 5, I can compare how long it took Clare and Jada to run 15 miles. Clare could run 15 miles in 135 minutes, and Jada could run 15 miles in 120 minutes. Therefore, Jada ran faster than Clare.

8. Shawn paid \$11.25 for 2.5 pounds of chicken. Lin paid \$25.50 for 6 pounds of chicken. Lin says she paid less per pound of chicken than Shawn. Is Lin correct? Explain your thinking.

Yes. Explanations vary. Shawn paid \$4.50 per pound, and Lin paid \$4.25 per pound.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.2.08

1. Priya has 2 white marbles and 4 blue marbles. Lin has 4 white marbles and 16 blue marbles. Who has the greater ratio of white marbles to blue marbles? Explain your thinking.

A. Priya **B.** Lin **C.** They have the same ratio of white to blue marbles.

Explanations vary. If Priya had $4 \div 4 = 1$ blue marbles, she would have $2 \div 4 = 0.5$ white marbles, which is more white marbles than what Lin has.

2. Kiran shops at a bulk store where he can purchase dry goods by the ounce.

a. Complete the table.

Item	Price (\$)	Number of Ounces	Price per Ounce (\$)
Banana chips	9	36	0.25
Macadamia nuts	16	8	2
Dried mango	8	8	1
Cinnamon pecans	18	24	0.75
Pistachios	3.75	5	0.75

b. Which item has the lowest price per ounce? Explain your thinking.

Banana chips. **Explanations vary.** The ratio of price to ounces of banana chips gives the lowest price per ounce out of all of the items.

3. Here are 3 possible mixtures to make pink paint.

- Mixture A is made with 18 teaspoons of white paint and 12 teaspoons of red paint.
- Mixture B is made with 6 teaspoons of white paint and 3 teaspoons of red paint.
- Mixture C is made with 12 teaspoons of white paint and 10 teaspoons of red paint.

Which mixture is the lightest shade of pink paint? Explain your thinking.

Mixture A. **Explanations vary.** Mixtures B and C are equivalent ratios, so their color is the same. I made equivalent ratios to compare Mixture A with Mixture B. When they both have 18 teaspoons of white paint, Mixture A has 12 teaspoons of red paint and Mixture B has 15 teaspoons of red paint. The mixture with less red paint makes a lighter pink.

Unit 2 Lesson 8

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Name: _____ Date: _____ Period: _____

4. Noah's team won 10 games and lost 5 games. Tyler's team won 12 games and lost 4 games. Whose team had the greater ratio of wins to losses? Explain your thinking.

A. Noah's team **B.** Tyler's team **C.** Both teams have the same ratio of wins to losses.
Explanations vary. $\frac{12}{4} = 3$ is greater than $\frac{10}{5} = 2$.

5. Movie tickets cost different amounts depending on the day and time of the show.

- Bard paid \$39 for 6 tickets.
- Andre paid \$21.75 for 3 tickets.
- Mai paid \$31.25 for 5 tickets.

Who paid the lowest price per ticket? Explain your thinking.

A. Bard **B.** Andre **C.** Mai

Explanations vary. Bard paid $\frac{39}{6} = \$6.50$ per ticket. Andre paid $\frac{21.75}{3} = \$7.25$ per ticket, and Mai paid $\frac{31.25}{5} = \$6.25$ per ticket.

6. Diego swam 200 meters in 58 seconds. Han swam 300 meters in 87 seconds. Both swam at a constant speed. Who swam at a faster constant speed? Explain your thinking.

A. Diego **B.** Han **C.** They swam at the same speed.

Explanations vary. $\frac{200}{58} \approx 3.45$ and $\frac{300}{87} \approx 3.45$.

7. Clare ran 3 miles in 27 minutes. Jada ran 5 miles in 40 minutes. Both ran at a constant speed. Did they run at the same constant speed? Explain your thinking.

No. **Explanations vary.** Clare and Jada did not run at the same constant speed. If I use the LCM of 3 and 5, I can compare how long it took Clare and Jada to run 15 miles. Clare could run 15 miles in 135 minutes, and Jada could run 15 miles in 120 minutes. Therefore, Jada ran faster than Clare.

8. Shawn paid \$11.25 for 2.5 pounds of chicken. Lin paid \$25.50 for 6 pounds of chicken. Lin says she paid less per pound of chicken than Shawn. Is Lin correct? Explain your thinking.

Yes. **Explanations vary.** Shawn paid \$4.50 per pound, and Lin paid \$4.25 per pound.

Unit 2 Lesson 8

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	6.RP.A.3
2	2	6.RP.A.3
3	2	6.RP.A.3
4	2	6.RP.A.3
5	2	6.RP.A.3
6	2	6.RP.A.3
7	2	6.RP.A.3
8	2	6.RP.A.3

Notes:

Additional Practice

ACC6.2.09

1. To make a fruit smoothie, Andre uses $\frac{1}{2}$ cup of almond milk and $1\frac{1}{2}$ cups of frozen berries. The table shows equivalent ratios of almond milk and frozen berries.

Almond Milk (cups)	Frozen Berries (cups)
$\frac{1}{2}$	$1\frac{1}{2}$
1	3
$1\frac{1}{2}$	$4\frac{1}{2}$
2	6

Select *all* of the true statements about the ratio of almond milk to frozen berries.

- ☒ A. For every 3 cups of berries, there is 1 cup of almond milk.
- ☐ B. For every 2 cups of berries, there are 6 cups of almond milk.
- ☐ C. For every $1\frac{1}{2}$ cups of berries, there is 1 cup of almond milk.
- ☒ D. For every 1 cup of almond milk, there are 3 cups of berries.
- ☒ E. For every $\frac{1}{2}$ cup of almond milk, there are $1\frac{1}{2}$ cups of berries.
2. A particular pink paint is made by mixing 3 parts of red paint with 9 parts of white paint. Complete the table with the amount of red paint and white paint needed to make different amounts of the same shade of pink paint.

Red Paint	White Paint
1	3
$\frac{1}{2}$	$1\frac{1}{2}$
4	12
2	6

3. In a recipe for waffles, there are $2\frac{1}{4}$ cups of flour for every 2 cups of milk. A family is making several batches of waffles. Determine how much of each ingredient the family will need. Use a table if it helps with your thinking.

a How many cups of milk are needed to make 3 batches of waffles?

6 cups

b How many cups of flour are needed to make 3 batches of waffles?

$6\frac{3}{4}$ cups (or equivalent)

c How many cups of milk are needed to make 5 batches of waffles?

10 cups

d How many cups of flour are needed to make 6 batches of waffles?

$13\frac{1}{2}$ cups (or equivalent)

4. A car travels at a constant speed and its distance traveled in 1, 2, and 3 hours is shown in the table. How far does the car travel in 8 hours? Show or explain your thinking.

520 miles. Explanations vary. The car travels 520 miles in 8 hours because $8 \cdot 65 = 520$.

Time (hours)	Distance (miles)
1	65
2	130
3	195
8	520

5. Complete the table by writing three equivalent ratios to 24 : 15.

Any 3 ratios that are equivalent to 24 : 15 are considered correct. Sample shown in table.

x	y
24	15
8	5
16	10
32	20

6. A recipe for a batch of bran muffins calls for $\frac{3}{4}$ cups of brown sugar and 3 cups of bran cereal. Andre says that, if he triples the recipe, he will need $3\frac{3}{4}$ cups of brown sugar and 9 cups of bran cereal. Is Andre correct? Show or explain your thinking.

No. Explanations vary. If Andre triples the recipe, he will need $\frac{3}{4} \cdot 3 = 2\frac{1}{4}$ cups of brown sugar.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.2.09

1. To make a fruit smoothie, Andre uses $\frac{1}{2}$ cup of almond milk and $1\frac{1}{2}$ cups of frozen berries. The table shows equivalent ratios of almond milk and frozen berries.

Almond Milk (cups)	Frozen Berries (cups)
$\frac{1}{2}$	$1\frac{1}{2}$
1	3
$1\frac{1}{2}$	$4\frac{1}{2}$
2	6

Select all of the true statements about the ratio of almond milk to frozen berries.

- ☒ A. For every 3 cups of berries, there is 1 cup of almond milk.
☐ B. For every 2 cups of berries, there are 6 cups of almond milk.
☐ C. For every $1\frac{1}{2}$ cups of berries, there is 1 cup of almond milk.
☒ D. For every 1 cup of almond milk, there are 3 cups of berries.
☒ E. For every $\frac{1}{2}$ cup of almond milk, there are $1\frac{1}{2}$ cups of berries.

2. A particular pink paint is made by mixing 3 parts of red paint with 9 parts of white paint. Complete the table with the amount of red paint and white paint needed to make different amounts of the same shade of pink paint.

Red Paint	White Paint
1	3
$\frac{1}{2}$	$1\frac{1}{2}$
4	12
2	6

Unit 2 Lesson 9

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Name: _____ Date: _____ Period: _____

3. In a recipe for waffles, there are $2\frac{1}{2}$ cups of flour for every 2 cups of milk. A family is making several batches of waffles. Determine how much of each ingredient the family will need. Use a table if it helps with your thinking.

- a. How many cups of milk are needed to make 3 batches of waffles?
6 cups
b. How many cups of flour are needed to make 3 batches of waffles?
 $6\frac{1}{2}$ cups (or equivalent)
c. How many cups of milk are needed to make 5 batches of waffles?
10 cups
d. How many cups of flour are needed to make 6 batches of waffles?
 $13\frac{1}{2}$ cups (or equivalent)

4. A car travels at a constant speed and its distance traveled in 1, 2, and 3 hours is shown in the table. How far does the car travel in 8 hours? Show or explain your thinking.

Time (hours)	Distance (miles)
1	65
2	130
3	195
8	520

5. Complete the table by writing three equivalent ratios to 24 : 15.

Any 3 ratios that are equivalent to 24 : 15 are considered correct. Sample shown in table.

x	y
24	15
8	5
16	10
32	20

6. A recipe for a batch of bran muffins calls for $\frac{3}{4}$ cups of brown sugar and 3 cups of bran cereal. Andre says that, if he triples the recipe, he will need $3\frac{3}{4}$ cups of brown sugar and 9 cups of bran cereal. Is Andre correct? Show or explain your thinking.

No. Explanations vary. If Andre triples the recipe, he will need $\frac{3}{4} \cdot 3 = 2\frac{1}{4}$ cups of brown sugar.

Unit 2 Lesson 9

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	6.RP.A.3
2	2	6.RP.A.3, 6.RP.A.3.A
3	2	6.RP.A.3
4	2	6.RP.A.3, 6.RP.A.3.A
5	2	6.RP.A.3, 6.RP.A.3.A
6	2	6.RP.A.3

Notes:

Additional Practice

ACC6.2.10

1. Determine the missing number in the ratio table.

Number of Knives	Number of Spoons
7	12
28	48

2. A recipe calls for 2 pounds of chicken and $3\frac{1}{2}$ cups of tomato sauce. Andre makes 3 batches of the recipe. What amount of ingredients are needed for 3 batches?
- A. 5 pounds of chicken, 9 cups of tomato sauce
 - B. 5 pounds of chicken, $10\frac{1}{2}$ cups of tomato sauce
 - C. 6 pounds of chicken, 9 cups of tomato sauce
 - ☒ D. 6 pounds of chicken, $10\frac{1}{2}$ cups of tomato sauce
3. A chef is using vinegar to pickle onions. The restaurant supply store sells 2 gallons of vinegar for \$4.88. Which ratios correctly represent the price of the vinegar? Select *all* that apply.
- ☐ A. 7 gallons to \$14.64
 - ☒ B. 14 gallons to \$34.16
 - ☒ C. 1 gallon to \$2.44
 - ☐ D. 20 gallons to \$43.92
 - ☒ E. 5 gallons to \$12.20
4. A full-grown elephant drinks about 1,330 liters of water each week. Complete each ratio based on this information.
- a. _____ **190** _____ liters to 1 day
 - b. 570 liters to _____ **3** _____ days
 - c. _____ **3,990** _____ liters to 3 weeks
 - d. 2,280 liters to _____ **12** _____ days
 - e. _____ **15,960** _____ liters to 12 weeks

5. A caterer needs to buy 38 pounds of potatoes for an event. It costs \$6 for 5 pounds of potatoes.

a Write a ratio for the price of the potatoes per pound.

\$6 : 5 or \$1.20 : 1

b If all potatoes are sold at this rate, how much will the caterer pay for the potatoes they need? Explain your thinking.

\$45.60. Explanations vary. The caterer will pay \$45.60 for 38 pounds of potatoes. I used equivalent ratios to determine that the cost of the potatoes is \$1.20 per pound. Because the caterer needs 38 pounds, I multiplied the cost of 1 pound by 38: $1.20 \cdot 38 = \$45.60$.

6. A caterer also needs to buy 130 rolls for the event. A bakery sells 12 rolls for \$4.80.

a Write a ratio to describe the cost of the rolls.

\$4.80 : 12 or \$0.40 : 1

b If all rolls are sold at the same rate, how much will the caterer pay for the rolls they need? Explain your thinking.

\$52. Explanations vary. The caterer will pay \$52 for 130 rolls. I used equivalent ratios to determine that the cost of 1 roll is \$0.40. Because the caterer needs 130 rolls, I multiplied the cost of 1 roll by 130: $0.40 \cdot 130 = \$52$.

7. The directions for a shade of green paint say to mix 7 milliliters of yellow paint with 3 milliliters of blue paint. Han has 42 milliliters of yellow paint and 25 milliliters of blue paint. If Han wants to use all of the yellow paint, how much blue paint, if any, will he have left? Show or explain your thinking. Use the ratio table if it helps you show your thinking.

7 milliliters. Explanations vary.

Yellow (mL)	Blue (mL)
7	3
42	18

I used equivalent ratios to determine that Han will use 18 milliliters of blue paint with 42 milliliters of yellow paint. Then I subtracted to find how much blue paint was left: $25 - 18 = 7$.

8. Mai is reading a 55-page book. She read the first 35 pages in 30 minutes. If she continues to read at the same rate, will she be able to complete this book in less than 1 hour? Show or explain your thinking.

Yes. Explanations vary. There are 2 sets of 30 minutes in 1 hour, so Mai can read $35 \cdot 2 = 70$ pages in 1 hour. Therefore, she will be able to complete her 55-page book within 1 hour.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.2.10

1. Determine the missing number in the ratio table.

Number of Knives	Number of Spoons
7	12
28	48

2. A recipe calls for 2 pounds of chicken and $3\frac{1}{2}$ cups of tomato sauce. Andre makes 3 batches of the recipe. What amount of ingredients are needed for 3 batches?
- A. 5 pounds of chicken, 9 cups of tomato sauce
 B. 5 pounds of chicken, $10\frac{1}{2}$ cups of tomato sauce
 C. 6 pounds of chicken, 9 cups of tomato sauce
 D. 6 pounds of chicken, $10\frac{1}{2}$ cups of tomato sauce
3. A chef is using vinegar to pickle onions. The restaurant supply store sells 2 gallons of vinegar for \$4.88. Which ratios correctly represent the price of the vinegar? Select all that apply.
- ☐ A. 7 gallons to \$14.64
☒ B. 14 gallons to \$34.16
☒ C. 1 gallon to \$2.44
☐ D. 20 gallons to \$43.92
☒ E. 5 gallons to \$12.20
4. A full-grown elephant drinks about 1,330 liters of water each week. Complete each ratio based on this information.
- a. 190 liters to 1 day
 b. 570 liters to 3 days
 c. 3,990 liters to 3 weeks
 d. 2,280 liters to 12 days
 e. 15,960 liters to 12 weeks

Unit 2 Lesson 10

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Name: _____ Date: _____ Period: _____

5. A caterer needs to buy 38 pounds of potatoes for an event. It costs \$6 for 5 pounds of potatoes.
- a. Write a ratio for the price of the potatoes per pound.
\$6 : 5 or \$1.20 : 1
- b. If all potatoes are sold at this rate, how much will the caterer pay for the potatoes they need? Explain your thinking.
\$45.60. Explanations vary. The caterer will pay \$45.60 for 38 pounds of potatoes. I used equivalent ratios to determine that the cost of the potatoes is \$1.20 per pound. Because the caterer needs 38 pounds, I multiplied the cost of 1 pound by 38: $1.20 \times 38 = \$45.60$.
6. A caterer also needs to buy 130 rolls for the event. A bakery sells 12 rolls for \$4.80.
- a. Write a ratio to describe the cost of the rolls.
\$4.80 : 12 or \$0.40 : 1
- b. If all rolls are sold at the same rate, how much will the caterer pay for the rolls they need? Explain your thinking.
\$52. Explanations vary. The caterer will pay \$52 for 130 rolls. I used equivalent ratios to determine that the cost of 1 roll is \$0.40. Because the caterer needs 130 rolls, I multiplied the cost of 1 roll by 130: $0.40 \times 130 = \$52$.
7. The directions for a shade of green paint say to mix 7 milliliters of yellow paint with 3 milliliters of blue paint. Han has 42 milliliters of yellow paint and 25 milliliters of blue paint. If Han wants to use all of the yellow paint, how much blue paint, if any, will he have left? Show or explain your thinking. Use the ratio table it helps you show your thinking. **7 milliliters. Explanations vary.**

Yellow (mL)	Blue (mL)
7	3
42	18

I used equivalent ratios to determine that Han will use 18 milliliters of blue paint with 42 milliliters of yellow paint. Then I subtracted to find how much blue paint was left: $25 - 18 = 7$.

8. Mai is reading a 55-page book. She read the first 35 pages in 30 minutes. If she continues to read at the same rate, will she be able to complete this book in less than 1 hour? Show or explain your thinking.
Yes. Explanations vary. There are 2 sets of 30 minutes in 1 hour, so Mai can read $35 \times 2 = 70$ pages in 1 hour. Therefore, she will be able to complete her 55-page book within 1 hour.

Unit 2 Lesson 10

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	6.RP.A.3
2	2	6.RP.A.3
3	2	6.RP.A.3
4	2	6.RP.A.3
5	2	6.RP.A.3
6	2	6.RP.A.3
7	2	6.RP.A.3
8	2	6.RP.A.3

Notes:

Additional Practice

ACC6.2.11

Problems 1–2: The ratio of cats to dogs at a boarding facility is 3 : 8. There are 33 total cats and dogs staying at the boarding facility.

1. How many cats are at the boarding facility?

9 cats

2. How many dogs are at the boarding facility?

24 dogs

Problems 3–4: Here is a tape diagram representing the ratio of yellow paint to blue paint in a mixture of green paint.



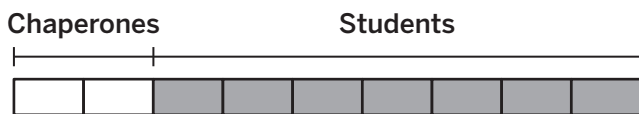
3. What is the ratio of yellow paint to blue paint?

3 : 4

4. Complete the table to show the amount of yellow and blue paint needed to make each quantity of green paint.

Green (gal)	Yellow (gal)	Blue (gal)
21	9	12
35	15	20

Problems 5–6: Here is a tape diagram representing the ratio of students to chaperones on a school trip.



5. What is the ratio of chaperones to students?

2 : 7

6. Complete the table to show the number of chaperones and students for each amount of people on the school trip.

Number of People	Chaperones	Students
18	4	14
90	20	70

7. Jada has a bag of 100 marbles. The ratio of red marbles to blue marbles is 4 : 1. How many blue marbles does she have? Show or explain your thinking.

20 blue marbles. Explanations vary.

Red Marbles	Blue Marbles	Total
4	1	5
80	20	100

8. A veterinarian examined 54 pets at her clinic in one day. The ratio of dogs to other pets was 8 : 1. How many dogs were examined? Show or explain your thinking.

48 dogs. Explanations vary.

Dogs	Other Pets	Total
8	1	9
48	6	54

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.2.11

Problems 1–2: The ratio of cats to dogs at a boarding facility is 3 : 8. There are 33 total cats and dogs staying at the boarding facility.

1. How many cats are at the boarding facility?

9 cats

2. How many dogs are at the boarding facility?

24 dogs

Problems 3–4: Here is a tape diagram representing the ratio of yellow paint to blue paint in a mixture of green paint.



3. What is the ratio of yellow paint to blue paint?

3 : 4

4. Complete the table to show the amount of yellow and blue paint needed to make each quantity of green paint.

Green (gal)	Yellow (gal)	Blue (gal)
21	9	12
35	15	20

Unit 2 Lesson 11

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Name: _____ Date: _____ Period: _____

Problems 5–6: Here is a tape diagram representing the ratio of students to chaperones on a school trip.



5. What is the ratio of chaperones to students?

2 : 7

6. Complete the table to show the number of chaperones and students for each amount of people on the school trip.

Number of People	Chaperones	Students
18	4	14
90	20	70

7. Jada has a bag of 100 marbles. The ratio of red marbles to blue marbles is 4 : 1. How many blue marbles does she have? Show or explain your thinking.

20 blue marbles. *Explanations vary.*

Red Marbles	Blue Marbles	Total
4	1	5
80	20	100

8. A veterinarian examined 54 pets at her clinic in one day. The ratio of dogs to other pets was 8 : 1. How many dogs were examined? Show or explain your thinking.

48 dogs. *Explanations vary.*

Dogs	Other Pets	Total
8	1	9
48	6	54

Unit 2 Lesson 11

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	6.RP.A.3
2	2	6.RP.A.3
3	2	6.RP.A.3
4	1	6.RP.A.3
5	2	6.RP.A.3
6	1	6.RP.A.3
7	1	6.RP.A.3
8	1	6.RP.A.3

Notes:

Additional Practice

ACC6.2.12

Problems 1–2: Pasta is made from 2 parts flour and 5 parts water.

1. How much flour is needed to make 28 ounces of pasta?

8 ounces

2. How much water is needed to make 28 ounces of pasta?

20 ounces

Problems 3–4: A smoothie is made from 6 parts fruit and 2 parts milk.

3. How much fruit is needed to make 24 ounces of smoothie?

18 ounces

4. How much milk is needed to make 24 ounces of smoothie?

6 ounces

Problems 5–6: One batch of purple paint uses 6 teaspoons of blue paint for every 9 teaspoons of red paint.

5. Complete the table to show the amount of blue and red paint needed to make each amount of purple paint.

Blue Paint (tsp)	Red Paint (tsp)	Purple Paint (tsp)
18	27	45
4	6	10

6. Shawn made 5 teaspoons of purple by combining blue paint and red paint. Shawn used 3 teaspoons of blue paint. Is Shawn using the same recipe? Explain your thinking.

No. Explanations vary. Shawn's mixture uses a ratio of 3 teaspoons of blue paint to 2 teaspoons of red paint. If I double that mixture, the ratio is 6 : 4. The original mixture uses 9 teaspoons of red paint for the same amount of blue paint.

7. A hospital requires there to be 2 nurses for every 8 patients. If the hospital has a total of 80 patients and nurses on one of the floors, how many nurses and how many patients are there? Show or explain your thinking.

16 nurses and 64 patients. Explanations vary. If there were 2 nurses for every 8 patients, that's a ratio of 2 : 8, representing a total of 10 people. 80 people is 8 times as many, so I multiplied the 2 and 8 each by 8 to get the total number of nurses and patients.

Nurses	Patients	Total
2	8	10
16	64	80

8. A teacher is planning a class trip to the zoo. The zoo requires 3 adults to join for every 10 students. If the teacher orders 78 tickets, how many tickets are for adults and how many are for students? Show or explain your thinking.

18 tickets were for adults and 60 tickets were for students. Explanations vary. If there are 3 adults for every 10 students, that's a ratio of 3 : 10, representing a total of 13 tickets. 78 tickets is 6 times as many tickets, so I multiplied the 3 and 10 each by 6 to get the total number of tickets for the adults and the students.

Adults	Students	Total Tickets
3	10	13
18	60	78

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.2.12

Problems 1–2: Pasta is made from 2 parts flour and 5 parts water.

1. How much flour is needed to make 28 ounces of pasta?

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2. How much water is needed to make 28 ounces of pasta?

20 ounces

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3. How much fruit is needed to make 24 ounces of smoothie?

18 ounces

4. How much milk is needed to make 24 ounces of smoothie?

6 ounces

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Blue Paint (tsp)	Red Paint (tsp)	Purple Paint (tsp)
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6. Shawn made 5 teaspoons of purple by combining blue paint and red paint. Shawn used 3 teaspoons of blue paint. Is Shawn using the same recipe? Explain your thinking.

No. Explanations vary. Shawn's mixture uses a ratio of 3 teaspoons of blue paint to 2 teaspoons of red paint. If I double that mixture, the ratio is 6:4. The original mixture uses 9 teaspoons of red paint for the same amount of blue paint.

Unit 2 Lesson 12

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Name: _____ Date: _____ Period: _____

7. A hospital requires there to be 2 nurses for every 8 patients. If the hospital has a total of 80 patients and nurses on one of the floors, how many nurses and how many patients are there? Show or explain your thinking.

16 nurses and 64 patients. Explanations vary. If there were 2 nurses for every 8 patients, that's a ratio of 2:8, representing a total of 10 people. 80 people is 8 times as many, so I multiplied the 2 and 8 each by 8 to get the total number of nurses and patients.

Nurses	Patients	Total
2	8	10
16	64	80

8. A teacher is planning a class trip to the zoo. The zoo requires 3 adults to join for every 10 students. If the teacher orders 78 tickets, how many tickets are for adults and how many are for students? Show or explain your thinking.

18 tickets were for adults and 60 tickets were for students. Explanations vary. If there are 3 adults for every 10 students, that's a ratio of 3:10, representing a total of 13 tickets. 78 tickets is 6 times as many tickets, so I multiplied the 3 and 10 each by 6 to get the total number of tickets for the adults and the students.

Adults	Students	Total Tickets
3	10	13
18	60	78

Unit 2 Lesson 12

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	6.RP.A.3
2	1	6.RP.A.3
3	1	6.RP.A.3
4	1	6.RP.A.3
5	2	6.RP.A.3
6	2	6.RP.A.3
7	2	6.RP.A.3
8	2	6.RP.A.3

Notes:

Additional Practice

ACC6.2.13

Problems 1–3: For each pair, circle the larger unit of measurement.

1. Mile Meter

2. Inch Foot

3. Pound Gram

4. For each unit of measurement, select whether it measures length, volume, or weight.

Unit	Length	Volume	Weight
Millimeter	X		
Pound			X
Cup		X	
Quart		X	
Yard	X		
Gram			X

Problems 5–7: Identify a unit that can be used to measure:

5. The length of a hammer.

Responses vary. Inches

6. The volume of an orange soda can.

Responses vary. Fluid ounces

7. The weight of a truck.

Responses vary. Tons

8. For each pair, select the unit you would most likely use to measure a box of cereal.

- | | | |
|----------|---------|-----------|
| a | Inches | Miles |
| b | Gallons | Cups |
| c | Ounces | Kilograms |

9. Match each object with the unit you would most likely use to measure it.

- | | |
|--|----------------------|
| a. The height of a house. | d Kilometers |
| b. The length of a nail. | e Pounds |
| c. The weight of a staple. | c Grams |
| d. The distance between two towns. | b Centimeters |
| e. The weight of a piggy bank. | a Feet |
| f. The volume of a pitcher of lemonade. | f Gallons |

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.2.13

Problems 1–3: For each pair, circle the larger unit of measurement.

1. Mile Meter

2. Inch Foot

3. Pound Gram

4. For each unit of measurement, select whether it measures length, volume, or weight.

Unit	Length	Volume	Weight
Millimeter	X		
Pound			X
Cup		X	
Quart		X	
Yard	X		
Gram			X

Problems 5–7: Identify a unit that can be used to measure:

5. The length of a hammer.

Responses vary. Inches

6. The volume of an orange soda can.

Responses vary. Fluid ounces

7. The weight of a truck.

Responses vary. Tons

Unit 2 Lesson 13

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Name: _____ Date: _____ Period: _____

8. For each pair, select the unit you would most likely use to measure a box of cereal.

- a. Inches Miles
b. Gallons Cups
c. Ounces Kilograms

9. Match each object with the unit you would most likely use to measure it.

- a. The height of a house. d Kilometers
b. The length of a nail. e Pounds
c. The weight of a staple. c Grams
d. The distance between two towns. b Centimeters
e. The weight of a piggy bank. a Feet
f. The volume of a pitcher of lemonade. f Gallons

Unit 2 Lesson 13

54

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	6.RP.A.3.D
2	1	6.RP.A.3.D
3	1	6.RP.A.3.D
4	1	6.RP.A.3.D
5	2	6.RP.A.3.D
6	2	6.RP.A.3.D
7	2	6.RP.A.3.D
8	2	6.RP.A.3.D
9	2	6.RP.A.3.D

Notes:

Additional Practice

ACC6.2.14

1. Use the conversion rate 1 inch $\approx$ 2.54 centimeters to complete the table.

Inches	Centimeters
1	2.54
6	15.24
100	254
50	127
20	50.8

2. There are 60 seconds in 1 minute. Use this conversion rate to complete the table.

Seconds	Minutes
60	100
30	0.5
210	3.5
300	5
480	8

3. There are 3,785 milliliters in 1 gallon, and there are 4 quarts in 1 gallon. Determine the following. Show or explain your thinking.

- a How many quarts are in 20 gallons?
80 quarts. Explanations vary. There are 4 quarts in 1 gallon. I can multiply 4 by 20 to find the amount in 20 gallons, and $4 \cdot 20 = 80$.
- b How many milliliters are in 20 quarts?
18,925 milliliters. Explanations vary. 20 quarts is equal to 5 gallons, so I multiplied 3785 by 5.

4. There are 5,280 feet in 1 mile. A dolphin swam 3 miles. How many feet did the dolphin swim?

15,840 feet

5. Clare is making spaghetti sauce that calls for 46 oz of tomatoes. There are 16 ounces in one pound. Determine the approximate weight of the tomatoes, in pounds.

Responses vary. A little less than 3 pounds.

6. A large jar holds 4 gallons of milk, which is about the same as 15.2 liters of milk. A smaller jar holds 2 gallons of milk. About how many liters does the small jar hold?

- A. 3.8 liters
- B. 7.6 liters**
- C. 11.4 liters
- D. 30.4 liters

7. Noah's recipe for bread calls for 8 pounds of flour. He has a scale that measures in kilograms. 1 kilogram is approximately 2.2 pounds.

- a** About how many kilograms of flour does Noah need?

Responses vary. About 3.6 kilograms

- b** Noah also has a 1-cup measuring cup. If it takes about 4 cups of flour to equal 1 pound, how many cups of flour does Noah need?

About 32 cups

8. There are 16 tablespoons in 1 cup. Priya says that $1\frac{1}{4}$ cups of chopped onions is equivalent to 60 tablespoons. Is Priya correct? Explain your thinking.

No. Explanations vary. If it takes 16 tablespoons to equal 1 cup, then 4 tablespoons would equal $\frac{1}{4}$ cup. So, $1\frac{1}{4}$ cups would equal $16 + 4 = 20$ tablespoons, not 60 tablespoons.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.2.14

1. Use the conversion rate 1 inch $\approx$ 2.54 centimeters to complete the table.

Inches	Centimeters
1	2.54
6	15.24
100	254
50	127
20	50.8

2. There are 60 seconds in 1 minute. Use this conversion rate to complete the table.

Seconds	Minutes
60	100
30	0.5
210	3.5
300	5
480	8

3. There are 3,785 milliliters in 1 gallon, and there are 4 quarts in 1 gallon. Determine the following. Show or explain your thinking.

- a. How many quarts are in 20 gallons?
80 quarts. Explanations vary. There are 4 quarts in 1 gallon. I can multiply 4 by 20 to find the amount in 20 gallons, and $4 \cdot 20 = 80$.
- b. How many milliliters are in 20 quarts?
18,925 milliliters. Explanations vary. 20 quarts is equal to 5 gallons, so I multiplied 3,785 by 5.

4. There are 5,280 feet in 1 mile. A dolphin swam 3 miles. How many feet did the dolphin swim?
15,840 feet

Unit 2 Lesson 14

55

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Name: _____ Date: _____ Period: _____

5. Clare is making spaghetti sauce that calls for 46 oz of tomatoes. There are 16 ounces in one pound. Determine the approximate weight of the tomatoes, in pounds.
Responses vary. A little less than 3 pounds.

6. A large jar holds 4 gallons of milk, which is about the same as 15.2 liters of milk. A smaller jar holds 2 gallons of milk. About how many liters does the small jar hold?

- A. 3.8 liters
B. 7.6 liters
 C. 11.4 liters
 D. 30.4 liters

7. Noah's recipe for bread calls for 8 pounds of flour. He has a scale that measures in kilograms. 1 kilogram is approximately 2.2 pounds.

- a. About how many kilograms of flour does Noah need?
Responses vary. About 3.6 kilograms
- b. Noah also has a 1-cup measuring cup. If it takes about 4 cups of flour to equal 1 pound, how many cups of flour does Noah need?
About 32 cups

8. There are 16 tablespoons in 1 cup. Priya says that $1\frac{1}{2}$ cups of chopped onions is equivalent to 60 tablespoons. Is Priya correct? Explain your thinking.

No. Explanations vary. If it takes 16 tablespoons to equal 1 cup, then 4 tablespoons would equal $\frac{1}{4}$ cup. So, $1\frac{1}{2}$ cups would equal $16 \cdot 4 = 20$ tablespoons, not 60 tablespoons.

Unit 2 Lesson 14

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	6.RP.A.3.D
2	2	6.RP.A.3.D
3	2	6.RP.A.3.D
4	2	6.RP.A.3.D
5	2	6.RP.A.3.D
6	2	6.RP.A.3.D
7	2	6.RP.A.3.D
8	3	6.RP.A.3.D

Notes:

Additional Practice

ACC6.2.16

Problems 1–2: Turtle A walks 1.5 feet in 5 seconds. Turtle B walks 3 feet in 9 seconds. Each turtle keeps walking at those speeds.

1. How far does each turtle walk in 45 seconds?

Turtle A walks 13.5 feet in 45 seconds. Turtle B walks 15 feet in 45 seconds.

2. If the two turtles start at the same place and walk in the same direction, how far apart will the two turtles be after 3 minutes? Show or explain your thinking.

6 feet apart. Explanations vary. 3 minutes is equal to 180 seconds, which is 4 times as long as 45 seconds. In 180 seconds, turtle A would walk $13.5 \cdot 4 = 54$ feet, and turtle B would walk $15 \cdot 4 = 60$ feet. Since $60 - 54 = 6$, turtle B will be 6 feet ahead of turtle A.

3. A cheetah runs 100 meters in 6 seconds.

- a At this rate, how long will it take the cheetah to run 150 meters?

9 seconds

- b How far will the cheetah run in 27 seconds?

450 meters

4. Snail A travels 4 inches in 7 minutes. Snail B travels 6 inches in 10 minutes. Each snail continues traveling at a constant speed.

- a How far does Snail A travel in 35 minutes? Show your thinking.

20 inches

- b How far does Snail B travel in 35 minutes? Show your thinking.

21 inches

- c If the two snails start at the same place and travel in the same direction, how far apart will the two snails be after 84 minutes?

2.4 inches

5. Elena reads 15 pages in 25 minutes. Shawn reads 12 pages in 15 minutes. Both read at a constant rate.

- a How many pages can Elena read in 1 hour?

36 pages

- b How many pages can Shawn read in 1 hour?

48 pages

- c Who reads more quickly?

Shawn

Problems 6–8: Priya types 1,100 words in 20 minutes. Tyler types 600 words in 10 minutes. Andre types 300 words in 6 minutes. Each person types at a constant rate.

6. Complete the tables to represent the number of words each person can type in different amounts of time.

Priya		Tyler		Andre	
Words	Time (min)	Words	Time (min)	Words	Time (min)
55	1	60	1	50	1
1,650	30	1,800	30	1,500	30
2,475	45	2,700	45	2,250	45

7. Which person is typing at the fastest rate? Show or explain your thinking.

Tyler. *Explanations vary.* Tyler is typing 60 words per minute, which is 5 words per minute faster than Priya and 10 words per minute faster than Andre.

8. How long would it take Priya to type 3,300 words? Explain your thinking.

60 minutes. *Explanations vary.* Priya can type 55 words per minute, and $3300 \div 55 = 60$ minutes.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.2.16

Problems 1–2: Turtle A walks 1.5 feet in 5 seconds. Turtle B walks 3 feet in 9 seconds. Each turtle keeps walking at those speeds.

1. How far does each turtle walk in 45 seconds?

Turtle A walks 13.5 feet in 45 seconds. Turtle B walks 15 feet in 45 seconds.

2. If the two turtles start at the same place and walk in the same direction, how far apart will the two turtles be after 3 minutes? Show or explain your thinking.

6 feet apart. Explanations vary. 3 minutes is equal to 180 seconds, which is 4 times as long as 45 seconds. In 180 seconds, turtle A would walk $1.5 \times 4 = 6$ feet, and turtle B would walk $3 \times 4 = 12$ feet. Since $12 - 6 = 6$, turtle B will be 6 feet ahead of turtle A.

3. A cheetah runs 100 meters in 6 seconds.

- a. At this rate, how long will it take the cheetah to run 150 meters?

9 seconds

- b. How far will the cheetah run in 27 seconds?

450 meters

4. Snail A travels 4 inches in 7 minutes. Snail B travels 6 inches in 10 minutes. Each snail continues traveling at a constant speed.

- a. How far does Snail A travel in 35 minutes? Show your thinking.

20 inches

- b. How far does Snail B travel in 35 minutes? Show your thinking.

21 inches

- c. If the two snails start at the same place and travel in the same direction, how far apart will the two snails be after 84 minutes?

2.4 inches

Unit 2 Lesson 16

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Name: _____ Date: _____ Period: _____

5. Elena reads 15 pages in 25 minutes. Shawn reads 12 pages in 15 minutes. Both read at a constant rate.

- a. How many pages can Elena read in 1 hour?

36 pages

- b. How many pages can Shawn read in 1 hour?

48 pages

- c. Who reads more quickly?

Shawn

Problems 6–8: Priya types 1,100 words in 20 minutes. Tyler types 600 words in 10 minutes. Andre types 300 words in 6 minutes. Each person types at a constant rate.

6. Complete the tables to represent the number of words each person can type in different amounts of time.

Priya		Tyler		Andre	
Words	Time (min)	Words	Time (min)	Words	Time (min)
55	1	60	1	50	1
1,650	30	1,800	30	1,500	30
2,475	45	2,700	45	2,250	45

7. Which person is typing at the fastest rate? Show or explain your thinking.

Tyler. Explanations vary. Tyler is typing 60 words per minute, which is 5 words per minute faster than Priya and 10 words per minute faster than Andre.

8. How long would it take Priya to type 3,300 words? Explain your thinking.

60 minutes. Explanations vary. Priya can type 55 words per minute, and $3300 \div 55 = 60$ minutes.

Unit 2 Lesson 16

60

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	6.RP.A.2
2	2	6.RP.A.2, 6.RP.A.3.B
3	2	6.RP.A.2, 6.RP.A.3.B
4	2	6.RP.A.2, 6.RP.A.3.B
5	2	6.RP.A.2, 6.RP.A.3.B
6	2	6.RP.A.2, 6.RP.A.3.B
7	3	6.RP.A.2, 6.RP.A.3.B
8	3	6.RP.A.2, 6.RP.A.3.B

Notes:

Additional Practice

ACC6.2.17

Problems 1–3: Tomas and Leslie purchased a new printer for their office. The printer can print 250 pages every 2 minutes.

1. How many pages can the printer print per minute?

125 pages

2. How many minutes does it take to print one page?

0.008 minutes

3. How many pages can be printed in 10 minutes?

1,250 pages

4. A farm lets you pick 2 pints of blueberries for \$6.00.

- a What is the cost per pint?

\$3

- b How many pints can you buy per dollar?

$\frac{1}{3}$ pint (or equivalent)

- c At this rate, how many pints can you buy for \$15?

5 pints

- d At this rate, how much will 7 pints of blueberries cost?

\$21

5. A farm lets you pick 4 pounds of strawberries for \$16.00.

- a What is the cost per pound of strawberries?

\$4

- b How many pounds of strawberries can you buy per dollar?

$\frac{1}{4}$ pound (or equivalent)

- c At this rate, how many pounds can you buy for \$22?

$5\frac{1}{2}$ pounds

- d At this rate, how much will 10 pounds of strawberries cost?

\$40

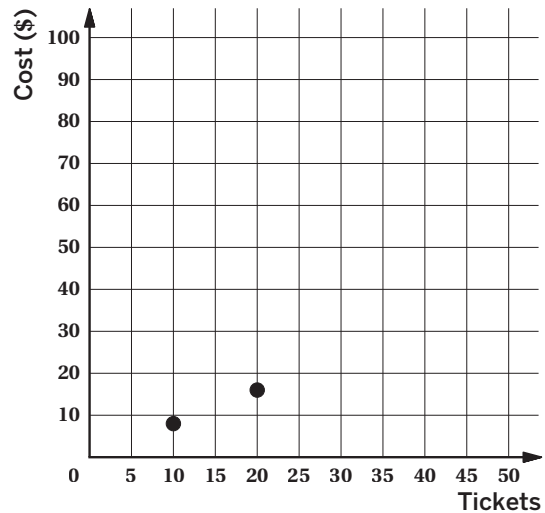
Problems 6–8: Clare paid \$10 for 8 game tickets at her school carnival.

6. Complete the table to show the cost of different numbers of tickets.

Tickets	Cost (\$)
8	10
1	1.25
10	12.50
15	18.75
24	30

7. Clare says that this graph represents two pairs of numbers of tickets and cost in dollars. Is Clare’s graph correct? Explain your thinking.

No. Explanations vary. Clare’s graph shows that 10 tickets cost a little less than \$10, when they actually cost \$12.50.



8. Kiran says that if he pays \$6 for 4 tickets, he’s getting a better deal than Clare. Is Kiran correct? Explain your thinking.

No. Explanations vary. Clare paid \$10 for 8 tickets, which is equivalent to paying \$5 for 4 tickets. Since she is paying less than Kiran for the same number of tickets, Clare is getting a better deal.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.2.17

Problems 1–3: Tomas and Leslie purchased a new printer for their office. The printer can print 250 pages every 2 minutes.

1. How many pages can the printer print per minute?

125 pages

2. How many minutes does it take to print one page?

0.008 minutes

3. How many pages can be printed in 10 minutes?

1,250 pages

4. A farm lets you pick 2 pints of blueberries for \$6.00.

- a. What is the cost per pint?

\$3

- b. How many pints can you buy per dollar?

$\frac{1}{3}$ pint (or equivalent)

- c. At this rate, how many pints can you buy for \$15?

5 pints

- d. At this rate, how much will 7 pints of blueberries cost?

\$21

5. A farm lets you pick 4 pounds of strawberries for \$16.00.

- a. What is the cost per pound of strawberries?

\$4

- b. How many pounds of strawberries can you buy per dollar?

$\frac{1}{4}$ pound (or equivalent)

- c. At this rate, how many pounds can you buy for \$22?

$5\frac{1}{2}$ pounds

- d. At this rate, how much will 10 pounds of strawberries cost?

\$40

Unit 2 Lesson 17

61

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Name: _____ Date: _____ Period: _____

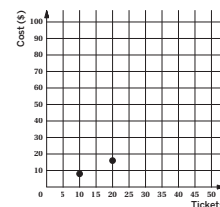
Problems 6–8: Clare paid \$10 for 8 game tickets at her school carnival.

6. Complete the table to show the cost of different numbers of tickets.

Tickets	Cost (\$)
8	10
1	1.25
10	12.50
15	18.75
24	30

7. Clare says that this graph represents two pairs of numbers of tickets and cost in dollars. Is Clare's graph correct? Explain your thinking.

No. Explanations vary. Clare's graph shows that 10 tickets cost a little less than \$10, when they actually cost \$12.50.



8. Kiran says that if he pays \$6 for 4 tickets, he's getting a better deal than Clare. Is Kiran correct? Explain your thinking.

No. Explanations vary. Clare paid \$10 for 8 tickets, which is equivalent to paying \$5 for 4 tickets. Since she is paying less than Kiran for the same number of tickets, Clare is getting a better deal.

Unit 2 Lesson 17

62

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	6.RP.A.3.B
2	1	6.RP.A.3.B
3	1	6.R.P.A.3.B
4	2	6.RP.A.3.B
5	2	6.RP.A.3.B
6	2	6.RP.A.3.B
7	3	6.RP.A.3.B
8	3	6.RP.A.3.B

Notes:

Additional Practice

ACC6.2.18

- A train travels 8 miles in 6 minutes. Which of these is *not* an equivalent rate?
 - 4 miles in 3 minutes
 - 1 mile in 45 seconds
 - $\frac{4}{3}$ miles in 1 minute
 - $\frac{3}{4}$ miles in 1 minute
- A 2-pound squash costs \$8 at a grocery store. Which of these represent the same price per pound of squash? Select *all* that apply.
 - ☒ A. A 1-pound squash for \$4
 - ☐ B. A 1-pound squash for \$2
 - ☒ C. A $\frac{1}{4}$ -pound squash for \$1
 - ☐ D. A 1-pound squash for \$0.25
 - ☒ E. 8 pounds of squash for \$32
- Lin bought a 40-pound bag of dog food for \$48.80. Shawn bought an 8-pound bag of dog food for \$9.88.

- a Complete this table to determine the unit rate for each bag of dog food.

	Dog food (lb)	Price (\$)	Unit rate (\$ per lb)
Lin	40	48.80	1.22
Shawn	8	9.88	1.24

- b Which size bag of dog food offers the best deal? Explain your thinking.

40-pound bag. Explanations vary. The 40-pound bag of dog food offers the best deal because it has a lower unit rate.

- Which brand of peanut butter offers the best deal? Explain your thinking.
 - Brand A: 28 ounces for \$4.76
 - Brand B: 48 ounces for \$8.64
 - Brand C: 16 ounces for \$3.52

Brand A. Explanations vary. Brand A offers the best deal. Brand A costs \$0.17 per ounce, Brand B costs \$0.18 per ounce, and Brand C costs \$0.22 per ounce.

5. A grocery store sells four kinds of apples in bags.

- a Complete the table to determine the unit price for each variety of apple.

Apple Variety	Price (\$)	Weight (lb)	Unit Rate (\$ per lb)
Gala	6.45	5	1.29
Fuji	3.08	4	0.77
Honeycrisp	5.37	3	1.79
Granny Smith	7.74	6	1.29

- b Which type of apple is the best deal? Explain your thinking.

Fuji apples. Explanations vary. Fuji apples are the best deal because their unit price is the least, \$0.77 per pound.

6. Jada reads 15 pages in 12 minutes. Bard reads 21 pages in 28 minutes. Who reads faster? Explain your thinking.

Jada. Explanations vary. Jada can read 5 pages in 4 minutes, and Bard reads 3 pages in 4 minutes.

7. Mai uses 8 cups of apples to make 3 pies. Mai says, “At this same rate, I can make 5 pies using 16 cups of apples.” Is Mai correct? Explain your thinking.

No. Explanations vary. That's twice as many apples, so Mai could make twice as many pies. $3 \cdot 2 = 6$, not 5.

8. Plane A travels 2,260 miles in 4 hours. Plane B travels 810 miles in $1\frac{1}{2}$ hours. Plane C travels 3,692 miles in $6\frac{1}{2}$ hours. Tyler claims that Plane C has the fastest speed. Is Tyler correct? Explain your thinking.

Yes. Explanations vary. Plane A travels 565 miles per hour, Plane B travels 540 miles per hour, and Plane C travels 568 miles per hour. Plane C is traveling the fastest.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.2.18

- A train travels 8 miles in 6 minutes. Which of these is *not* an equivalent rate?
 - 4 miles in 3 minutes
 - 1 mile in 45 seconds
 - $\frac{4}{3}$ miles in 1 minute
 - $\frac{3}{4}$ miles in 1 minute
- A 2-pound squash costs \$8 at a grocery store. Which of these represent the same price per pound of squash? Select *all* that apply.
 - A 1-pound squash for \$4
 - A 1-pound squash for \$2
 - $\frac{1}{2}$ -pound squash for \$1
 - A 1-pound squash for \$0.25
 - 8 pounds of squash for \$32

- Lin bought a 40-pound bag of dog food for \$48.80. Shawn bought an 8-pound bag of dog food for \$9.88.

- Complete this table to determine the unit rate for each bag of dog food.

	Dog food (lb)	Price (\$)	Unit rate (\$ per lb)
Lin	40	48.80	1.22
Shawn	8	9.88	1.24

- Which size bag of dog food offers the best deal? Explain your thinking.
40-pound bag. Explanations vary. The 40-pound bag of dog food offers the best deal because it has a lower unit rate.

- Which brand of peanut butter offers the best deal? Explain your thinking.

- Brand A: 28 ounces for \$4.76
- Brand B: 48 ounces for \$8.64
- Brand C: 16 ounces for \$3.52

Brand A. Explanations vary. Brand A offers the best deal. Brand A costs \$0.17 per ounce, Brand B costs \$0.18 per ounce, and Brand C costs \$0.22 per ounce.

Unit 2 Lesson 18

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Name: _____ Date: _____ Period: _____

- A grocery store sells four kinds of apples in bags.

- Complete the table to determine the unit price for each variety of apple.

Apple Variety	Price (\$)	Weight (lb)	Unit Rate (\$ per lb)
Gala	6.45	5	1.29
Fuji	3.08	4	0.77
Honeycrisp	5.37	3	1.79
Granny Smith	7.74	6	1.29

- Which type of apple is the best deal? Explain your thinking.

Fuji apples. Explanations vary. Fuji apples are the best deal because their unit price is the least, \$0.77 per pound.

- Jada reads 15 pages in 12 minutes. Bard reads 21 pages in 28 minutes. Who reads faster? Explain your thinking.

Jada. Explanations vary. Jada can read 5 pages in 4 minutes, and Bard reads 3 pages in 4 minutes.

- Mai uses 8 cups of apples to make 3 pies. Mai says, "At this same rate, I can make 5 pies using 16 cups of apples." Is Mai correct? Explain your thinking.

No. Explanations vary. That's twice as many apples, so Mai could make twice as many pies. $3 \cdot 2 = 6$, not 5.

- Plane A travels 2,260 miles in 4 hours. Plane B travels 810 miles in $1\frac{1}{2}$ hours.

Plane C travels 3,692 miles in $6\frac{1}{2}$ hours. Tyler claims that Plane C has the fastest speed. Is Tyler correct? Explain your thinking.

Yes. Explanations vary. Plane A travels 565 miles per hour, Plane B travels 540 miles per hour, and Plane C travels 568 miles per hour. Plane C is traveling the fastest.

Unit 2 Lesson 18

64

Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	6.RP.A.3.B
2	1	6.RP.A.3.B
3	2	6.RP.A.2, 6.RP.A.3.B
4	2	6.RP.A.2, 6.RP.A.3.B
5	2	6.RP.A.2, 6.RP.A.3.B
6	2	6.RP.A.2, 6.RP.A.3.B
7	2	6.RP.A.2, 6.RP.A.3.B
8	3	6.RP.A.2, 6.RP.A.3.B

Notes:

Additional Practice

4.12

1. How many groups of $\frac{5}{6}$ are in each of the following quantities? Show your thinking.

a $\frac{2}{3}$

$\frac{12}{15}$ or $\frac{4}{5}$; Students' explanations may vary.

b $1\frac{1}{2}$

$\frac{18}{10}$ or $1\frac{4}{5}$; Students' explanations may vary.

c $4\frac{1}{6}$

$\frac{150}{30}$ or 5; Students' explanations may vary.

2. How many groups of $1\frac{3}{4}$ are in each of the following quantities? Show your thinking.

a $2\frac{1}{2}$

$\frac{10}{7}$ or $1\frac{3}{7}$; Students' explanations may vary.

b 4

$\frac{16}{7}$ or $2\frac{2}{7}$; Students' explanations may vary.

c $\frac{3}{4}$

$\frac{3}{7}$; Students' explanations may vary.

3. Shawn's golf club is $3\frac{3}{4}$ ft tall. Shawn's hockey stick is $5\frac{1}{2}$ ft tall.

- a What fraction of the golf club's height is the hockey stick? Show your thinking.

$1\frac{7}{15}$ times as tall; Sample response: $5\frac{1}{2} \div 3\frac{3}{4} = \frac{11}{2} \div \frac{15}{4} = \frac{11}{2} \cdot \frac{4}{15} = \frac{22}{15}$ or $1\frac{7}{15}$

- b How many times as tall is the hockey stick than the golf club? Show your thinking.

$\frac{15}{22}$ of the height; Sample response: $3\frac{3}{4} \div 5\frac{1}{2} = \frac{15}{4} \div \frac{11}{2} = \frac{15}{4} \cdot \frac{2}{11} = \frac{15}{22}$

4. Jada is $5\frac{1}{4}$ ft tall. Her little sister is 4 ft tall. How many times taller is Jada than her sister?

A. $\frac{4}{5}$

B. $\frac{16}{21}$

C. $1\frac{1}{4}$

D. $1\frac{5}{16}$

Clare and Diego each went for a bike ride. Clare rode $1\frac{1}{2}$ mi, and Diego rode $2\frac{3}{4}$ mi. Use this information for Problems 5–6.

5. How many times as far as Clare did Diego ride?

- a Write a division expression to represent this situation.

$$2\frac{3}{4} \div 1\frac{1}{2}$$

- b Determine the solution. Show your thinking.

Diego rode $1\frac{5}{6}$ times as far as Clare. Sample response: $\frac{11}{4} \div \frac{3}{2} = \frac{11}{4} \cdot \frac{2}{3} = \frac{11}{6}$

6. What fraction of Diego's distance did Clare ride?

- a Write a division expression to represent this situation.

$$1\frac{1}{2} \div 2\frac{3}{4}$$

- b Determine the solution. Show your thinking.

Clare rode $\frac{6}{11}$ of Diego's distance. Sample response: $\frac{3}{2} \div \frac{11}{4} = \frac{3}{2} \cdot \frac{4}{11} = \frac{6}{11}$

7. Andre uses $1\frac{5}{6}$ lb of tomatoes for a batch of homemade spaghetti sauce. He has $9\frac{1}{2}$ lb of tomatoes from his garden. Does Andre have enough tomatoes to make 6 batches of spaghetti sauce? Explain your thinking.

No; Sample response: Andre does not have enough to make 6 batches of spaghetti sauce, but he has enough to make $5\frac{2}{11}$ batches because $9\frac{1}{2} \div 1\frac{5}{6} = 5\frac{2}{11}$, or he can make 5 whole batches.

8. Mai has worked $3\frac{3}{4}$ hours of her $7\frac{3}{4}$ hour shift. She says that she has worked more than half of her shift. Is Mai correct? Explain your thinking.

No; Sample response: Mai has not quite worked half of her shift. Mai worked $3\frac{3}{4} \div 7\frac{3}{4} = \frac{15}{31}$ of her shift.

Name: _____ Date: _____ Period: _____

Additional Practice

4.12

1. How many groups of $\frac{5}{8}$ are in each of the following quantities? Show your thinking.

a. $\frac{2}{3}$
 $\frac{12}{15}$ or $\frac{4}{5}$; Students' explanations may vary.

b. $1\frac{1}{2}$
 $\frac{16}{10}$ or $1\frac{4}{5}$; Students' explanations may vary.

c. $4\frac{1}{6}$
 $\frac{150}{36}$ or 5; Students' explanations may vary.

2. How many groups of $1\frac{3}{4}$ are in each of the following quantities? Show your thinking.

a. $2\frac{1}{2}$
 $\frac{10}{7}$ or $1\frac{3}{7}$; Students' explanations may vary.

b. $\frac{4}{7}$
 $\frac{16}{7}$ or $2\frac{2}{7}$; Students' explanations may vary.

c. $\frac{3}{4}$
 $\frac{3}{7}$; Students' explanations may vary.

3. Shawn's golf club is $3\frac{3}{4}$ ft tall. Shawn's hockey stick is $5\frac{1}{2}$ ft tall.

a. What fraction of the golf club's height is the hockey stick? Show your thinking.
 $1\frac{7}{15}$ times as tall; Sample response: $5\frac{1}{2} \div 3\frac{3}{4} = \frac{11}{2} \div \frac{15}{4} = \frac{11}{2} \cdot \frac{4}{15} = \frac{22}{15}$ or $1\frac{7}{15}$

b. How many times as tall is the hockey stick than the golf club? Show your thinking.
 $\frac{15}{22}$ of the height; Sample response: $3\frac{3}{4} \div 5\frac{1}{2} = \frac{15}{4} \div \frac{11}{2} = \frac{15}{4} \cdot \frac{2}{11} = \frac{15}{22}$

4. Jada is $5\frac{1}{5}$ ft tall. Her little sister is 4 ft tall. How many times taller is Jada than her sister?

A. $\frac{4}{5}$ B. $\frac{16}{21}$ C. $1\frac{1}{4}$ D. $1\frac{5}{16}$

Unit 4 Lesson 12

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Name: _____ Date: _____ Period: _____

Clare and Diego each went for a bike ride. Clare rode $1\frac{1}{2}$ mi, and Diego rode $2\frac{3}{4}$ mi. Use this information for Problems 5–6.

5. How many times as far as Clare did Diego ride?

a. Write a division expression to represent this situation.

$2\frac{3}{4} \div 1\frac{1}{2}$

b. Determine the solution. Show your thinking.

Diego rode $1\frac{5}{6}$ times as far as Clare. Sample response: $\frac{11}{4} \div \frac{3}{2} = \frac{11}{4} \cdot \frac{2}{3} = \frac{11}{6}$

6. What fraction of Diego's distance did Clare ride?

a. Write a division expression to represent this situation.

$1\frac{1}{2} \div 2\frac{3}{4}$

b. Determine the solution. Show your thinking.

Clare rode $\frac{6}{11}$ of Diego's distance. Sample response: $\frac{3}{2} \div \frac{11}{4} = \frac{3}{2} \cdot \frac{4}{11} = \frac{6}{11}$

7. Andre uses $1\frac{5}{6}$ lb of tomatoes for a batch of homemade spaghetti sauce. He has $9\frac{1}{2}$ lb of tomatoes from his garden. Does Andre have enough tomatoes to make 6 batches of spaghetti sauce? Explain your thinking.

No; Sample response: Andre does not have enough to make 6 batches of spaghetti sauce, but he has enough to make $5\frac{2}{11}$ batches because $9\frac{1}{2} \div 1\frac{5}{6} = 5\frac{2}{11}$, or he can make 5 whole batches.

8. Mai has worked $3\frac{3}{4}$ hours of her $7\frac{3}{4}$ hour shift. She says that she has worked more than half of her shift. Is Mai correct? Explain your thinking.

No; Sample response: Mai has not quite worked half of her shift. Mai worked $3\frac{3}{4} \div 7\frac{3}{4} = \frac{15}{31}$ of her shift.

Unit 4 Lesson 12

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	6.NS.A.1
2	2	6.NS.A.1
3	2	6.NS.A.1
4	2	6.NS.A.1
5	2	6.NS.A.1
6	2	6.NS.A.1
7	3	6.NS.A.1
8	3	6.NS.A.1

Notes:

Additional Practice

ACC6.3.10

1. A rectangular coffee table has a length of $46\frac{1}{2}$ inches, a width of w inches, and an area of 1,395 square inches. Select *all* the equations that represent the relationship between the dimensions of the coffee table.

☐ A. $w \cdot 1395 = 46\frac{1}{2}$

☐ B. $w \div 1395 = 46\frac{1}{2}$

☒ C. $46\frac{1}{2} \cdot w = 1395$

☐ D. $46\frac{1}{2} \cdot 1395 = w$

☒ E. $1395 \div w = 46\frac{1}{2}$

☒ F. $1395 \div 46\frac{1}{2} = w$

2. The triangle has a height of 8 feet and an area of 42 square feet. Select *all* the equations that represent the relationship between the dimensions of the triangle.

☒ A. $42 = \frac{1}{2} \cdot b \cdot 8$

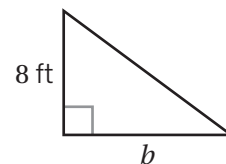
☒ B. $42 \div b = \frac{1}{2} \cdot 8$

☐ C. $42 \div b = 8$

☒ D. $4 \cdot b = 42$

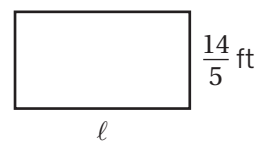
☐ E. $8 \cdot b = 42$

☐ F. $b \div 8 = 42$



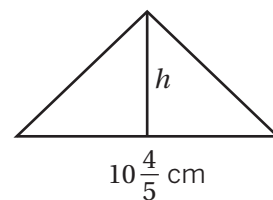
3. The area of the rectangle is 28 square feet. What is the length of the rectangle? Show your thinking.

10 feet. Work varies. $28 \div \frac{14}{5} = 28 \cdot \frac{5}{14} = \frac{140}{14} = 10$.



4. The area of the triangle is 27 square centimeters. What is the missing height h ? Show your thinking.

5 centimeters. Work varies. Since the area of a triangle is $\frac{1}{2}$ base times height, the product of the triangle's base and height is $27 \cdot 2 = 54$ square centimeters, and $54 \div 10\frac{4}{5} = 5$ centimeters.

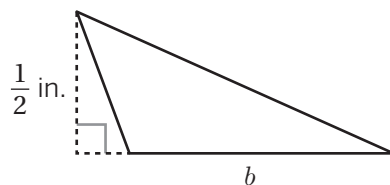


5. Han is tiling a shower wall that is 8 feet by 6 feet. The tiles are squares with a side length of $1\frac{1}{3}$ feet. How many tiles are needed to cover the entire shower wall? Show or explain your thinking.

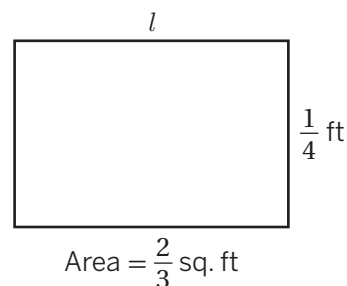
27 tiles. Explanations vary. Han needs 27 tiles because $(8 \div 1\frac{1}{3}) \cdot (6 \div 1\frac{1}{3}) = 27$.

6. The area of the triangle is $\frac{2}{7}$ square inches. What is the missing length b ? Show your thinking.

$1\frac{1}{7}$ inches. Work varies. Since the area of a triangle is $\frac{1}{2}bh$, the product of the base and height would be $\frac{2}{7} \cdot 2 = \frac{4}{7}$ square inches. Then, divide by the height to determine the base: $\frac{4}{7} \div \frac{1}{2} = \frac{8}{7}$ or $1\frac{1}{7}$ inches.

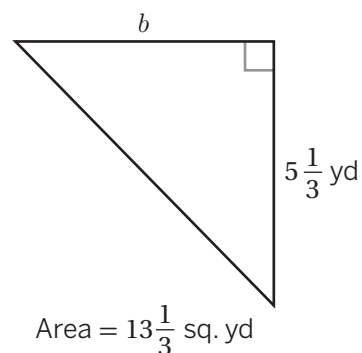


7. Lin wrote the equation $\frac{2}{3} = l \cdot \frac{1}{4}$ to determine the missing length of the rectangle. Use Lin's equation to solve for l .
 $\frac{8}{3}$ or $2\frac{2}{3}$ ft



8. Bard wrote the equation $\frac{1}{2} \cdot b \cdot 13\frac{1}{3} = 5\frac{1}{3}$ to determine the unknown base of the triangle. Is Bard's equation correct? If yes, solve the equation to determine the unknown base. If no, write and solve a new equation for b .

No. Work varies. The values for the area and the height have been switched. The equation Bard should use is $\frac{1}{2} \cdot b \cdot 5\frac{1}{3} = 13\frac{1}{3}$. The base is 5 units long.



Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.3.10

1. A rectangular coffee table has a length of $46\frac{1}{2}$ inches, a width of w inches, and an area of 1,395 square inches. Select *all* the equations that represent the relationship between the dimensions of the coffee table.

- ☐ A. $w \cdot 1395 = 46\frac{1}{2}$ ☐ B. $w \div 1395 = 46\frac{1}{2}$
☒ C. $46\frac{1}{2} \cdot w = 1395$ ☐ D. $46\frac{1}{2} \cdot 1395 = w$
☒ E. $1395 \div w = 46\frac{1}{2}$ ☒ F. $1395 \div 46\frac{1}{2} = w$

2. The triangle has a height of 8 feet and an area of 42 square feet. Select *all* the equations that represent the relationship between the dimensions of the triangle.

- ☒ A. $42 = \frac{1}{2} \cdot b \cdot 8$ ☒ B. $42 \div b = \frac{1}{2} \cdot 8$
☐ C. $42 \div b = 8$ ☒ D. $4 \cdot b = 42$
☐ E. $8 \cdot b = 42$ ☐ F. $b \div 8 = 42$



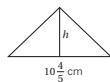
3. The area of the rectangle is 28 square feet. What is the length of the rectangle? Show your thinking.

10 feet. *Work varies.* $28 \div \frac{14}{5} = 28 \cdot \frac{5}{14} = \frac{140}{14} = 10$.



4. The area of the triangle is 27 square centimeters. What is the missing height h ? Show your thinking.

5 centimeters. *Work varies.* Since the area of a triangle is $\frac{1}{2}$ base times height, the product of the triangle's base and height is $27 \cdot 2 = 54$ square centimeters, and $54 \div 10\frac{4}{5} = 5$ centimeters.



Unit 3 Lesson 10

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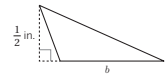
Name: _____ Date: _____ Period: _____

5. Han is tiling a shower wall that is 8 feet by 6 feet. The tiles are squares with a side length of $1\frac{1}{3}$ feet. How many tiles are needed to cover the entire shower wall? Show or explain your thinking.

27 tiles. *Explanations vary. Han needs 27 tiles because $(8 \div 1\frac{1}{3}) \cdot (6 \div 1\frac{1}{3}) = 27$.*

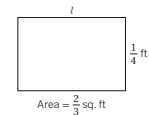
6. The area of the triangle is $\frac{2}{3}$ square inches.

What is the missing length h ? Show your thinking.
 $1\frac{1}{2}$ inches. *Work varies.* Since the area of a triangle is $\frac{1}{2}bh$, the product of the base and height would be $\frac{2}{3} \cdot 2 = \frac{4}{3}$ square inches. Then, divide by the height to determine the base: $\frac{4}{3} \div \frac{1}{2} = \frac{8}{3}$ or $2\frac{2}{3}$ inches.



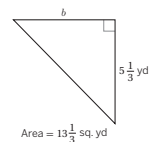
7. Lin wrote the equation $\frac{2}{3} = l \cdot \frac{1}{4}$ to determine the missing length of the rectangle. Use Lin's equation to solve for l .

$\frac{8}{3}$ or $2\frac{2}{3}$ ft



8. Bard wrote the equation $\frac{1}{2} \cdot b \cdot 13\frac{1}{3} = 5\frac{1}{3}$ to determine the unknown base of the triangle. Is Bard's equation correct? If yes, solve the equation to determine the unknown base. If no, write and solve a new equation for b .

No. *Work varies.* The values for the area and the height have been switched. The equation Bard should use is $\frac{1}{2} \cdot b \cdot 5\frac{1}{3} = 13\frac{1}{3}$. The base is 5 units long.



Unit 3 Lesson 10

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	6.NS.A.1, 6.G.A.1
2	1	6.NS.A.1, 6.G.A.1
3	2	6.NS.A.1, 6.G.A.1
4	2	6.NS.A.1, 6.G.A.1
5	2	6.NS.A.1, 6.G.A.1
6	2	6.NS.A.1, 6.G.A.1
7	3	6.NS.A.1, 6.G.A.1
8	3	6.NS.A.1, 6.G.A.1

Notes:

Additional Practice

ACC6.3.12

1. Select *all* the expressions that have a value greater than 1.

☐ A. $0.61 + 0.25$

☐ B. $0.8 + 0.03$

☒ C. $0.85 + 0.36$

☒ D. $0.53 + 0.55$

☐ E. $0.1 + 0.098$

2. This diagram represents a square backyard with a garden and a pond. The shaded area represents the pond, and the unshaded area represents the garden. Each small square represents 1 hundredth of the total backyard area.

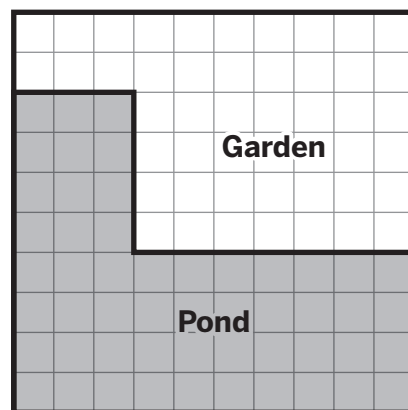
Which equation represents the difference between the size of the pond and the garden?

A. $1 - 0.52 = 0.48$

B. $0.48 + 0.52 = 1$

☒ C. $0.52 - 0.48 = 0.04$

D. $1 + 0.48 = 1.48$



Problems 3–4: Determine each sum. Use a hundredths chart if it helps with your thinking.

3. $0.26 + 0.725 = 0.985$

4. $38.70 + 0.212 = 38.912$

5. Determine the sum: $2.6 + 0.31$. Show or explain your thinking.

Work varies.

$$\begin{array}{r} 2.60 \\ + 0.31 \\ \hline 2.91 \end{array}$$

6. Determine the difference: $5.972 - 2.87$. Show or explain your thinking.

Work varies.

$$\begin{array}{r} 5.972 \\ - 2.870 \\ \hline 3.102 \end{array}$$

7. Mai wrote a vertical calculation for the expression $8.75 + 3.1$. Her work is shown.

$$\begin{array}{r} 8.75 \\ + 3.01 \\ \hline 11.76 \end{array}$$

Did Mai correctly determine the sum? Explain your thinking.

No. Explanations vary. Mai changed the second addend from 3.1 to 3.01 because the second addend did not have the same amount of digits after the decimal point. The correct sum is 11.85.

$$\begin{array}{r} 8.75 \\ + 3.10 \\ \hline 11.85 \end{array}$$

8. Priya wrote a vertical calculation for the expression $0.91 - 0.8$. Her work is shown.

$$\begin{array}{r} 0.91 \\ - 0.8 \\ \hline 0.83 \end{array}$$

Did Priya correctly determine the difference? Explain your thinking.

No. Explanations vary. Priya did not line up digits with the same place value. The correct difference is 0.11.

$$\begin{array}{r} 0.91 \\ - 0.80 \\ \hline 0.11 \end{array}$$

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.3.12

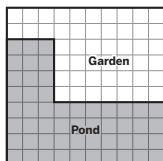
1. Select *all* the expressions that have a value greater than 1.

☐ A. $0.61 + 0.25$ ☐ B. $0.8 + 0.03$ ☒ C. $0.85 + 0.36$
☒ D. $0.53 + 0.55$ ☐ E. $0.1 + 0.098$

2. This diagram represents a square backyard with a garden and a pond. The shaded area represents the pond, and the unshaded area represents the garden. Each small square represents 1 hundredth of the total backyard area.

Which equation represents the difference between the size of the pond and the garden?

A. $1 - 0.52 = 0.48$
 B. $0.48 + 0.52 = 1$
☒ C. $0.52 - 0.48 = 0.04$
 D. $1 + 0.48 = 1.48$



Problems 3–4: Determine each sum. Use a hundredths chart if it helps with your thinking.

3. $0.26 + 0.725 = 0.985$
4. $38.70 + 0.212 = 38.912$

Unit 3 Lesson 12

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Name: _____ Date: _____ Period: _____

5. Determine the sum: $2.6 + 0.31$. Show or explain your thinking.

Work varies.

$$\begin{array}{r} 2.60 \\ + 0.31 \\ \hline 2.91 \end{array}$$

6. Determine the difference: $5.972 - 2.87$. Show or explain your thinking.

Work varies.

$$\begin{array}{r} 5.972 \\ - 2.870 \\ \hline 3.102 \end{array}$$

7. Mai wrote a vertical calculation for the expression $8.75 + 3.1$. Her work is shown.

$$\begin{array}{r} 8.75 \\ + 3.01 \\ \hline 11.76 \end{array}$$

Did Mai correctly determine the sum? Explain your thinking.

No. Explanations vary. Mai changed the second addend from 3.1 to 3.01 because the second addend did not have the same amount of digits after the decimal point. The correct sum is 11.85.

$$\begin{array}{r} 8.75 \\ + 3.10 \\ \hline 11.85 \end{array}$$

8. Priya wrote a vertical calculation for the expression $0.91 - 0.8$. Her work is shown.

$$\begin{array}{r} 0.91 \\ - 0.8 \\ \hline 0.83 \end{array}$$

Did Priya correctly determine the difference? Explain your thinking.

No. Explanations vary. Priya did not line up digits with the same place value. The correct difference is 0.11.

$$\begin{array}{r} 0.91 \\ - 0.80 \\ \hline 0.11 \end{array}$$

Unit 3 Lesson 12

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Additional Practice

Practice Problem Analysis

Problems	DOK	Standard(s)
1	1	6.NS.B.3
2	1	6.NS.B.3
3	2	6.NS.B.3
4	2	6.NS.B.3
5	2	6.NS.B.3
6	2	6.NS.B.3
7	3	6.NS.B.3
8	3	6.NS.B.3

Notes:

Additional Practice

ACC6.3.13

1. Determine which decimals have the same value as 0.81. Select *all* that apply.

☐ A. 0.80

☐ B. 0.801

☒ C. 0.810

☐ D. 0.8010

☒ E. 0.8100

2. Determine which expressions have the same value as $1.9 + 0.653$. Select *all* that apply.

☐ A. $1.9 + 6.5$

☒ B. $1.9 + 0.653$

☐ C. $1.009 + 0.653$

☒ D. $1.900 + 0.653$

☒ E. $1.9000 + 0.653$

3. Determine which expressions have the same value as $0.05 - 0.0036$. Select *all* that apply.

☒ A. $0.0500 - 0.0036$

☐ B. $0.0005 - 0.0036$

☐ C. $0.050 - 0.003$

☐ D. $0.05 - 0.36$

☒ E. $0.05000 - 0.00360$

4. Determine each sum. Use a hundredths chart if it helps with your thinking.

a $0.513 + 0.168 = 0.681$

b $0.951 + 0.04 = 0.991$

c $6.4 + 0.399 = 6.799$

5. Determine each difference. Use a hundredths chart if it helps with your thinking.

a $310.8 - 6.3 = 304.5$

b $14.39 - 4.1 = 10.29$

c $1.8 - 0.54 = 1.26$

Problems 6–7: The table shows the times, in seconds, of the top three teams in a snowboarding competition.

Rank	Medal	Run 1	Run 2
1	Gold	18.865	15.916
2	Silver	15.867	19.02
3	Bronze	16.56	18.81

- 6.** Each team had two runs. Determine each team's total time by adding their times for the two runs.

- a** Gold
34.781 seconds
- b** Silver
34.887 seconds
- c** Bronze
35.37 seconds

- 7.** Determine the difference in times for each team's two runs by subtracting their fastest time from their slowest time.

- a** Gold
2.949 seconds
- b** Silver
3.153 seconds
- c** Bronze
2.25 seconds

- 8.** Diego wants to subtract 1.89 from 3, but he is not sure how to proceed because 3 is a whole number and 1.89 is a decimal. Determine the difference, and explain how Diego could solve this problem.

1.11. Explanations vary. Diego can think of 3 as 300 hundredths and 1.89 as 189 hundredths, then subtract $300 - 189$. That equals 111 hundredths, or 1.11.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.3.13

- Determine which decimals have the same value as 0.81. Select *all* that apply.
 - ☐ A. 0.80
 - ☒ C. 0.810
 - ☒ E. 0.8100
 - ☐ B. 0.801
 - ☐ D. 0.8010
- Determine which expressions have the same value as $1.9 + 0.653$. Select *all* that apply.
 - ☐ A. $1.9 + 6.5$
 - ☒ B. $1.9 + 0.653$
 - ☐ C. $1.009 + 0.653$
 - ☒ D. $1.900 + 0.653$
 - ☒ E. $1.9000 + 0.653$
- Determine which expressions have the same value as $0.05 - 0.0036$. Select *all* that apply.
 - ☒ A. $0.0500 - 0.0036$
 - ☐ B. $0.0005 - 0.0036$
 - ☐ C. $0.050 - 0.003$
 - ☐ D. $0.05 - 0.36$
 - ☒ E. $0.05000 - 0.00360$
- Determine each sum. Use a hundredths chart if it helps with your thinking.
 - a $0.513 + 0.168 = 0.681$
 - b $0.951 + 0.04 = 0.991$
 - c $6.4 + 0.399 = 6.799$
- Determine each difference. Use a hundredths chart if it helps with your thinking.
 - a $310.8 - 6.3 = 304.5$
 - b $14.39 - 4.1 = 10.29$
 - c $1.8 - 0.54 = 1.26$

Unit 3 Lesson 13

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Name: _____ Date: _____ Period: _____

Problems 6–7: The table shows the times, in seconds, of the top three teams in a snowboarding competition.

Rank	Medal	Run 1	Run 2
1	Gold	18.865	15.916
2	Silver	15.867	19.02
3	Bronze	16.56	18.81

- Each team had two runs. Determine each team's total time by adding their times for the two runs.
 - a Gold
34.781 seconds
 - b Silver
34.887 seconds
 - c Bronze
35.37 seconds
- Determine the difference in times for each team's two runs by subtracting their fastest time from their slowest time.
 - a Gold
2.949 seconds
 - b Silver
3.153 seconds
 - c Bronze
2.25 seconds
- Diego wants to subtract 1.89 from 3, but he is not sure how to proceed because 3 is a whole number and 1.89 is a decimal. Determine the difference, and explain how Diego could solve this problem.

1.11. Explanations vary. Diego can think of 3 as 300 hundredths and 1.89 as 189 hundredths, then subtract $300 - 189$. That equals 111 hundredths, or 1.11.

Unit 3 Lesson 13

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Additional Practice

Practice Problem Analysis

Problems	DOK	Standard(s)
1	1	6.NS.B.3
2	1	6.NS.B.3
3	1	6.NS.B.3
4	2	6.NS.B.3
5	2	6.NS.B.3
6	2	6.NS.B.3
7	2	6.NS.B.3
8	3	6.NS.B.3

Notes:

Additional Practice

ACC6.3.15

1. Select *all* expressions that are equivalent to $(0.5) \cdot (0.8)$.

☐ A. $5 \cdot (0.1) \cdot 8 \cdot (0.01)$

☐ B. $5 \cdot (0.01) \cdot 8 \cdot (0.1)$

☒ C. $5 \cdot (0.1) \cdot 8 \cdot (0.1)$

☐ D. $5 \cdot (0.01) \cdot 8 \cdot (0.01)$

☒ E. $5 \cdot \frac{1}{10} \cdot 8 \cdot \frac{1}{10}$

☒ F. $5 \cdot 8 \cdot \frac{1}{10} \cdot \frac{1}{10}$

☐ G. $5 \cdot 8 \cdot \frac{1}{10}$

☒ H. $\frac{5}{10} \cdot \frac{8}{10}$

2. Determine each product.

a $15 \cdot (0.1)$

1.5

b $38 \cdot \frac{1}{10}$

3.8

c $3.1 \cdot 0.1$

0.31

d $1.84 \cdot \frac{1}{10}$

0.184

3. Determine each product.

a $151 \cdot \frac{1}{100}$

1.51

b $8.3 \cdot (0.01)$

0.083

c $1.95 \cdot \frac{1}{100}$

0.0195

d $9.436 \cdot (0.01)$

0.09436

4. Which expressions have the same value as $(0.316) \cdot (0.9)$? Select *all* that apply.

☐ A. 0.02844

☒ B. $316 \cdot 9 \cdot (0.0001)$

☒ C. $316 \cdot \frac{1}{1000} \cdot 9 \cdot \frac{1}{10}$

☐ D. $316 \cdot \frac{1}{1000} \cdot 9 \cdot \frac{1}{100}$

☒ E. $316 \cdot 9 \cdot \frac{1}{10000}$

5. Calculate the value of each expression. Show your thinking.

a $(5.7) \cdot 3$

17.1. Work varies.

$$\frac{57}{10} \cdot \frac{3}{1} = \frac{171}{10} = 17.1$$

b $(0.8) \cdot (6.4)$

5.12. Work varies.

$$\frac{8}{10} \cdot \frac{64}{10} = \frac{512}{100} = 5.12$$

c $(0.1) \cdot (0.23)$

0.023. Work varies.

$$\frac{1}{10} \cdot \frac{23}{100} = \frac{23}{1000} = 0.023$$

d $(0.48) \cdot (0.29)$

0.1392. Work varies.

$$\frac{48}{100} \cdot \frac{29}{100} = \frac{1392}{10000} = 0.1392$$

6. Calculate the value of each expression. Show your thinking.

a $(1.6) \cdot (0.006)$

0.0096. Work varies.

$$\frac{16}{10} \cdot \frac{6}{1,000} = \frac{96}{10,000} = 0.0096$$

b $(0.007) \cdot (0.038)$

0.000266. Work varies.

$$\frac{7}{1,000} \cdot \frac{38}{1,000} = \frac{266}{1,000,000} = 0.000266$$

7. Noah calculated the value of the expression $(0.18) \cdot (0.09)$ by first writing the decimal factors as fractions, and then writing their product as a decimal. Noah's work is shown: $\frac{18}{100} \cdot \frac{9}{10} = \frac{162}{1000} = 0.162$. Is Noah correct? Explain your thinking.

No. Explanations vary. Noah did not correctly write the second factor. 0.09 should be written as $\frac{9}{100}$, not $\frac{9}{10}$. The value of the expression $(0.18) \cdot (0.09)$ is 0.0162.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.3.15

1. Select *all* expressions that are equivalent to $(0.5) \cdot (0.8)$.

☐ A. $5 \cdot (0.1) \cdot 8 \cdot (0.01)$

☐ B. $5 \cdot (0.01) \cdot 8 \cdot (0.1)$

☒ C. $5 \cdot (0.1) \cdot 8 \cdot (0.1)$

☐ D. $5 \cdot (0.01) \cdot 8 \cdot (0.01)$

☒ E. $5 \cdot \frac{1}{10} \cdot 8 \cdot \frac{1}{10}$

☒ F. $5 \cdot 8 \cdot \frac{1}{10} \cdot \frac{1}{10}$

☐ G. $5 \cdot 8 \cdot \frac{1}{10}$

☒ H. $\frac{5}{10} \cdot \frac{8}{10}$

2. Determine each product.

a. $15 \cdot (0.1)$

1.5

b. $38 \cdot \frac{1}{10}$

3.8

c. $3.1 \cdot 0.1$

0.31

d. $1.84 \cdot \frac{1}{10}$

0.184

3. Determine each product.

a. $151 \cdot \frac{1}{100}$

1.51

b. $8.3 \cdot (0.01)$

0.083

c. $1.95 \cdot \frac{1}{100}$

0.0195

d. $9.436 \cdot (0.01)$

0.09436

Unit 3 Lesson 15

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Name: _____ Date: _____ Period: _____

4. Which expressions have the same value as $(0.316) \cdot (0.9)$? Select *all* that apply.

☐ A. 0.02844

☒ B. $316 \cdot 9 \cdot (0.0001)$

☒ C. $316 \cdot \frac{1}{1000} \cdot 9 \cdot \frac{1}{10}$

☐ D. $316 \cdot \frac{1}{1000} \cdot 9 \cdot \frac{1}{100}$

☒ E. $316 \cdot 9 \cdot \frac{1}{10000}$

5. Calculate the value of each expression. Show your thinking.

a. $(5.7) \cdot 3$

17.1. Work varies.
 $\frac{57}{10} \cdot \frac{3}{1} = \frac{171}{10} = 17.1$

b. $(0.8) \cdot (6.4)$

5.12. Work varies.
 $\frac{8}{10} \cdot \frac{64}{10} = \frac{512}{100} = 5.12$

c. $(0.1) \cdot (0.23)$

0.023. Work varies.
 $\frac{1}{10} \cdot \frac{23}{100} = \frac{23}{1000} = 0.023$

d. $(0.48) \cdot (0.29)$

0.1392. Work varies.
 $\frac{48}{100} \cdot \frac{29}{100} = \frac{1392}{10000} = 0.1392$

6. Calculate the value of each expression. Show your thinking.

a. $(1.6) \cdot (0.006)$

0.0096. Work varies.
 $\frac{16}{10} \cdot \frac{6}{1000} = \frac{96}{10000} = 0.0096$

b. $(0.007) \cdot (0.038)$

0.000266. Work varies.
 $\frac{7}{1000} \cdot \frac{38}{1000} = \frac{266}{1,000,000} = 0.000266$

7. Noah calculated the value of the expression $(0.18) \cdot (0.09)$ by first writing the decimal factors as fractions, and then writing their product as a decimal. Noah's work is shown: $\frac{18}{100} \cdot \frac{9}{10} = \frac{162}{1000} = 0.162$. Is Noah correct? Explain your thinking.

No. Explanations vary. Noah did not correctly write the second factor. 0.09 should be written as $\frac{9}{100}$, not $\frac{9}{10}$. The value of the expression $(0.18) \cdot (0.09)$ is 0.0162.

Unit 3 Lesson 15

96

Additional Practice

Practice Problem Analysis

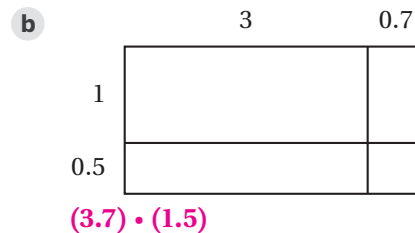
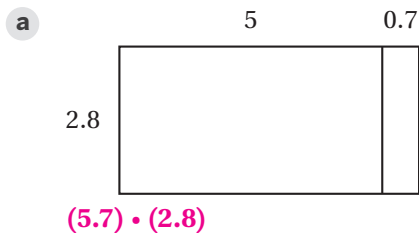
Problems	DOK	Standard(s)
1	1	6.NS.B.3
2	1	6.NS.B.3
3	2	6.NS.B.3
4	2	6.NS.B.3
5	2	6.NS.B.3
6	2	6.NS.B.3
7	3	6.NS.B.3

Notes:

Additional Practice

ACC6.3.16

1. Write a multiplication expression that represents each area diagram.



2. The rectangle has an area in square units and has been partitioned into four smaller rectangles. For each expression, write the name of the smaller rectangle whose area matches that expression. Then determine the area of each smaller rectangle.

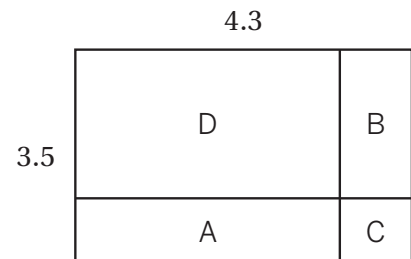
a $(0.3) \cdot 3$
B; 0.9 square units

b $3 \cdot 4$
D; 12 square units

c $(0.3) \cdot (0.5)$
C; 0.15 square units

d $4 \cdot (0.5)$
A; 2 square units

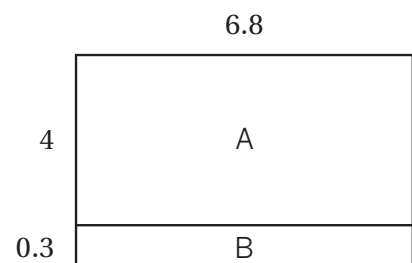
e Determine the total area of the rectangle.
15.05 square units



3. The area diagram represents $(6.8) \cdot (4.3)$.

a Determine the areas of Rectangles A and B.
Rectangle A: 27.2 square units
Rectangle B: 2.04 square units

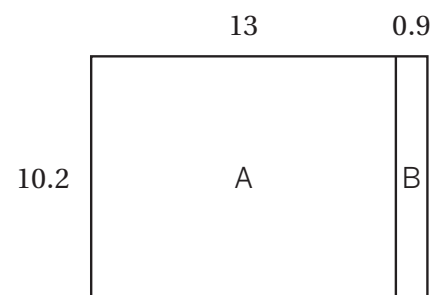
b What is the area of the larger 6.8 by 4.3 rectangle?
29.24 square units



4. The area diagram represents $(13.9) \cdot (10.2)$.

a Determine the areas of Rectangles A and B.
Rectangle A: 132.6 square units
Rectangle B: 9.18 square units

b What is the area of the larger 13.9 by 10.2 rectangle?
141.78 square units



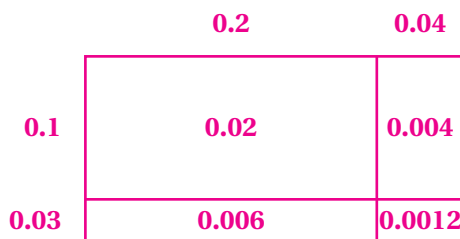
5. Consider the expression $(0.24) \cdot (0.13)$.

- a Draw an area diagram to represent this expression.

Drawings vary.

- b Determine the product of the expression. Use your area diagram if it helps with your thinking.

0.0312



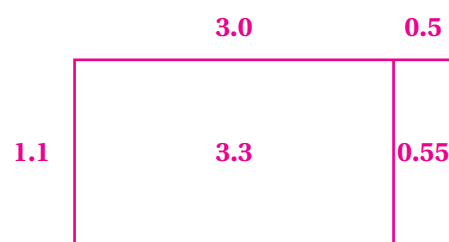
6. Consider the expression $(3.5) \cdot (1.1)$.

- a Draw an area diagram to represent this expression.

Drawings vary.

- b Determine the product of the expression. Use your area diagram if it helps with your thinking.

3.85



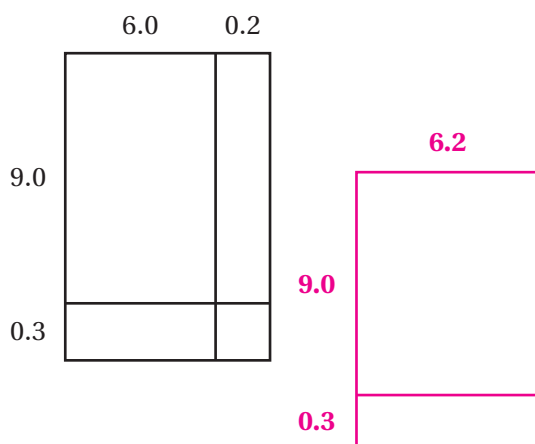
7. Tyler drew an area diagram to represent the expression $(6.2) \cdot (9.3)$.

- a Tyler's teacher says that there is another way he could use an area diagram to represent the same expression. Draw another area diagram that Tyler could use.

Drawings vary.

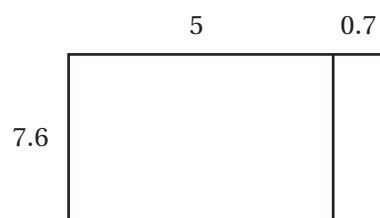
- b Determine the product of the expression. Use an area diagram if it helps with your thinking.

57.66



8. Bard drew this area diagram to represent the expression $(5.7) \cdot (7.6)$, and claimed to be able to use it to determine the area of the rectangle. Kiran said Bard's area diagram is not correct because it only shows two partitioned rectangles, when instead it should show four partitioned rectangles. Who is correct? Explain your thinking.

Both. Explanations vary. Both Bard and Kiran are correct. There are different area models that can be used to determine the area of the expression.

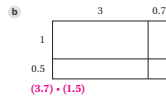
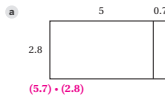


Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.3.16

1. Write a multiplication expression that represents each area diagram.



2. The rectangle has an area in square units and has been partitioned into four smaller rectangles. For each expression, write the name of the smaller rectangle whose area matches that expression. Then determine the area of each smaller rectangle.

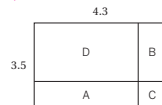
a $(0.3) \cdot 3$
B: 0.9 square units

b $3 \cdot 4$
D: 12 square units

c $(0.3) \cdot (0.5)$
C: 0.15 square units

d $4 \cdot (0.5)$
A: 2 square units

e Determine the total area of the rectangle.
15.05 square units



3. The area diagram represents $(6.8) \cdot (4.3)$.

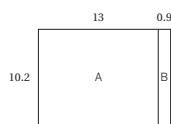
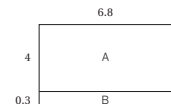
a Determine the areas of Rectangles A and B.
Rectangle A: 27.2 square units
Rectangle B: 2.04 square units

b What is the area of the larger 6.8 by 4.3 rectangle?
29.24 square units

4. The area diagram represents $(13.9) \cdot (10.2)$.

a Determine the areas of Rectangles A and B.
Rectangle A: 132.6 square units
Rectangle B: 9.18 square units

b What is the area of the larger 13.9 by 10.2 rectangle?
141.78 square units



Unit 3 Lesson 16

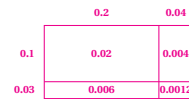
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5. Consider the expression $(0.24) \cdot (0.13)$.

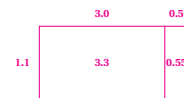
a Draw an area diagram to represent this expression.
Drawings vary.



b Determine the product of the expression. Use your area diagram if it helps with your thinking.
0.0312

6. Consider the expression $(3.5) \cdot (1.1)$.

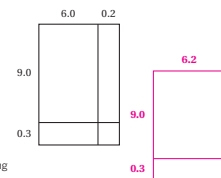
a Draw an area diagram to represent this expression.
Drawings vary.



b Determine the product of the expression. Use your area diagram if it helps with your thinking.
3.85

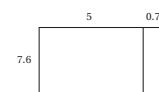
7. Tyler drew an area diagram to represent the expression $(6.2) \cdot (9.3)$.

a Tyler's teacher says that there is another way he could use an area diagram to represent the same expression. Draw another area diagram that Tyler could use.
Drawings vary.



b Determine the product of the expression. Use an area diagram if it helps with your thinking.
57.66

8. Bard drew this area diagram to represent the expression $(5.7) \cdot (7.6)$, and claimed to be able to use it to determine the area of the rectangle. Kiran said Bard's area diagram is not correct because it only shows two partitioned rectangles, when instead it should show four partitioned rectangles. Who is correct? Explain your thinking.



Both. Explanations vary. Both Bard and Kiran are correct. There are different area models that can be used to determine the area of the expression.

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Additional Practice

Practice Problem Analysis

Problems	DOK	Standard(s)
1	1	6.NS.B.3
2	2	6.NS.B.3
3	2	6.NS.B.3
4	2	6.NS.B.3
5	2	6.NS.B.3
6	2	6.NS.B.3
7	3	6.NS.B.3
8	3	6.NS.B.3

Notes:

Additional Practice

ACC6.3.17

1. Determine the value of each expression.

a $3 \cdot (0.5)$

1.5

b $(6.2) \cdot 3$

18.6

c $2 \cdot (16.3)$

32.6

d $(0.9) \cdot 5$

4.5

2. A pound of turkey lunch meat at the deli costs \$7.49 per pound. Lin wants to purchase 2.25 pounds of turkey for the week. Determine how much Lin will pay, rounded to the nearest cent. Show your thinking.

\$16.85. *Work varies.*

$$\begin{array}{r} 7.49 \\ \times 2.25 \\ \hline 3745 \\ 14980 \\ + 149800 \\ \hline 16.8525 \end{array}$$

Problems 3–6: Determine the value of each expression using any strategy. Show or explain your thinking.

3. $(3.4) \cdot (7.6)$ *Work varies.*

$$\begin{array}{r} 3.4 \\ \times 7.6 \\ \hline 204 \\ + 2380 \\ \hline 25.84 \end{array}$$

4. $(1.2) \cdot (9.9)$ *Work varies.*

$$\begin{aligned} 12 \cdot 99 \cdot \frac{1}{10} \cdot \frac{1}{10} &= \\ 1188 \cdot \frac{1}{100} &= 11.88 \end{aligned}$$

5. $(0.54) \cdot (0.8)$ *Work varies.*

$$54 \cdot 8 \cdot \frac{1}{100} \cdot \frac{1}{10} = 432 \cdot \frac{1}{1000} = 0.432$$

6. $(8.79) \cdot (6.04)$ *Work varies.*

$$\begin{array}{r} 8.79 \\ \times 6.04 \\ \hline 3516 \\ 0000 \\ + 527400 \\ \hline 53.0916 \end{array}$$

- 7.** A dog weighs 5.4 kilograms and a cat weighs 10.25 pounds. Which weighs more and by how much? Note: 1 kilogram = 2.2 pounds. Show your thinking.

The dog weighs more, by 1.63 lb or 0.74 kg. Explanations vary.

- 8.** A pound of strawberries costs \$4.29 and a pound of bananas costs \$0.69. What is the total cost of 1.5 pounds of strawberries and 0.8 pounds of bananas? Round your answer to the nearest cent. Show your thinking.

\$6.99. Explanations vary. 1.5 lb of strawberries costs 6.435 and 0.8 lb of bananas costs 0.552. $6.435 + 0.552 = 6.987$

- 9.** Three polygons with their dimensions are given.

- A rectangle with a base of 6.437 centimeters and a height of 5.5 centimeters.
- A square with side lengths of 5.99 centimeters.
- A rectangle that is 14.25 centimeters wide and 2.5 centimeters long.

Noah says that the square has the greatest area. Is Noah correct? Explain your thinking.

Yes. Explanations vary. The first rectangle has an area of $(6.437) \cdot (5.5) = 35.4035 \text{ cm}^2$, the square has an area of $(5.99) \cdot (5.99) = 35.8801 \text{ cm}^2$, and the second rectangle has an area of $(14.25) \cdot (2.5) = 35.625 \text{ cm}^2$. Therefore, the square has the greatest area.

- 10.** There are 2.75 grams of sugar in 1 ounce of applesauce. Shawn says this means that there are 165 grams of sugar in 6 ounces of applesauce. Is Shawn correct? Explain your thinking.

No. Explanations vary. Shawn placed the decimal point in the incorrect place after multiplying. There are $(2.75) \cdot 6 = 16.5 \text{ g}$ of sugar in 6 oz of applesauce.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.3.17

1. Determine the value of each expression.

a. $3 \cdot (0.5)$
1.5

b. $(6.2) \cdot 3$
18.6

c. $2 \cdot (16.3)$
32.6

d. $(0.9) \cdot 5$
4.5

2. A pound of turkey lunch meat at the deli costs \$7.49 per pound. Lin wants to purchase 2.25 pounds of turkey for the week. Determine how much Lin will pay, rounded to the nearest cent. Show your thinking.

\$16.85. *Work varies.*

$$\begin{array}{r} 7.49 \\ \times 2.25 \\ \hline 3745 \\ 14980 \\ + 149800 \\ \hline 16.8525 \end{array}$$

Problems 3–6: Determine the value of each expression using any strategy. Show or explain your thinking.

3. $(3.4) \cdot (7.6)$ *Work varies.*

$$\begin{array}{r} 3.4 \\ \times 7.6 \\ \hline 204 \\ + 2380 \\ \hline 25.84 \end{array}$$

4. $(1.2) \cdot (9.9)$ *Work varies.*

$$12 \cdot 99 \cdot \frac{1}{10} \cdot \frac{1}{10} = 11.88$$

5. $(0.54) \cdot (0.8)$ *Work varies.*

$$54 \cdot 8 \cdot \frac{1}{100} \cdot \frac{1}{10} = 432 \cdot \frac{1}{1000} = 0.432$$

6. $(8.79) \cdot (6.04)$ *Work varies.*

$$\begin{array}{r} 8.79 \\ \times 6.04 \\ \hline 3516 \\ 0000 \\ + 527400 \\ \hline 53.0916 \end{array}$$

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7. A dog weighs 5.4 kilograms and a cat weighs 10.25 pounds. Which weighs more and by how much? Note: 1 kilogram = 2.2 pounds. Show your thinking.
The dog weighs more, by 1.63 lb or 0.74 kg. Explanations vary.

8. A pound of strawberries costs \$4.29 and a pound of bananas costs \$0.69. What is the total cost of 1.5 pounds of strawberries and 0.8 pounds of bananas? Round your answer to the nearest cent. Show your thinking.

\$6.99. *Explanations vary. 1.5 lb of strawberries costs 6.435 and 0.8 lb of bananas costs 0.552. 6.435 + 0.552 = 6.987*

9. Three polygons with their dimensions are given.

- A rectangle with a base of 6.437 centimeters and a height of 5.5 centimeters.
- A square with side lengths of 5.99 centimeters.
- A rectangle that is 14.25 centimeters wide and 2.5 centimeters long.

Noah says that the square has the greatest area. Is Noah correct? Explain your thinking.

Yes. *Explanations vary. The first rectangle has an area of $(6.437) \cdot (5.5) = 35.4035 \text{ cm}^2$, the square has an area of $(5.99) \cdot (5.99) = 35.8801 \text{ cm}^2$, and the second rectangle has an area of $(14.25) \cdot (2.5) = 35.625 \text{ cm}^2$. Therefore, the square has the greatest area.*

10. There are 2.75 grams of sugar in 1 ounce of applesauce. Shawn says this means that there are 165 grams of sugar in 6 ounces of applesauce. Is Shawn correct? Explain your thinking.

No. *Explanations vary. Shawn placed the decimal point in the incorrect place after multiplying. There are $(2.75) \cdot 6 = 16.5 \text{ g}$ of sugar in 6 oz of applesauce.*

Unit 3 Lesson 17

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Additional Practice

Practice Problem Analysis

Problems	DOK	Standard(s)
1	1	6.NS.B.3
2	2	6.NS.B.3
3	2	6.NS.B.3
4	2	6.NS.B.3
5	2	6.NS.B.3
6	2	6.NS.B.3
7	2	6.NS.B.3
8	2	6.NS.B.3
9	3	6.NS.B.3
10	3	6.NS.B.3

Notes:

Additional Practice

ACC6.3.18

1. Match each division problem with its quotient.

Division Expression

Decimal

a. $1 \div 4$

..... **b** 0.8

b. $4 \div 5$

..... **c** 0.3

c. $3 \div 10$

..... **d** 0.5

d. $4 \div 8$

..... **a** 0.25

2. Use long division to show that each equation is true.

a $7 \div 10 = 0.7$

$$\begin{array}{r} 0.7 \\ 10 \overline{)7.0} \\ \underline{-70} \\ 0 \end{array}$$

b $9 \div 50 = 0.18$

$$\begin{array}{r} 0.18 \\ 50 \overline{)9.00} \\ \underline{-50} \\ 400 \\ \underline{-400} \\ 0 \end{array}$$

c $12 \div 25 = 0.48$

$$\begin{array}{r} 0.48 \\ 25 \overline{)12.00} \\ \underline{-100} \\ 200 \\ \underline{-200} \\ 0 \end{array}$$

3. Use long division to determine each quotient. Write your answer as a decimal.

a $26 \div 5$

5.2

b $75 \div 8$

9.375

c $79 \div 4$

19.75

4. Use long division to determine each quotient. Write your answer as a decimal.

a $324 \div 5$

64.8

b $509 \div 8$

63.625

c $951 \div 6$

158.5

5. Use long division to determine each quotient. Write your answer as a decimal.

a $240 \div 32$

7.5

b $1650 \div 16$

103.125

c $8415 \div 20$

420.75

6. Salinity is a measurement of how much salt is in water. It is measured in grams of salt per liter of water. Use division to complete the table to show the salinity of each ocean.

Water source	Salt (g)	Water (L)	Salinity (g/L)
Indian Ocean	1,235	38	32.5
Pacific Ocean	1,352	40	33.8
Arctic Ocean	658	20	32.9

7. Mai reasoned, " $\frac{22}{25}$ is equivalent to $\frac{44}{50}$ and to $\frac{88}{100}$, so the decimal of $\frac{22}{25}$ is 0.88."

a Use long division to show that Mai is correct.

$$\begin{array}{r} 0.88 \\ 25 \overline{)22.00} \\ \underline{-20\ 0} \\ 2\ 00 \\ \underline{-2\ 00} \\ 0 \end{array}$$

b Use long division to determine whether the decimal of $\frac{44}{50}$ is also 0.88.

$$\begin{array}{r} 0.88 \\ 50 \overline{)44.00} \\ \underline{-40\ 0} \\ 4\ 00 \\ \underline{-4\ 00} \\ 0 \end{array}$$

8. Noah calculated $1124 \div 20$. Priya said he made a mistake using long division. Is Priya correct? Explain your thinking.

Noah's work:

$$\begin{array}{r} 5.62 \\ 20 \overline{)1124.00} \\ \underline{-100} \\ 124 \\ \underline{-120} \\ 40 \\ \underline{-40} \\ 0 \end{array}$$

Yes. Explanations vary. Priya is correct, Noah made a mistake when dividing because he did not place the 5 over the tens place. The correct quotient is 56.2, not 5.62.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.3.18

1. Match each division problem with its quotient.

Division Expression

Decimal

- a. $1 \div 4$ **b** 0.8
b. $4 \div 5$ **c** 0.3
c. $3 \div 10$ **d** 0.5
d. $4 \div 8$ **a** 0.25

2. Use long division to show that each equation is true.

- a. $7 \div 10 = 0.7$ b. $9 \div 50 = 0.18$ c. $12 \div 25 = 0.48$
- $$\begin{array}{r} 0.7 \\ 10 \overline{) 7.0} \\ \underline{- 70} \\ 0 \end{array}$$
- $$\begin{array}{r} 0.18 \\ 50 \overline{) 9.00} \\ \underline{- 50} \\ 400 \\ \underline{- 400} \\ 0 \end{array}$$
- $$\begin{array}{r} 0.48 \\ 25 \overline{) 12.00} \\ \underline{- 100} \\ 200 \\ \underline{- 200} \\ 0 \end{array}$$

3. Use long division to determine each quotient. Write your answer as a decimal.

- a. $26 \div 5$
5.2
b. $75 \div 8$
9.375
c. $79 \div 4$
19.75

4. Use long division to determine each quotient. Write your answer as a decimal.

- a. $324 \div 5$
64.8
b. $509 \div 8$
63.625
c. $951 \div 6$
158.5

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Name: _____ Date: _____ Period: _____

5. Use long division to determine each quotient. Write your answer as a decimal.

- a. $240 \div 32$
7.5
b. $1650 \div 16$
103.125
c. $8415 \div 20$
420.75

6. Salinity is a measurement of how much salt is in water. It is measured in grams of salt per liter of water. Use division to complete the table to show the salinity of each ocean.

Water source	Salt (g)	Water (L)	Salinity (g/L)
Indian Ocean	1,235	38	32.5
Pacific Ocean	1,352	40	33.8
Arctic Ocean	658	20	32.9

7. Mai reasoned, " $\frac{22}{25}$ is equivalent to $\frac{44}{50}$ and to $\frac{88}{100}$, so the decimal of $\frac{22}{25}$ is 0.88."

- a. Use long division to show that Mai is correct.

$$\begin{array}{r} 0.88 \\ 25 \overline{) 22.00} \\ \underline{- 200} \\ 200 \\ \underline{- 200} \\ 0 \end{array}$$

- b. Use long division to determine whether the decimal of $\frac{22}{25}$ is also 0.88.

$$\begin{array}{r} 0.88 \\ 50 \overline{) 44.00} \\ \underline{- 400} \\ 400 \\ \underline{- 400} \\ 0 \end{array}$$

8. Noah calculated $1124 \div 20$. Priya said he made a mistake using long division. Is Priya correct? Explain your thinking.

Noah's work:

$$\begin{array}{r} 5.62 \\ 20 \overline{) 1124.00} \\ \underline{- 100} \\ 124 \\ \underline{- 120} \\ 40 \\ \underline{- 40} \\ 0 \end{array}$$

Yes. Explanations vary. Priya is correct. Noah made a mistake when dividing because he did not place the 5 over the tens place. The correct quotient is 56.2, not 5.62.

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Additional Practice

Practice Problem Analysis

Problems	DOK	Standard(s)
1	1	6.NS.B.2
2	2	6.NS.B.2
3	2	6.NS.B.2
4	2	6.NS.B.2
5	2	6.NS.B.2
6	2	6.NS.B.2
7	2	6.NS.B.2
8	3	6.NS.B.2

Notes:

Additional Practice

ACC6.3.19

1. Select *all* the expressions that are equivalent to $35 \div 10$.

- ☐ A. $0.35 \div 10$
- ☒ B. $3.5 \div 1$
- ☒ C. $350 \div 100$
- ☐ D. $350 \div 1,000$
- ☐ E. $3500 \div 100$

2. Select *all* the expressions that are equivalent to $2100 \div 30$.

- ☐ A. $210 \div 0.3$
- ☐ B. $21 \div 3$
- ☒ C. $2.1 \div 0.03$
- ☒ D. $21 \div 0.3$
- ☒ E. $210 \div 3$

3. Calculate $560.7 \div 5$. Show or explain your thinking.

112.14. Work varies.

4. Calculate $4295.9 \div 7$. Show or explain your thinking.

613.7. Work varies.

5. Use long division to determine each quotient.

a $15.4 \div 28$
0.55

b $6.86 \div 2$
3.43

c $529 \div 0.5$
1,058

6. Consider the expression $84 \div 1.5$.

- a Write a related expression with the same value that uses only whole numbers. Explain your thinking.

Responses vary. $840 \div 15$. Explanations vary. I can multiply both the dividend and divisor by 10 so that both are whole numbers.

- b Evaluate your expression.

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7. Tyler paid \$37.36 for four 1-pound containers of protein powder. How much is each container of protein powder? Show your thinking.

\$9.34. Work varies.

$$\begin{array}{r} 9.34 \\ 4 \overline{)37.36} \\ \underline{-36} \\ 13 \\ \underline{-12} \\ 16 \\ \underline{-16} \\ 0 \end{array}$$

8. Lin used long division to determine the quotient of $42 \div 1.4$. Her work is shown. Is Lin correct? Explain your thinking.

$$\begin{array}{r} 0.3 \\ 1.4 \overline{)42} \\ \underline{-42} \\ 0 \end{array}$$

No. Explanations vary. Lin divided by 14, not 1.4. Lin could use a related expression of $420 \div 14$ to determine the quotient, which is 30.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.3.19

1. Select *all* the expressions that are equivalent to $35 \div 10$.

- ☐ A. $0.35 \div 10$
☒ B. $3.5 \div 1$
☒ C. $350 \div 100$
☐ D. $350 \div 1,000$
☐ E. $3500 \div 100$

2. Select *all* the expressions that are equivalent to $2100 \div 30$.

- ☐ A. $210 \div 0.3$
☐ B. $21 \div 3$
☒ C. $2.1 \div 0.03$
☒ D. $21 \div 0.3$
☒ E. $210 \div 3$

3. Calculate $560.7 \div 5$. Show or explain your thinking.

112.14. Work varies.

4. Calculate $4295.9 \div 7$. Show or explain your thinking.

613.7. Work varies.

5. Use long division to determine each quotient.

- a. $15.4 \div 28$
 0.55
 b. $6.86 \div 2$
 3.43
 c. $529 \div 0.5$
 1,058

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Name: _____ Date: _____ Period: _____

6. Consider the expression $84 \div 1.5$.

- a. Write a related expression with the same value that uses only whole numbers. Explain your thinking.

Responses vary. $840 \div 15$. Explanations vary. I can multiply both the dividend and divisor by 10 so that both are whole numbers.

- b. Evaluate your expression.

56

7. Tyler paid \$37.36 for four 1-pound containers of protein powder. How much is each container of protein powder? Show your thinking.

\$9.34. Work varies.

$$\begin{array}{r} 9.34 \\ 4 \overline{)37.36} \\ \underline{-36} \\ 13 \\ \underline{-12} \\ 16 \\ \underline{-16} \\ 0 \end{array}$$

8. Lin used long division to determine the quotient of $42 \div 1.4$. Her work is shown. Is Lin correct? Explain your thinking.

No. Explanations vary: Lin divided by 14, not 1.4. Lin could use a related expression of $420 \div 14$ to determine the quotient, which is 30.

$$\begin{array}{r} 0.3 \\ 1.4 \overline{)42} \\ \underline{-42} \\ 0 \end{array}$$

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Additional Practice

Practice Problem Analysis

Problems	DOK	Standard(s)
1	1	6.NS.B.3
2	1	6.NS.B.3
3	2	6.NS.B.3
4	2	6.NS.B.3
5	2	6.NS.B.3
6	2	6.NS.B.3
7	2	6.NS.B.3
8	3	6.NS.B.3

Notes:

Additional Practice

ACC6.3.20

1. Select *all* the expressions that are equivalent to $54.4 \div 1.7$.

☒ A. $544 \div 17$

☐ B. $540 \div 17$

☒ C. $5440 \div 170$

☐ D. $544 \div 0.17$

☒ E. $5.44 \div 0.17$

2. Consider the expression $0.0063 \div 0.009$.

- a Write two different division expressions that have the same quotient as $0.0063 \div 0.009$.

Responses vary. $6.3 \div 9$, $63 \div 90$ (or equivalent)

- b Evaluate $0.0063 \div 0.009$. Show or explain your thinking.

0.7. Explanations vary. Since the expression $0.0063 \div 0.009$ is equivalent to $6.3 \div 9$, I can evaluate that expression to get 0.7.

3. Bard said, "To determine the value of $5.064 \div 8$, I can divide 5,064 by 80."

- a Do you agree with Bard? Explain your thinking.

No. Explanations vary. I do not agree with Bard because Bard did not multiply both the dividend and divisor by the same power of 10.

- b Calculate the quotient of $5.064 \div 8$. Show your thinking.

0.633. Work varies.

$$\begin{array}{r} 0.633 \\ 8 \overline{)5.064} \\ \underline{-48} \\ 26 \\ \underline{-24} \\ 24 \\ \underline{-24} \\ 0 \end{array}$$

4. Clare said, "To determine the value of $730.2 \div 0.6$, I can divide 7,302 by 6.

- a Do you agree with Clare? Explain your thinking.

Yes. Explanations vary. I agree with Clare because both the dividend and divisor were multiplied by the same power of 10.

- b Calculate the quotient of $730.2 \div 0.6$.

1,217

5. A bag of quarters weighs 8.5 kilograms. Each quarter weighs 5.67 grams. Which is the best estimate for the number of quarters in the bag? Show or explain your thinking. (1 kilogram = 1000 grams)

A. 15,000

B. 1,500

C. 150

D. 15

Explanations vary. 8.5 kilograms = 8500 grams. $8500 \div 5.67$ is close to $8500 \div 5$, which is 1700 and closest to 1500.

6. Different brands of water bottles containing 16.9 ounces are sold in different quantities for different prices at a local grocery store. Complete the table to determine the unit cost for each brand.

Brand	Price (\$)	Quantity	Unit cost (\$)
Purely Water	4.40	40	0.11
Spring Water	4.80	32	0.15
Mountain Water	3.12	12	0.26

- 7.** Evaluate each expression.

a $11.9 \div 1.7$

7

b $0.036 \div 0.02$

1.8

- 8.** Diego says, “To determine the value of $180 \div 1.2$, I can divide 18,000 by 120.” Mai says, “To determine the value of $180 \div 1.2$, I can divide 1,800 by 12.”

a Who is correct? Explain your thinking.

Both. Explanations vary. Diego multiplied both the dividend and divisor by 100, and Mai multiplied both the divisor and dividend by 10.

b Calculate the quotient of $180 \div 1.2$. Show or explain your thinking.

150. *Explanations vary.* Since the expression $180 \div 1.2$ is equivalent to $1800 \div 12$, I can evaluate that expression to get 150.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.3.20

1. Select *all* the expressions that are equivalent to $54.4 \div 1.7$.

☒ A. $544 \div 17$ ☐ B. $540 \div 17$
☒ C. $5440 \div 170$ ☐ D. $544 \div 0.17$
☒ E. $5.44 \div 0.17$

2. Consider the expression $0.0063 \div 0.009$.

a Write two different division expressions that have the same quotient as $0.0063 \div 0.009$.
Responses vary. $6.3 \div 9$, $63 \div 90$ (or equivalent)

b Evaluate $0.0063 \div 0.009$. Show or explain your thinking.
0.7. Explanations vary. Since the expression $0.0063 \div 0.009$ is equivalent to $6.3 \div 9$, I can evaluate that expression to get 0.7.

3. Bard said, "To determine the value of $5.064 \div 8$, I can divide 5,064 by 80."

a Do you agree with Bard? Explain your thinking.
No. Explanations vary. I do not agree with Bard because Bard did not multiply both the dividend and divisor by the same power of 10.

b Calculate the quotient of $5.064 \div 8$. Show your thinking.
0.633. Work varies.

```

0.633
8 | 5.064
  - 48
  ---
    26
   - 24
   ---
     24
    - 24
    ---
      0
  
```

4. Clare said, "To determine the value of $730.2 \div 0.6$, I can divide 7,302 by 6.

a Do you agree with Clare? Explain your thinking.
Yes. Explanations vary. I agree with Clare because both the dividend and divisor were multiplied by the same power of 10.

b Calculate the quotient of $730.2 \div 0.6$.
1,217

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5. A bag of quarters weighs 8.5 kilograms. Each quarter weighs 5.67 grams. Which is the best estimate for the number of quarters in the bag? Show or explain your thinking. (1 kilogram = 1000 grams)

A. 15,000 ☐ B. 1,500
C. 150 ☒ D. 15

Explanations vary. $8.5 \text{ kilograms} = 8500 \text{ grams}$. $8500 \div 5.67$ is close to $8500 \div 5$, which is 1700 and closest to 1500.

6. Different brands of water bottles containing 16.9 ounces are sold in different quantities for different prices at a local grocery store. Complete the table to determine the unit cost for each brand.

Brand	Price (\$)	Quantity	Unit cost (\$)
Purely Water	4.40	40	0.11
Spring Water	4.80	32	0.15
Mountain Water	3.12	12	0.26

7. Evaluate each expression.

a $11.9 \div 1.7$
7

b $0.036 \div 0.02$
1.8

8. Diego says, "To determine the value of $180 \div 1.2$, I can divide 18,000 by 120." Mai says, "To determine the value of $180 \div 1.2$, I can divide 1,800 by 12."

a Who is correct? Explain your thinking.
Both. Explanations vary. Diego multiplied both the dividend and divisor by 100, and Mai multiplied both the divisor and dividend by 10.

b Calculate the quotient of $180 \div 1.2$. Show or explain your thinking.
150. Explanations vary. Since the expression $180 \div 1.2$ is equivalent to $1800 \div 12$, I can evaluate that expression to get 150.

Unit 3 Lesson 20

106

Additional Practice

Practice Problem Analysis

Problems	DOK	Standard(s)
1	1	6.NS.B.3
2	2	6.NS.B.2, 6.NS.B.3
3	2	6.NS.B.2, 6.NS.B.3
4	2	6.NS.B.2, 6.NS.B.3
5	2	6.NS.B.2, 6.NS.B.3
6	2	6.NS.B.2, 6.NS.B.3
7	2	6.NS.B.2, 6.NS.B.3
8	3	6.NS.B.2, 6.NS.B.3

Notes:

Additional Practice

5.16

1. Circle the expression that has a greater value.

9% of 180

1.8% of 90

They have the same value.

Problems 2–5. Alicia sells art prints on her online shop. Every week, she sells \$550 of art prints. She tries to spend no more than 6% of her weekly art print sales on art supplies.

2. Write an expression to represent how much money Alicia spends on art supplies every week.

550×0.06

3. How much money, at most, does Alicia spend on art supplies each week?

\$33

4. Alicia puts 11% of her weekly art print sales into a savings account for an upcoming art fair. How much money does Alicia save each week?

Alicia saves \$60.50 each week.

5. Alicia recently had to spend \$99.00 to renew her website hosting platform. What percent of her weekly art print sales did she spend on website maintenance?

A. 0.18%

B. 9%

C. 18%

D. 90%

Problems 6–7. Capri is remodeling their kitchen. They went to the hardware store and purchased these items.

Items	Cost (\$)
Hammer	13.57
Wood	27.25
Nails	5.25
Screwdriver	9.50
Paint	12.66
Paintbrush	4.80
Floor tiles	35.97
Total	109.00

6. Floor tiles are the most expensive item. What percent of the total cost are the floor tiles?

- A. 30%
- B. 31%
- C. 32%
- D. 33%**

7. What percentage of the total hardware store bill is wood? Show or explain your thinking.

25%

Explanations vary. Capri spent \$27.25 on wood. Since the total bill was \$109, Capri spent $\frac{27.25}{109} = 0.25 = 25\%$ on wood.

Name: _____ Date: _____ Period: _____

Additional Practice

5.16

1. Circle the expression that has a greater value.

9% of 180 1.8% of 90 They have the same value.

Problems 2–5. Alicia sells art prints on her online shop. Every week, she sells \$550 of art prints. She tries to spend no more than 6% of her weekly art print sales on art supplies.

2. Write an expression to represent how much money Alicia spends on art supplies every week.

550×0.06

3. How much money, at most, does Alicia spend on art supplies each week?

\$33

4. Alicia puts 11% of her weekly art print sales into a savings account for an upcoming art fair. How much money does Alicia save each week?

Alicia saves \$60.50 each week.

5. Alicia recently had to spend \$99.00 to renew her website hosting platform. What percent of her weekly art print sales did she spend on website maintenance?

- A. 0.18%
B. 9%
C. 18%
D. 90%

Unit 5 Lesson 16

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Name: _____ Date: _____ Period: _____

Problems 6–7. Capri is remodeling their kitchen. They went to the hardware store and purchased these items.

Items	Cost (\$)
Hammer	13.57
Wood	27.25
Nails	5.25
Screwdriver	9.50
Paint	12.66
Paintbrush	4.80
Floor tiles	35.97
Total	109.00

6. Floor tiles are the most expensive item. What percent of the total cost are the floor tiles?

- A. 30%
B. 31%
C. 32%
D. 33%

7. What percentage of the total hardware store bill is wood? Show or explain your thinking.

25%

Explanations vary. Capri spent \$27.25 on wood. Since the total bill was \$109, Capri spent $\frac{27.25}{109} = 0.25 = 25\%$ on wood.

Unit 5 Lesson 16

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Additional Practice

Practice Problem Analysis

Problems	DOK	Standard(s)
1	1	6.RP.A.3
2	2	6.RP.A.3
3	1	6.RP.A.3
4	1	6.RP.A.3
5	1	6.RP.A.3
6	1	6.RP.A.3
7	2	6.RP.A.3

Notes:

Additional Practice

ACC6.4.10

1. Rewrite each product expression as a sum expression.

a $3(x + 4)$

$3x + 12$

b $(5 + 7) \cdot x$

$5x + 7x$

2. Rewrite each sum expression as a product expression.

a $6x + 9$

$3(2x + 3)$

b $8x + 12y + 4z$

$4(2x + 3y + z)$ or $2(4x + 6y + 2z)$

3. Select *all* the expressions that are equivalent to the expression $12x + 24$.

☒ **A.** $2(6x + 12)$

☐ **B.** $4(4x + 6)$

☒ **C.** $6(2x + 4)$

☒ **D.** $12(x + 2)$

☐ **E.** $24(2x + 1)$

4. Select *all* the expressions that are equivalent to the expression $14a - 28b + 42c$.

☒ **A.** $14(a - 2b + 3c)$

☒ **B.** $7(2a - 4b + 6c)$

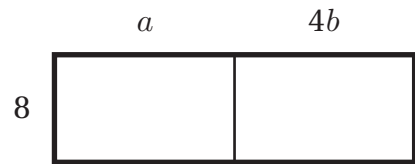
☐ **C.** $2(14a - 14b + 21c)$

☐ **D.** $14(a - 2b + 6c)$

☐ **E.** $7(2a - 4b + 7c)$

☒ **F.** $2(7a - 14b + 21c)$

5. Priya writes the area of this rectangle as the expression $8a + 32b$. Tyler writes the area of this rectangle as the expression $4(2a + 8b)$.



- a Who is correct? Explain your thinking.

Both. Explanations vary. The expressions are equivalent, so both Priya and Tyler correctly wrote the area of the partitioned rectangle.

- b Write two more equivalent expressions that represent the area of this rectangle.

Responses vary. $8(a + 4b)$, $2(4a + 16b)$

6. Select *all* the expressions that are equivalent to the expression $3(8x + 6)$.

- ☐ A. $2(8x + 9)$ ☒ B. $6(4x + 3)$
☐ C. $6 + 24x$ ☐ D. $11x + 18$
☒ E. $24x + 18$

7. Clare rewrites the expression $12bc + 8bd$ as $b(12c + 8d)$.
 Andre rewrites the expression $12bc + 8bd$ as $4(3bc + 2bd)$.

- a Are Clare's and Andre's expressions both equivalent to the expression $12bc + 8bd$?
 Explain your thinking.

Yes. Explanations vary. Clare divided $12bc$ and $8bd$ by b , which is a common factor. Andre divided $12bc$ and $8bd$ by 4 , which is a common factor. Because both are common factors, both expressions are equivalent to $12bc + 8bd$.

- b Write an additional equivalent expression to the expression $12bc + 8bd$.

Responses vary. $4b(3c + 2d)$

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.4.10

1. Rewrite each product expression as a sum expression.

a $3(x + 4)$
 $3x + 12$

b $(5 + 7) \cdot x$
 $5x + 7x$

2. Rewrite each sum expression as a product expression.

a $6x + 9$
 $3(2x + 3)$

b $8x + 12y + 4z$
 $4(2x + 3y + z)$ or $2(4x + 6y + 2z)$

3. Select *all* the expressions that are equivalent to the expression $12x + 24$.

- ☒ A. $2(6x + 12)$ ☐ B. $4(4x + 6)$
☒ C. $6(2x + 4)$ ☒ D. $12(x + 2)$
☐ E. $24(2x + 1)$

4. Select *all* the expressions that are equivalent to the expression $14a - 28b + 42c$.

- ☒ A. $14(a - 2b + 3c)$ ☒ B. $7(2a - 4b + 6c)$
☐ C. $2(14a - 14b + 21c)$ ☐ D. $14(a - 2b + 6c)$
☐ E. $7(2a - 4b + 7c)$ ☒ F. $2(7a - 14b + 21c)$

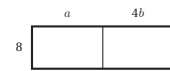
Unit 4 Lesson 10

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Name: _____ Date: _____ Period: _____

5. Priya writes the area of this rectangle as the expression $8a + 32b$. Tyler writes the area of this rectangle as the expression $4(2a + 8b)$.



- a Who is correct? Explain your thinking.

Both. Explanations vary. The expressions are equivalent, so both Priya and Tyler correctly wrote the area of the partitioned rectangle.

- b Write two more equivalent expressions that represent the area of this rectangle.

Responses vary. $8(a + 4b)$, $2(4a + 16b)$

6. Select *all* the expressions that are equivalent to the expression $3(8x + 6)$.

- ☐ A. $2(8x + 9)$ ☒ B. $6(4x + 3)$
☐ C. $6 + 24x$ ☐ D. $11x + 18$
☒ E. $24x + 18$

7. Clare rewrites the expression $12bc + 8bd$ as $b(12c + 8d)$.

Andre rewrites the expression $12bc + 8bd$ as $4(3bc + 2bd)$.

- a Are Clare's and Andre's expressions both equivalent to the expression $12bc + 8bd$? Explain your thinking.

Yes. Explanations vary. Clare divided $12bc$ and $8bd$ by b , which is a common factor. Andre divided $12bc$ and $8bd$ by 4 , which is a common factor. Because both are common factors, both expressions are equivalent to $12bc + 8bd$.

- b Write an additional equivalent expression to the expression $12bc + 8bd$.

Responses vary. $4b(3c + 2d)$

Unit 4 Lesson 10

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	6.EE.A.3, 6.EE.A.4
2	1	6.EE.A.3, 6.EE.A.4
3	2	6.EE.A.3, 6.EE.A.4
4	2	6.EE.A.3, 6.EE.A.4
5	3	6.EE.A.3, 6.EE.A.4
6	2	6.EE.A.3, 6.EE.A.4
7	3	6.EE.A.3, 6.EE.A.4

Notes:

Additional Practice

ACC6.4.11

1. Rewrite each expression using exponents.

a $9 \cdot 9 \cdot 9$

9^3

b $\frac{1}{2} \cdot \frac{1}{2}$

$\left(\frac{1}{2}\right)^2$

c $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$

2^7

2. Determine the value of each expression.

a 4^2

16

b 2^8

256

c $\left(\frac{1}{3}\right)^2$

$\frac{1}{9}$

3. Which value is equivalent to 8^2 ?

A. 6

B. 16

C. 32

D. 64

4. Using what you know about the properties of exponents, rewrite each expression using multiplication.

a 3^3

$3 \cdot 3 \cdot 3$

b 5^4

$5 \cdot 5 \cdot 5 \cdot 5$

c 7^5

$7 \cdot 7 \cdot 7 \cdot 7 \cdot 7$

5. Select *all* the expressions that are equivalent to the expression 5^4 .

☐ A. 20^2

☐ B. $5 + 5 + 5 + 5$

☐ C. 4^5

☒ D. 625

☒ E. $5 \cdot 5^3$

☒ F. 25^2

6. The expression 3^5 is equal to 243. Use this to determine the value of each expression.

a 3^6

729

b $3^3 \cdot 3^2$

243

c 3^4

81

7. The expression 6^4 is equal to 1,296. Use this to determine the value of each expression.

a 6^3

216

b $\frac{6^4}{6^2}$

36

c $6^2 \cdot 6^2$

1,296

8. The number of blades of grass in a field is about 8.4 million. Is this number closer to 10^6 or 10^7 ? Explain your thinking.

Closer to 10^7 . Explanations vary. The number of blades of grass is closer to 10 million (10^7) than 1 million (10^6).

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.4.11

1. Rewrite each expression using exponents.

a. $9 \cdot 9 \cdot 9$
9³

b. $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2}$
 $\left(\frac{1}{2}\right)^3$

c. $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$
2⁷

2. Determine the value of each expression.

a. 4^2
16

b. 2^8
256

c. $\left(\frac{1}{3}\right)^3$
 $\frac{1}{27}$

3. Which value is equivalent to 8²?

- A. 6
B. 16
C. 32
D. 64

4. Using what you know about the properties of exponents, rewrite each expression using multiplication.

a. 3^2
3 • 3

b. 5^4
5 • 5 • 5 • 5

c. 7^5
7 • 7 • 7 • 7 • 7

Unit 4 Lesson 11

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Name: _____ Date: _____ Period: _____

5. Select *all* the expressions that are equivalent to the expression 5⁴.

- ☐ A. 20² ☐ B. 5 + 5 + 5 + 5
☐ C. 4⁵ ☒ D. 625
☒ E. 5 • 5³ ☒ F. 25²

6. The expression 3⁵ is equal to 243. Use this to determine the value of each expression.

a. 3⁶
729

b. 3³ • 3²
243

c. 3⁴
81

7. The expression 6⁴ is equal to 1,296. Use this to determine the value of each expression.

a. 6³
216

b. $\frac{6^4}{6^2}$
36

c. 6⁴ • 6²
1,296

8. The number of blades of grass in a field is about 8.4 million. Is this number closer to 10⁶ or 10⁷? Explain your thinking.

Closer to 10⁷. Explanations vary. The number of blades of grass is closer to 10 million (10⁷) than 1 million (10⁶).

Unit 4 Lesson 11

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	6.EE.A.1
2	1	6.EE.A.1
3	2	6.EE.A.1
4	1	6.EE.A.1
5	2	6.EE.A.1
6	2	6.EE.A.1
7	2	6.EE.A.1
8	3	6.EE.A.1

Notes:

Additional Practice

ACC6.4.12

1. Evaluate each expression.

a $12 + 3^2$
21

b $6^2 \cdot 5$
180

c $35 - 4^2$
19

d $100 - 7^2$
51

e $5^2 \div 5$
5

f $15 \cdot \left(\frac{1}{3}\right)^2$
 $\frac{15}{9}$ (or equivalent)

2. Evaluate each expression.

a $10^3 - 8^2$
936

b $16 - 10^1$
6

c $20 + 2^4$
36

d $3^3 \cdot 5$
135

e $4 \cdot \left(\frac{1}{2}\right)^2$
1

f $\left(\frac{1}{6} \cdot 6\right)^6$
1

3. Determine whether the two expressions in each row are equivalent.
Write *equivalent* or *not equivalent* in the last column of the table.

Column A	Column B	Equivalent or not equivalent?
$8^2 + 10$	$10 + 2^6$	Equivalent
$4^2 + 2^2$	$45 - 5^2$	Equivalent
$(3 \cdot 4)^2$	$3^2 + 4^2$	Not equivalent
$9^3 + 9^2$	$3^4 \cdot 10$	Equivalent
$25^2 + 400$	$10^3 - 25$	Not equivalent
$6 \cdot 4^1$	$12 \cdot 2^2$	Not equivalent

4. Kiran's expression is $9 \cdot 4^3$. Select the expression that has the same value as Kiran's expression.

A. $(9 \cdot 4)^3$

B. $9 \cdot 3^4$

C. $9^3 \cdot 4$

☒ D. 576

5. Tyler's expression is $(6 + 7)^2$. Select the expression that has the same value as Tyler's expression.

A. $6^2 + 7^2$

B. $6 + 7^2$

☒ C. 13^2

D. 85

6. Shawn's expression is $3^5 + 50$. Select the expression that has the same value as Shawn's expression.

A. $3^5 \cdot 50$

☒ B. 293

C. 243

D. $50 + 3 \cdot 5$

7. Lin says the equation $(9 + 3)^2 = 9^2 + 3^2$ is false because the expressions on either side of the equal sign have different values. Is Lin correct? Explain your thinking.

Yes. Explanations vary. I think Lin is correct because $(9 + 3)^2 = 12^2 = 144$, and $9^2 + 3^2 = 81 + 9 = 90$.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.4.12

1. Evaluate each expression.

a $12 + 3^2$
21

b $6^2 \cdot 5$
180

c $35 - 4^2$
19

d $100 - 7^2$
51

e $5^2 \div 5$
5

f $15 \cdot \left(\frac{1}{3}\right)^2$
 $\frac{15}{9}$ (or equivalent)

2. Evaluate each expression.

a $10^2 - 8^2$
936

b $16 - 10^1$
6

c $20 + 2^4$
36

d $3^4 \cdot 5$
135

e $4 \cdot \left(\frac{1}{2}\right)^2$
1

f $\left(\frac{1}{6} \cdot 6\right)^5$
1

3. Determine whether the two expressions in each row are equivalent.
Write *equivalent* or *not equivalent* in the last column of the table.

Column A	Column B	Equivalent or not equivalent?
$8^2 + 10$	$10 + 2^6$	Equivalent
$4^2 + 2^2$	$45 - 5^2$	Equivalent
$(3 \cdot 4)^2$	$3^2 + 4^2$	Not equivalent
$9^2 + 9^2$	$3^4 \cdot 10$	Equivalent
$25^2 + 400$	$10^3 - 25$	Not equivalent
$6 \cdot 4^1$	$12 \cdot 2^2$	Not equivalent

Unit 4 Lesson 12

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Name: _____ Date: _____ Period: _____

4. Kiran's expression is $9 \cdot 4^3$. Select the expression that has the same value as Kiran's expression.

A. $(9 \cdot 4)^2$

B. $9 \cdot 3^4$

C. $9^3 \cdot 4$

D. 576

5. Tyler's expression is $(6 + 7)^2$. Select the expression that has the same value as Tyler's expression.

A. $6^2 + 7^2$

B. $6 + 7^2$

C. 13^2

D. 85

6. Shawn's expression is $3^3 + 50$. Select the expression that has the same value as Shawn's expression.

A. $3^3 \cdot 50$

B. 293

C. 243

D. $50 + 3 \cdot 5$

7. Lin says the equation $(9 + 3)^2 = 9^2 + 3^2$ is false because the expressions on either side of the equal sign have different values. Is Lin correct? Explain your thinking.

Yes. Explanations vary. I think Lin is correct because $(9 + 3)^2 = 12^2 = 144$, and $9^2 + 3^2 = 81 + 9 = 90$.

Unit 4 Lesson 12

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	6.EE.A.1, 6.EE.A.2.C
2	1	6.EE.A.1, 6.EE.A.2.C
3	2	6.EE.A.2.C, 6.EE.A.4
4	2	6.EE.A.2.C
5	2	6.EE.A.2.C
6	2	6.EE.A.2.C
7	3	6.EE.A.2.C

Notes:

Additional Practice

ACC6.4.13

1. Determine the value of each expression when $x = 4$.

Expression	$2^x + 6$	x^3	$x^2 - 8$	$(x + 2)^2$
Value When $x = 4$	22	64	8	36

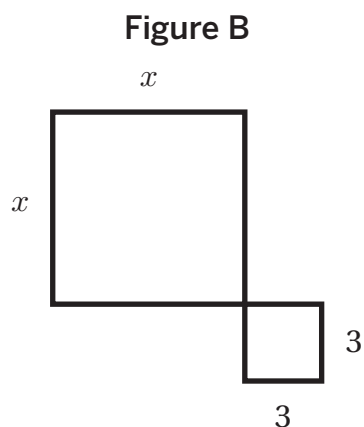
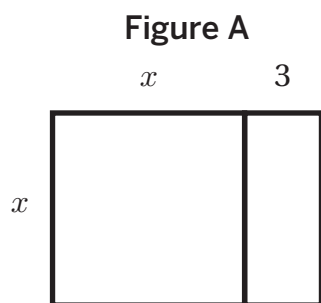
2. Determine the value of each expression when $x = 3$.

Expression	$x^3 + 6x$	$5x^2$	$(x - 1)^5$	$7 + 2x^2$
Value When $x = 3$	45	45	32	25

3. Determine the value of $3x^2 - 5x + 10$ when $x = 5$.

60

Problems 4–5: Here are two figures.



4. Match each figure with an expression that represents its area. One expression will not have a match.

$x^2 + 3x$

$(x + 3)^2$

$x^2 + 3^2$

Figure A

Figure B

5. Which of the three expressions in Problem 4 has the greatest value when $x = 5$?

$(x + 3)^2$

6. Evaluate each expression for the given value of x .

a 0.3^x , when x is 3

0.027

b $x^2 + 6$, when x is 4

22

c $2x^2 + 3x$, when x is 5

65

d $8x + x^2$, when x is 4

48

7. Clare evaluated the expression $6x^2 - 12$ when x is 2. She says the value of the expression is 1,716. Is Clare correct? Explain your thinking.

No. Explanations vary. Clare should have found 2^2 before multiplying by 6. The correct answer is 12.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.4.13

1. Determine the value of each expression when $x = 4$.

Expression	$2x + 6$	x^2	$x^2 - 8$	$(x + 2)^2$
Value When $x = 4$	22	64	8	36

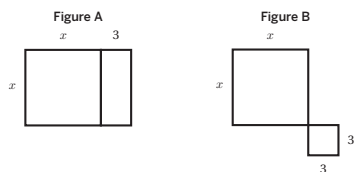
2. Determine the value of each expression when $x = 3$.

Expression	$x^2 + 6x$	$5x^2$	$(x - 1)^2$	$7 + 2x^2$
Value When $x = 3$	45	45	32	25

3. Determine the value of $3x^2 - 5x + 10$ when $x = 5$.

60

Problems 4–5: Here are two figures.



4. Match each figure with an expression that represents its area. One expression will not have a match.

$x^2 + 3x$

$(x + 3)^2$

$x^2 + 3^2$

Figure A

Figure B

5. Which of the three expressions in Problem 4 has the greatest value when $x = 5$?

$(x + 3)^2$

Unit 4 Lesson 13

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Name: _____ Date: _____ Period: _____

6. Evaluate each expression for the given value of x .

a. 0.3^x , when x is 3

0.027

b. $x^2 + 6$, when x is 4

22

c. $2x^2 + 3x$, when x is 5

65

d. $8x + x^2$, when x is 4

48

7. Clare evaluated the expression $6x^2 - 12$ when x is 2. She says the value of the expression is 1,716. Is Clare correct? Explain your thinking.

No. Explanations vary. Clare should have found 2^2 before multiplying by 6. The correct answer is 12.

Unit 4 Lesson 13

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	6.EE.A.2.C, 6.EE.B.5
2	2	6.EE.A.2.C
3	3	6.EE.A.2.C
4	1	6.EE.A.2.C, 6.EE.B.5
5	2	6.EE.A.2.C
6	2	6.EE.A.2.C
7	3	6.EE.A.2.C

Notes:

Additional Practice

ACC6.4.14

Problems 1–3: A florist is making bouquets of flowers. He puts 12 flowers in each bouquet.

1. Complete the table.

Number of Bouquets, b	Number of Flowers, f
1	12
2	24
6	72
8	96

2. Which variable is the independent variable, b or f ? Explain your thinking.

b. Explanations vary. The number of bouquets the florist is making determines how many flowers he needs to use.

3. Which of these equations represents the relationship? Circle one.

$b = 12f$ $f = 12b$ Neither

Explain your thinking

Explanations vary. The number of flowers is 12 times as much as the number of bouquets.

Problems 4–5: Adhira has a coupon for \$8 off any item at a store.

4. Complete the table. All values are in dollars.

Original Price, b	Price With Coupon, c
10	2
20	12
35	27
48	40

5. Which equation represents the relationship in the table?

A. $p = 8c$

B. $c = p - 8$

C. $c = 8p$

D. $p + 8 = c$

Problems 6–8: The table shows the number of robots a factory can create in different numbers of hours.

6. Write an equation to represent the relationship between the number of hours, h , and the number of robots, r , the factory produces.

$r = 6h$ (or equivalent)

Number of Hours, h	Number of Robots, r
1	6
2	12
6	36
15	90

7. Which variable is the dependent variable, r or h ? Explain your thinking.

r . *Explanations vary. The number of robots depends on how long the factory is open. You use the number of hours to calculate the number of robots.*

8. Kris says that the factory could produce 100 robots in less than 24 hours. Is Kris correct? Explain your thinking.

Yes. *Explanations vary. In 24 hours, the factory could produce $24 \cdot 6 = 144$ robots, so it would take less time than that to produce 100.*

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.4.14

Problems 1–3: A florist is making bouquets of flowers. He puts 12 flowers in each bouquet.

1. Complete the table.

Number of Bouquets, b	Number of Flowers, f
1	12
2	24
6	72
8	96

2. Which variable is the independent variable, b or f ? Explain your thinking.

b . *Explanations vary. The number of bouquets the florist is making determines how many flowers he needs to use.*

3. Which of these equations represents the relationship? Circle one.

$b = 12f$ $f = 12b$ Neither

Explain your thinking

Explanations vary. The number of flowers is 12 times as much as the number of bouquets.

Problems 4–5: Adhira has a coupon for \$8 off any item at a store.

4. Complete the table. All values are in dollars.

Original Price, b	Price With Coupon, c
10	2
20	12
35	27
48	40

5. Which equation represents the relationship in the table?

- A. $p = 8c$
 B. $c = p - 8$
 C. $c = 8p$
 D. $p + 8 = c$

Unit 4 Lesson 14

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Name: _____ Date: _____ Period: _____

Problems 6–8: The table shows the number of robots a factory can create in different numbers of hours.

6. Write an equation to represent the relationship between the number of hours, h , and the number of robots, r , the factory produces.

$r = 6h$ (or equivalent)

Number of Hours, h	Number of Robots, r
1	6
2	12
6	36
15	90

7. Which variable is the dependent variable, r or h ? Explain your thinking.

r . *Explanations vary. The number of robots depends on how long the factory is open. You use the number of hours to calculate the number of robots.*

8. Kris says that the factory could produce 100 robots in less than 24 hours. Is Kris correct? Explain your thinking.

Yes. *Explanations vary. In 24 hours, the factory could produce $24 \cdot 6 = 144$ robots, so it would take less time than that to produce 100.*

Unit 4 Lesson 14

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	6.RP.A.3.A
2	2	6.EE.C.9
3	2	6.EE.C.9, 6.RP.A.3.A
4	3	6.EE.C.9
5	1	6.RP.A.3.A
6	2	6.EE.C.9
7	2	6.EE.C.9, 6.RP.A.3.A
8	3	6.EE.C.9

Notes:

Additional Practice

ACC6.4.15

Problems 1–3: Han, Elena, and Clare are training for a bike race, and they bike at a constant speed.

1. Complete the table.

Time (hours), t	Han's Distance (miles), h	Elena's Distance (miles), e	Clare's Distance (miles), c
0.25	3	2.5	3.75
0.5	6	5	7.5
2	24	20	30
3	36	30	45
4	48	40	60

2. Write three equations that represent the distance d traveled, in miles, given the time t , in hours.

Han

Elena

Clare

$d = 12 \cdot t$ (or equivalent)

$d = 10 \cdot t$ (or equivalent)

$d = 15 \cdot t$ (or equivalent)

3. For your equations for Problem 2, which is the dependent variable and which is the independent variable?

The time is the independent variable and the distance is the dependent variable.

Problems 4–7: Bard reads 35 pages in an hour.

4. Assuming Bard reads at a constant rate, complete the table.

Time (hours), t	2	3.2	t	4	1.6
Pages, p	70	112	$35t$	140	56

5. Refer to the table.

- a** Write an equation that relates the number of pages read, p , to the time, t , in hours.

$35 \cdot t = p$ (or equivalent)

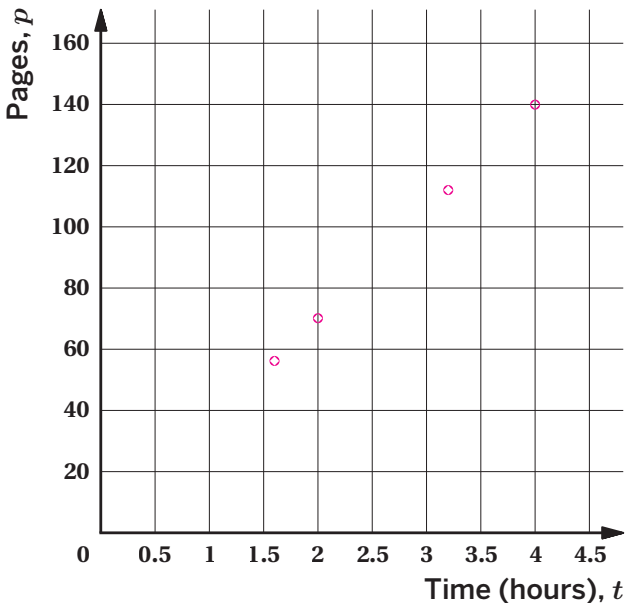
- b** Identify the independent and dependent variables in your equation.

Time is the independent variable. The number of pages read is the dependent variable.

6. Use the points in the table to create a graph that shows the relationship between time, t , and the number of pages, p , Bard reads.

7. Write an equation that Bard can use to determine the time, t , it will take to read 91 pages.

$t = \frac{1}{35}p$ (or equivalent)



Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.4.15

Problems 1–3: Han, Elena, and Clare are training for a bike race, and they bike at a constant speed.

1. Complete the table.

Time (hours), t	Han's Distance (miles), h	Elena's Distance (miles), e	Clare's Distance (miles), c
0.25	3	2.5	3.75
0.5	6	5	7.5
2	24	20	30
3	36	30	45
4	48	40	60

2. Write three equations that represent the distance d traveled, in miles, given the time t , in hours.

Han: $d = 12 \cdot t$ (or equivalent) Elena: $d = 10 \cdot t$ (or equivalent) Clare: $d = 15 \cdot t$ (or equivalent)

3. For your equations for Problem 2, which is the dependent variable and which is the independent variable?

The time is the independent variable and the distance is the dependent variable.

Unit 4 Lesson 15

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Name: _____ Date: _____ Period: _____

Problems 4–7: Bard reads 35 pages in an hour.

4. Assuming Bard reads at a constant rate, complete the table.

Time (hours), t	2	3.2	t	4	1.6
Pages, p	70	112	35 t	140	56

5. Refer to the table.

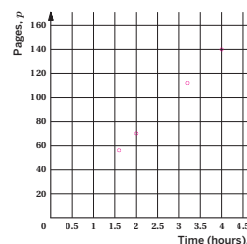
a. Write an equation that relates the number of pages read, p , to the time, t , in hours.

$35 \cdot t = p$ (or equivalent)

b. Identify the independent and dependent variables in your equation.

Time is the independent variable. The number of pages read is the dependent variable.

6. Use the points in the table to create a graph that shows the relationship between time, t , and the number of pages, p , Bard reads.



7. Write an equation that Bard can use to determine the time, t , it will take to read 91 pages.

$t = \frac{1}{35}p$ (or equivalent)

Unit 4 Lesson 15

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	6.RP.A.3.A
2	2	6.EE.C.9
3	2	6.EE.C.9
4	1	6.RP.A.3.A
5	2	6.EE.C.9
6	2	6.EE.C.9, 6.RP.A.3.A
7	2	6.RP.A.3.A, 6.RP.A.3.B

Notes:

Additional Practice

6.15

1. Maria is making lemonade for a picnic with friends. The equation $w = l + 6$ represents the relationship between the amount of lemonade concentrate (in cups), l , and water (in cups), w . Which table represents the same relationship?

A.

l	w
8	2
10	4
12	6

B.

l	w
2	8
4	10
6	12

C.

l	w
10	4
8	2
6	0

D.

l	w
6	10
8	8
10	6

Problems 2–4: Sanjeev is buying limes for \$0.60 each.

2. Write an equation that represents how much money Sanjeev spends, m , for buying a number of limes, l .

$$m = 0.60l$$

3. Complete the table that represents this situation.

l	m
6	\$3.60
7	\$4.20
9	\$5.40
11	\$6.60

4. Explain what $l = 9$ and $m = \$5.40$ means in this situation.

Explanations vary. This means that if Sanjeev buys 9 limes, it would cost \$5.40.

5. Sanjeev spends \$9.00 on limes. How many limes did he purchase?

- A. 12 limes
- B. 13 limes
- C. 14 limes
- D. 15 limes**

Name: _____ Date: _____ Period: _____

Additional Practice

6.15

1. Maria is making lemonade for a picnic with friends. The equation $w = l + 6$ represents the relationship between the amount of lemonade concentrate (in cups), l , and water (in cups), w . Which table represents the same relationship?

A.

l	w
8	2
10	4
12	6

B.

l	w
2	8
4	10
6	12

C.

l	w
10	4
8	2
6	0

D.

l	w
6	10
8	8
10	6

Unit 6 Lesson 15

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Name: _____ Date: _____ Period: _____

Problems 2–4: Sanjeev is buying limes for \$0.60 each.

2. Write an equation that represents how much money Sanjeev spends, m , for buying a number of limes, l .

$m = 0.60l$

3. Complete the table that represents this situation.

l	m
6	\$3.60
7	\$4.20
9	\$5.40
11	\$6.60

4. Explain what $l = 9$ and $m = \$5.40$ means in this situation.

Explanations vary. This means that if Sanjeev buys 9 limes, it would cost \$5.40.

5. Sanjeev spends \$9.00 on limes. How many limes did he purchase?

A. 12 limes

B. 13 limes

C. 14 limes

D. 15 limes

Unit 6 Lesson 15

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	6.NS.B.2, 6.NS.B.3
2	1	6.NS.B.2, 6.RP.A.3
3	1	6.NS.B.2, 6.RP.A.3
4	1	6.NS.B.2, 6.RP.A.3
5	1	6.NS.B.2, 6.RP.A.3

Notes:

Additional Practice

ACC6.4.16

1. Select *all* the expressions that are equivalent to the expression $3a + 6a$.

☒ A. $4a + 5a$

☒ B. $a(3 + 6)$

☒ C. $6a + 3a$

☐ D. $a \div \frac{1}{18}$

☒ E. $9a$

☐ F. $18a$

☐ G. $3(a + 6a)$

☒ H. $a \cdot 9$

2. Determine the value of b . Show your thinking.

a 45% of b is 225.
500

b 300% of b is 12.
4

c 12% of b is 4.8.
40

3. Solve each equation. Show your thinking.

a $180 = 15c$
 $c = 12$. Work varies.
 $180 \div 15 = 15c \div 15$
 $12 = c$

b $3\frac{1}{5} + c = 7$
 $c = 3\frac{4}{5}$. Work varies.
 $3\frac{1}{5} + c - 3\frac{1}{5} = 7 - 3\frac{1}{5}$
 $c = 3\frac{4}{5}$

c $c - 17.34 = 29.2$
 $c = 46.54$. Work varies.
 $c - 17.34 + 17.34 = 29.2 + 17.34$
 $c = 46.54$

d $\frac{9}{8} = 2\frac{1}{2}c$
 $c = \frac{9}{20}$. Work varies.
 $\frac{9}{8} \div 2\frac{1}{2} = 2\frac{1}{2}c \div 2\frac{1}{2}$
 $\frac{9}{8} \cdot \frac{2}{5} = c$
 $\frac{18}{40} = c$

4. Select *all* the expressions that are equal to $5 \cdot 5 \cdot 5 \cdot 5$.

☐ A. $5 \cdot 4$

☒ B. $5^2 \cdot 5^2$

☐ C. $5^3 + 5$

☐ D. 4^5

☒ E. 5^4

☒ F. $5 \cdot 5^3$

5. Evaluate each expression. Show your thinking.

a $30 - 3^3 = 3$

Work varies.

$30 - 27 = 3$

b $5^3 + 16 = 141$

Work varies.

$125 + 16 = 141$

c $12^2 - 8^2 = 80$

Work varies.

$144 - 64 = 80$

d $8 \cdot \left(\frac{1}{2}\right)^3 = 1$ (or equivalent)

Work varies.

$8 \cdot \frac{1}{8} = \frac{8}{8} = 1$

e $2^4 \cdot 2 = 32$

Work varies.

$16 \cdot 2 = 32$

f $4^4 \div 4 = 64$

Work varies.

$254 \div 4 = 64$

6. Select *all* the expressions that are equivalent to the expression $4(3x - 6)$.

☐ A. $3(4x - 6)$

☒ B. $2(6x - 12)$

☐ C. $8(2x - 3)$

☒ D. $6(2x - 4)$

☒ E. $12x - 24$

7. Shawn separately baked 6 batches of pretzels, for a total of 72 minutes. Each batch of pretzels baked for b minutes. Select *all* the equations that represent this situation.

☒ A. $6 \cdot b = 72$

☐ B. $b - 6 = 72$

☒ C. $b = 72 \div 6$

☐ D. $6 \cdot 72 = b$

☐ E. $72 = b + 6$

8. Priya says the solution to the equation $\frac{6}{5}x = \frac{1}{3}$ is $\frac{2}{5}$. Is Priya correct? Explain your thinking.

No. Explanations vary. Priya did not multiply by the reciprocal of $\frac{6}{5}$.

The solution to the equation is $x = \frac{1}{3} \cdot \frac{5}{6} = \frac{5}{18}$.

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.4.16

1. Select *all* the expressions that are equivalent to the expression $3a + 6a$.

- ☒ A. $4a + 5a$ ☒ B. $a(3 + 6)$
☒ C. $6a + 3a$ ☐ D. $a \div \frac{1}{18}$
☒ E. $9a$ ☐ F. $18a$
☐ G. $3(a + 6a)$ ☒ H. $a \cdot 9$

2. Determine the value of b . Show your thinking.

- a. 45% of b is 225.
500
 b. 300% of b is 12.
4
 c. 12% of b is 4.8.
40

3. Solve each equation. Show your thinking.

- a. $180 = 15c$
 $c = 12$. *Work varies.*
 $180 \div 15 = 15c \div 15$
 $12 = c$
 b. $\frac{1}{5} + c = 7$
 $c = 6\frac{4}{5}$. *Work varies.*
 $\frac{1}{5} + c - \frac{1}{5} = 7 - \frac{1}{5}$
 $c = 6\frac{4}{5}$
 c. $c - 17.34 = 29.2$
 $c = 46.54$. *Work varies.*
 $c - 17.34 + 17.34 = 29.2 + 17.34$
 $c = 46.54$
 d. $\frac{9}{8} = 2\frac{1}{2}c$
 $c = \frac{9}{20}$. *Work varies.*
 $\frac{9}{8} \div 2\frac{1}{2} = 2\frac{1}{2}c \div 2\frac{1}{2}$
 $\frac{9}{8} \cdot \frac{2}{5} = c$
 $\frac{18}{40} = c$

4. Select *all* the expressions that are equal to $5 \cdot 5 \cdot 5 \cdot 5$.

- ☐ A. $5 \cdot 4$ ☒ B. $5^2 \cdot 5^2$
☐ C. $5^3 + 5$ ☐ D. 4^5
☒ E. 5^4 ☒ F. $5 \cdot 5^3$

Unit 4 Lesson 16

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Name: _____ Date: _____ Period: _____

5. Evaluate each expression. Show your thinking.

- a. $30 - 3^2 = 3$
Work varies.
 $30 - 27 = 3$
 b. $5^2 + 16 = 141$
Work varies.
 $125 + 16 = 141$
 c. $12^2 - 8^2 = 80$
Work varies.
 $144 - 64 = 80$
 d. $8 \cdot \left(\frac{1}{2}\right)^3 = 1$ (or equivalent)
Work varies.
 $8 \cdot \frac{1}{8} = 1$
 e. $2^4 \cdot 2 = 32$
Work varies.
 $16 \cdot 2 = 32$
 f. $4^4 \div 4 = 64$
Work varies.
 $254 \div 4 = 64$

6. Select *all* the expressions that are equivalent to the expression $4(3x - 6)$.

- ☐ A. $3(4x - 6)$ ☒ B. $2(6x - 12)$
☐ C. $8(2x - 3)$ ☒ D. $6(2x - 4)$
☒ E. $12x - 24$

7. Shawn separately baked 6 batches of pretzels, for a total of 72 minutes. Each batch of pretzels baked for b minutes. Select *all* the equations that represent this situation.

- ☒ A. $6 \cdot b = 72$ ☐ B. $b - 6 = 72$
☒ C. $b = 72 \div 6$ ☐ D. $6 \cdot 72 = b$
☐ E. $72 = b + 6$

8. Priya says the solution to the equation $\frac{6}{5}x = \frac{1}{3}$ is $\frac{2}{5}$. Is Priya correct? Explain your thinking.

No. *Explanations vary. Priya did not multiply by the reciprocal of $\frac{6}{5}$.*
 The solution to the equation is $x = \frac{1}{3} \cdot \frac{5}{6} = \frac{5}{18}$.

Unit 4 Lesson 16

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	6.EE.A.4
2	1	6.RP.A.3.C
3	1	6.EE.B.7
4	1	6.EE.A.1
5	1	6.EE.A.1, 6.EE.A.2.C
6	2	6.EE.A.3, 6.EE.A.4
7	2	6.EE.B.6
8	3	6.EE.B.5

Notes:

Additional Practice

ACC6.5.01

Problems 1–5: The table shows an original mixture for orange paint, as well as four other mixtures.

Mixture	Yellow Paint (cups)	Red Paint (cups)
Original	9	6
Mixture A	6	8
Mixture B	6	4
Mixture C	3	2
Mixture D	12	18

1. Select *all* the mixtures that will create the same shade of orange as the original mixture.

- ☐ A. Mixture A
☒ B. Mixture B
☒ C. Mixture C
☐ D. Mixture D

2. Choose one mixture that creates the same shade of orange as the original. Explain how you know it creates the same shade of orange as the original.

Explanations vary. Mixture C uses 1.5 cups of yellow paint for every cup of red paint, just like the original.

3. Choose a mixture that does not have the same shade of orange as the original. How do the two mixtures compare?

Responses vary. Mixture A uses 1.5 cups of yellow paint for every 2 cups of red paint. Mixture A will be a “redder” shade of orange because there is more red to yellow in this mixture.

4. How much red paint would you need to mix with 1 cup of yellow paint to make a mixture that has the same shade of orange as the original mixture?

$\frac{2}{3}$ cups of red paint

5. Complete the table to create a different mixture that will also have the same shade of orange as the original mixture.

Responses vary. Sample shown in table.

Yellow Paint (cups)	Red Paint (cups)
12	8

6. Brenden mixed 4 cups of blue paint with 3 cups of yellow paint to make the perfect color green to paint his bedroom.

Complete the table to show several other ways to make this color.

Responses shown in table.

Yellow Paint (cups)	Blue Paint (cups)
3	4
9	12
12	16
15	20

7. Select *all* the ratios that are equivalent to 3 : 4.

☐ A. 50 : 100

☒ B. 18 : 24

☐ C. $\frac{1}{4} : \frac{1}{5}$

☒ D. 75 : 100

☐ E. 4 : 5

☒ F. $\frac{1}{3} : \frac{1}{4}$

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.5.01

Problems 1–5: The table shows an original mixture for orange paint, as well as four other mixtures.

Mixture	Yellow Paint (cups)	Red Paint (cups)
Original	9	6
Mixture A	6	8
Mixture B	6	4
Mixture C	3	2
Mixture D	12	18

1. Select *all* the mixtures that will create the same shade of orange as the original mixture.

☐ A. Mixture A
☒ B. Mixture B
☒ C. Mixture C
☐ D. Mixture D

2. Choose one mixture that creates the same shade of orange as the original. Explain how you know it creates the same shade of orange as the original.

Explanations vary. Mixture C uses 1.5 cups of yellow paint for every cup of red paint, just like the original.

3. Choose a mixture that does not have the same shade of orange as the original. How do the two mixtures compare?

Responses vary. Mixture A uses 1.5 cups of yellow paint for every 2 cups of red paint. Mixture A will be a “redder” shade of orange because there is more red to yellow in this mixture.

4. How much red paint would you need to mix with 1 cup of yellow paint to make a mixture that has the same shade of orange as the original mixture?

$\frac{2}{3}$ cups of red paint

5. Complete the table to create a different mixture that will also have the same shade of orange as the original mixture.

Responses vary. Sample shown in table.

Yellow Paint (cups)	Red Paint (cups)
12	8

Unit 5 Lesson 1

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Name: _____ Date: _____ Period: _____

6. Brenden mixed 4 cups of blue paint with 3 cups of yellow paint to make the perfect color green to paint his bedroom.

Complete the table to show several other ways to make this color.

Responses shown in table.

Yellow Paint (cups)	Blue Paint (cups)
3	4
9	12
12	16
15	20

7. Select *all* the ratios that are equivalent to 3 : 4.

☐ A. 50 : 100
☒ B. 18 : 24
☐ C. $\frac{1}{4} : \frac{1}{5}$
☒ D. 75 : 100
☐ E. 4 : 5
☒ F. $\frac{1}{3} : \frac{1}{4}$

Unit 5 Lesson 1

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	7.RP.A
2	2	7.RP.A
3	2	7.RP.A
4	2	7.RP.A
5	2	7.RP.A
6	2	7.RP.A
7	2	7.RP.A

Notes:

Additional Practice

ACC6.5.02

Problems 1–4: Complete the tables so that each relationship is proportional.

1.

x	y
16	4
20	5
36	9

2.

x	y
1	2.4
4	9.6
7.5	18

3.

x	y
12	36
1	3
0	0

4.

x	y
0.4	1
1	2.5
6	15

- 5.** DesKayaks rents one-person and two-person kayaks for a fee of \$10 an hour plus \$2.50 per person for gear. The table shows the cost for several groups of people.

Is the relationship between the number of people renting kayaks and the fees charged a proportional relationship? Explain your thinking.

No. Explanations vary. This is not a proportional relationship because when I multiply 2 by 12.50 from the first row, I get 24, not 15.00.

Number of People in Group	Total Fees (\$)
1	12.50
2	15.00
3	27.50
4	30.00

6. Sabaan’s favorite smoothie shop charges \$6.20 for 12 ounces. Complete the table so that it shows a proportional relationship between ounces of smoothie and cost. Show or explain your thinking.

Explanations vary. I divided 6.6 by 12 to find the cost per ounce of smoothie, which was \$0.55. Then, I multiplied 18 by that number to find the cost of an 18 ounce smoothie. Then, I divided 13.20 by that number to find how many ounces of smoothie I can buy for \$13.20.

Smoothie (oz)	Cost (\$)
12	6.60
18	9.90
24	13.20

7. At an airport in Mexico City, Sara exchanged 300 U.S. dollars for 6,156 Mexican pesos. Complete the table so that it shows a proportional relationship between U.S. dollars and Mexican pesos. Show or explain your thinking.

Explanations vary. I divided 6,156 by 300 to find how many pesos \$1 U.S. was worth, which was 20.52. Then I divided 10,260 pesos (and then 1 peso) by that number to find how many U.S. dollars that was worth.

U.S. Dollars	Mexican Pesos
300	6,156
1	20.52
500	10,260
≈0.05	1

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.5.02

Problems 1–4: Complete the tables so that each relationship is proportional.

1.

x	y
16	4
20	5
36	9

2.

x	y
1	2.4
4	9.6
7.5	18

3.

x	y
12	36
1	3
0	0

4.

x	y
0.4	1
1	2.5
6	15

5. DesKayaks rents one-person and two-person kayaks for a fee of \$10 an hour plus \$2.50 per person for gear. The table shows the cost for several groups of people.

Is the relationship between the number of people renting kayaks and the fees charged a proportional relationship? Explain your thinking.

No. *Explanations vary. This is not a proportional relationship because when I multiply 2 by 12.50 from the first row, I get 24, not 15.00.*

Number of People in Group	Total Fees (\$)
1	12.50
2	15.00
3	27.50
4	30.00

Unit 5 Lesson 2

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Name: _____ Date: _____ Period: _____

6. Sabaan's favorite smoothie shop charges \$6.20 for 12 ounces. Complete the table so that it shows a proportional relationship between ounces of smoothie and cost. Show or explain your thinking.

Smoothie (oz)	Cost (\$)
12	6.60
18	9.90
24	13.20

Explanations vary. I divided 6.6 by 12 to find the cost per ounce of smoothie, which was \$0.55. Then, I multiplied 18 by that number to find the cost of an 18 ounce smoothie. Then, I divided 13.20 by that number to find how many ounces of smoothie I can buy for \$13.20.

7. At an airport in Mexico City, Sara exchanged 300 U.S. dollars for 6,156 Mexican pesos. Complete the table so that it shows a proportional relationship between U.S. dollars and Mexican pesos. Show or explain your thinking.

U.S. Dollars	Mexican Pesos
300	6,156
1	20.52
500	10,260
≈0.05	1

Explanations vary. I divided 6,156 by 300 to find how many pesos \$1 U.S. was worth, which was 20.52. Then I divided 10,260 pesos (and then 1 peso) by that number to find how many U.S. dollars that was worth.

Unit 5 Lesson 2

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	1	7.RP.A.2
2	1	7.RP.A.2
3	1	7.RP.A.2
4	1	7.RP.A.2
5	2	7.RP.A.2.A
6	2	7.RP.A.2
7	2	7.RP.A.2

Notes:

Additional Practice

ACC6.5.04

1. Diego is running at a speed of 3.5 meters per second. Complete the table.

Time (seconds)	Distance (meters)
1	3.5
15	52.5
20	70
60	210
x	$3.5x$

2. The table shows a proportional relationship. Determine the constant of proportionality that completes the equation.

$$c = 2.5 w$$

w	c
2	5.00
0.5	1.25
0.25	0.63

3. The table shows the amount of money a tourist would receive in Egyptian pounds in exchange for different amounts of money in British pounds.

- a Complete the table with the missing values if the exchange rate is 1 British pound to 21.48 Egyptian pounds.

Responses shown in table.

British Pounds, x	Egyptian Pounds, y
100	2,148
250	5,370
4.66	100

- b Write an equation to represent the amount of Egyptian pounds, y , the tourist receives in exchange for x British pounds.

$$y = 21.48x$$

4. Han makes fruit punch following a recipe that uses one 48-ounce can of pineapple juice to one 6-ounce can of frozen lemonade.

- a Complete the table.
- b Write an equation to model how many ounces of pineapple juice, p , are needed for f ounces of frozen lemonade.

$$p = 8f$$

Pineapple Juice (oz)	Frozen Lemonade (oz)
48	6
16	2
12	$1\frac{1}{2}$ (or equivalent)
4	$\frac{1}{2}$

5. Bard buys a half pound of turkey breast for \$4.99 at the local deli.

- a What is the price per pound of the turkey breast?
- b Write an equation to represent the price in dollars of the turkey breast, y , given its weight, x , in pounds.

$$y = 9.98x$$

- c Use your equation to complete the table.

Weight (lb)	0.33	0.68	1.82	2.70
Price (\$)	3.29	6.75	18.16	26.94

6. Noah and Andre examine the table shown representing a proportional relationship. Noah writes the equation $y = 1.5x$ to represent the relationship. Andre writes the equation $y = \frac{2}{3}x$ to represent the relationship.

Who is correct? Explain your thinking.

x	y
12	8
39	26
57	38

Andre. Explanations vary. If you multiply each value of x in the table by $\frac{2}{3}$, the result is its corresponding value of y .

Name: _____ Date: _____ Period: _____

Additional Practice

ACC6.5.04

1. Diego is running at a speed of 3.5 meters per second. Complete the table.

Time (seconds)	Distance (meters)
1	3.5
15	52.5
20	70
60	210
x	$3.5x$

2. The table shows a proportional relationship. Determine the constant of proportionality that completes the equation.

$c = 2.5 \cdot w$

w	c
2	5.00
0.5	1.25
0.25	0.63

3. The table shows the amount of money a tourist would receive in Egyptian pounds in exchange for different amounts of money in British pounds.

British Pounds, x	Egyptian Pounds, y
100	2,148
250	5,370
4.66	100

- a. Complete the table with the missing values if the exchange rate is 1 British pound to 21.48 Egyptian pounds.

Responses shown in table.

- b. Write an equation to represent the amount of Egyptian pounds, y , the tourist receives in exchange for x British pounds.

$y = 21.48x$

Unit 5 Lesson 4

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Name: _____ Date: _____ Period: _____

4. Han makes fruit punch following a recipe that uses one 48-ounce can of pineapple juice to one 6-ounce can of frozen lemonade.

- a. Complete the table.
b. Write an equation to model how many ounces of pineapple juice, p , are needed for f ounces of frozen lemonade.
 $p = 8f$

Pineapple Juice (oz)	Frozen Lemonade (oz)
48	6
16	2
12	$1\frac{1}{2}$ (or equivalent)
4	$\frac{1}{2}$

5. Bard buys a half pound of turkey breast for \$4.99 at the local deli.

- a. What is the price per pound of the turkey breast?
 $\$9.98$
b. Write an equation to represent the price in dollars of the turkey breast, y , given its weight, x , in pounds.
 $y = 9.98x$
c. Use your equation to complete the table.

Weight (lb)	0.33	0.68	1.82	2.70
Price (\$)	3.29	6.75	18.16	26.94

6. Noah and Andre examine the table shown representing a proportional relationship. Noah writes the equation $y = 1.5x$ to represent the relationship. Andre writes the equation $y = \frac{2}{3}x$ to represent the relationship.
Who is correct? Explain your thinking.

x	y
12	8
39	26
57	38

Andre. Explanations vary. If you multiply each value of x in the table by $\frac{2}{3}$, the result is its corresponding value of y .

Unit 5 Lesson 4

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Additional Practice

Practice Problem Analysis

Problem	DOK	Standard(s)
1	2	7.RP.A.2, 7.RP.A.2.C
2	1	7.RP.A.2, 7.RP.A.2.C
3	2	7.RP.A.2, 7.RP.A.2.C
4	2	7.RP.A.2, 7.RP.A.2.C
5	2	7.RP.A.2, 7.RP.A.2.C
6	2	7.RP.A.2, 7.RP.A.2.C

Notes: